Lecture 13:

Global Illumination & Path Tracing

Computer Graphics and Imaging
UC Berkeley CS184/284A

Hard Shadows

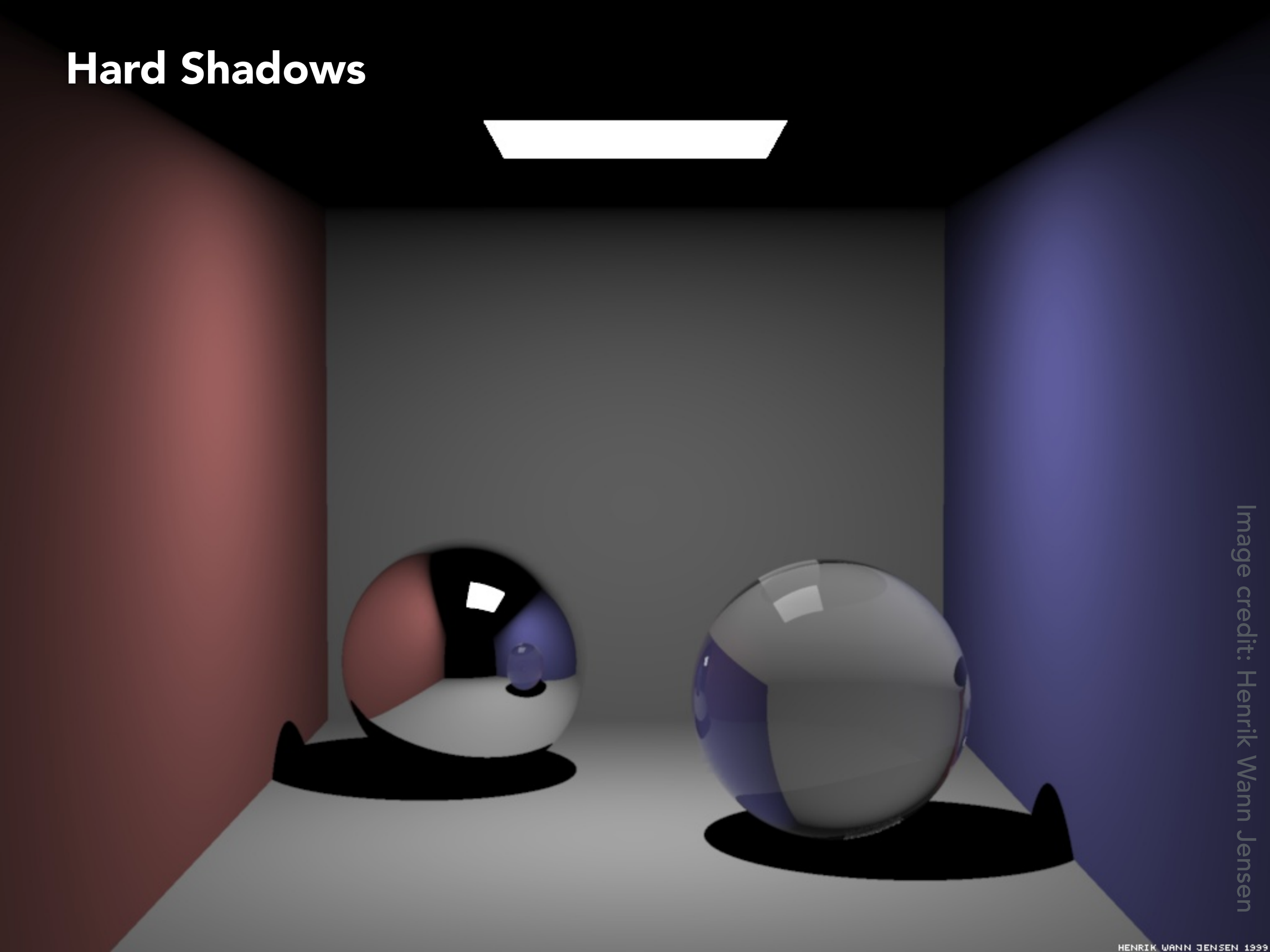


Image credit: Henrik Wann Jensen

Soft Shadows

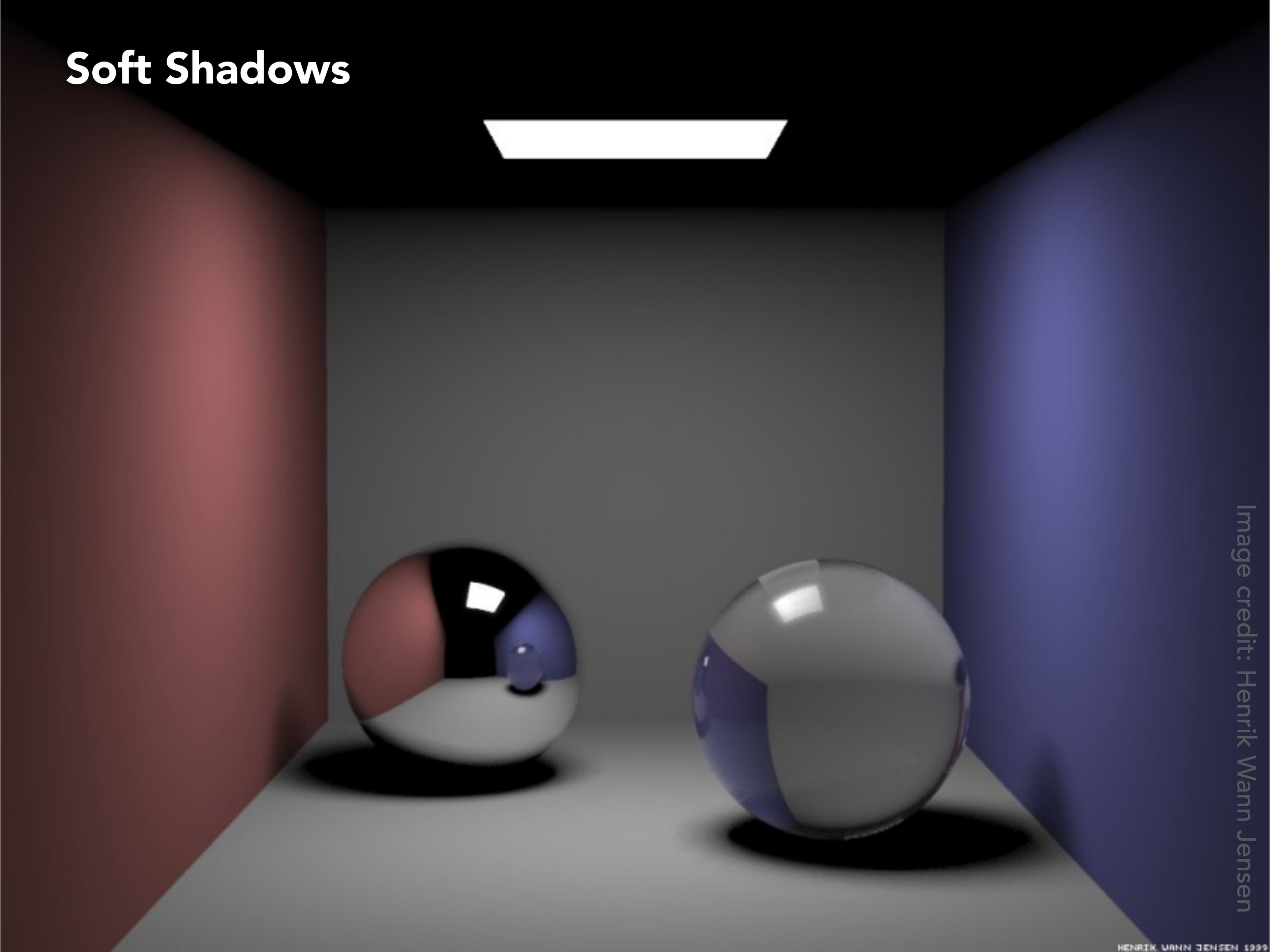


Image credit: Henrik Wann Jensen

+ Caustics

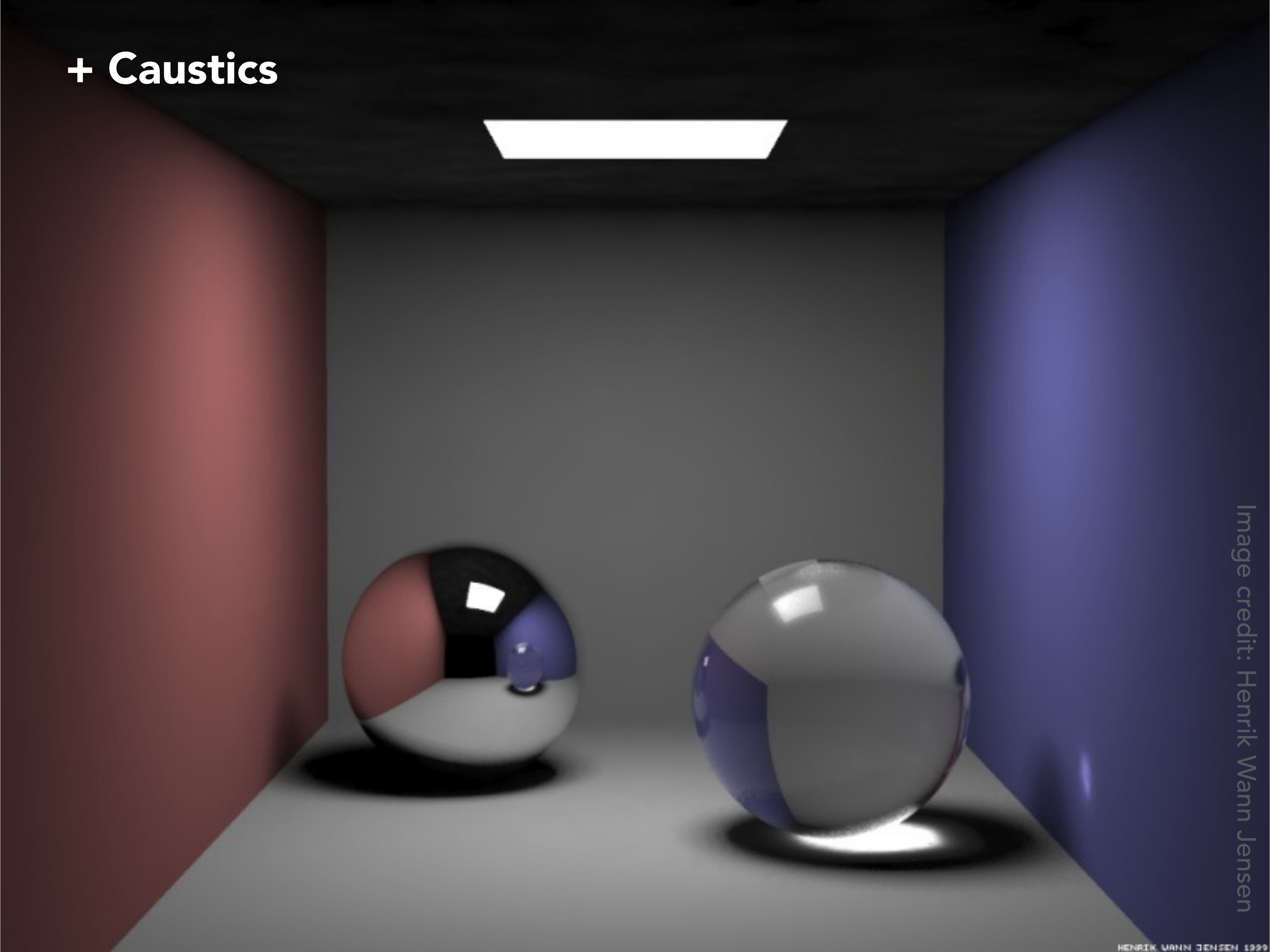


Image credit: Henrik Wann Jensen

+ Diffuse Inter-Reflections = Global Illumination

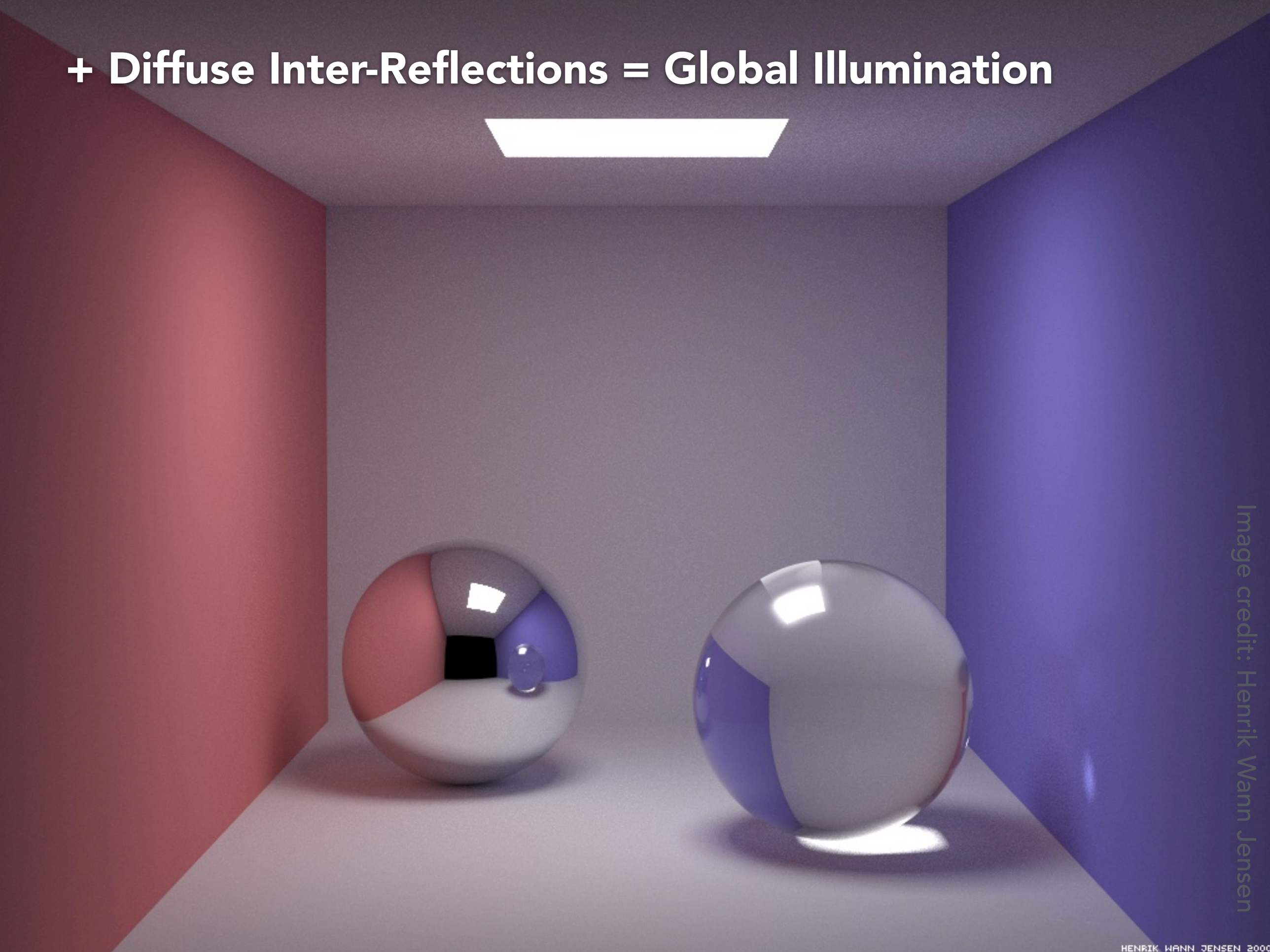
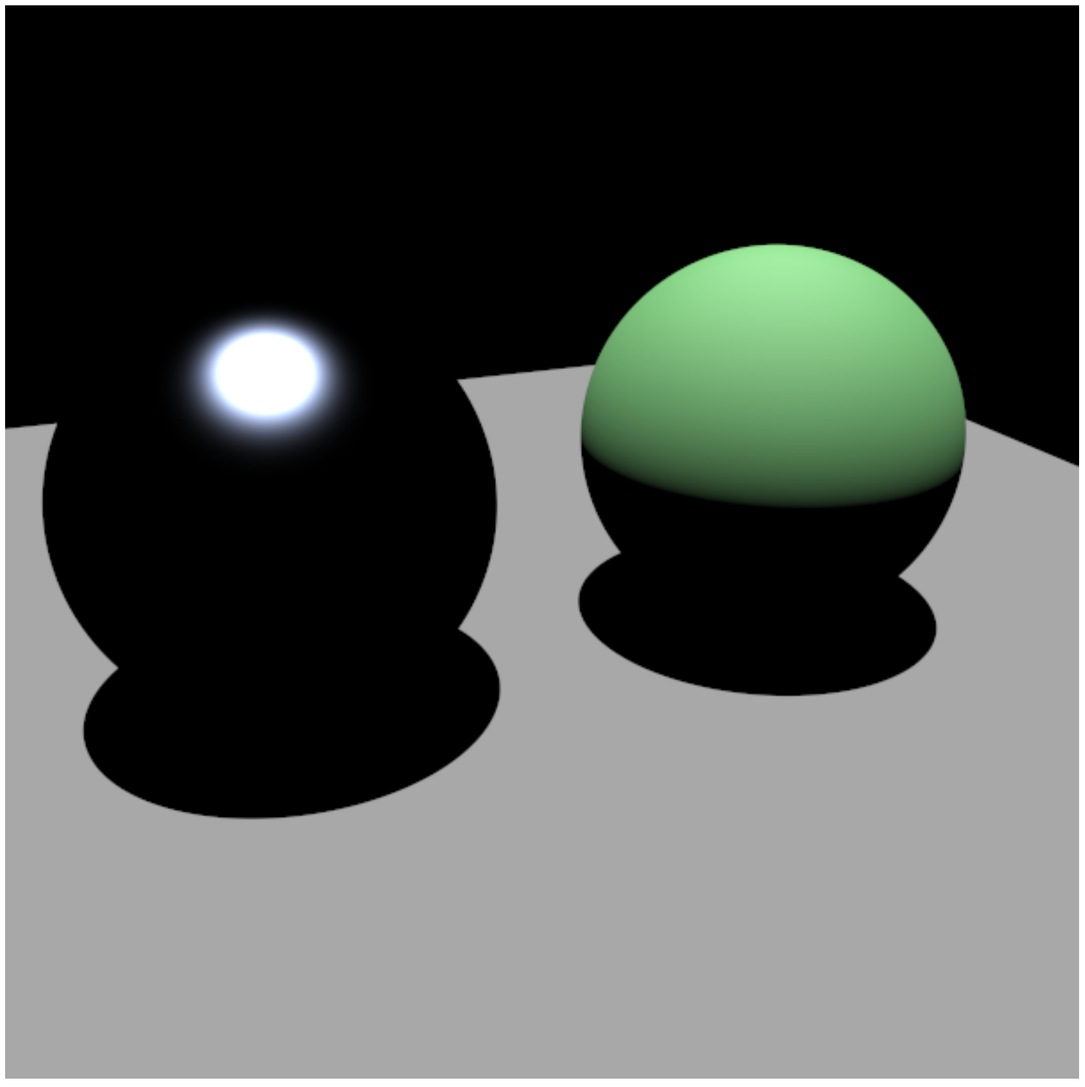


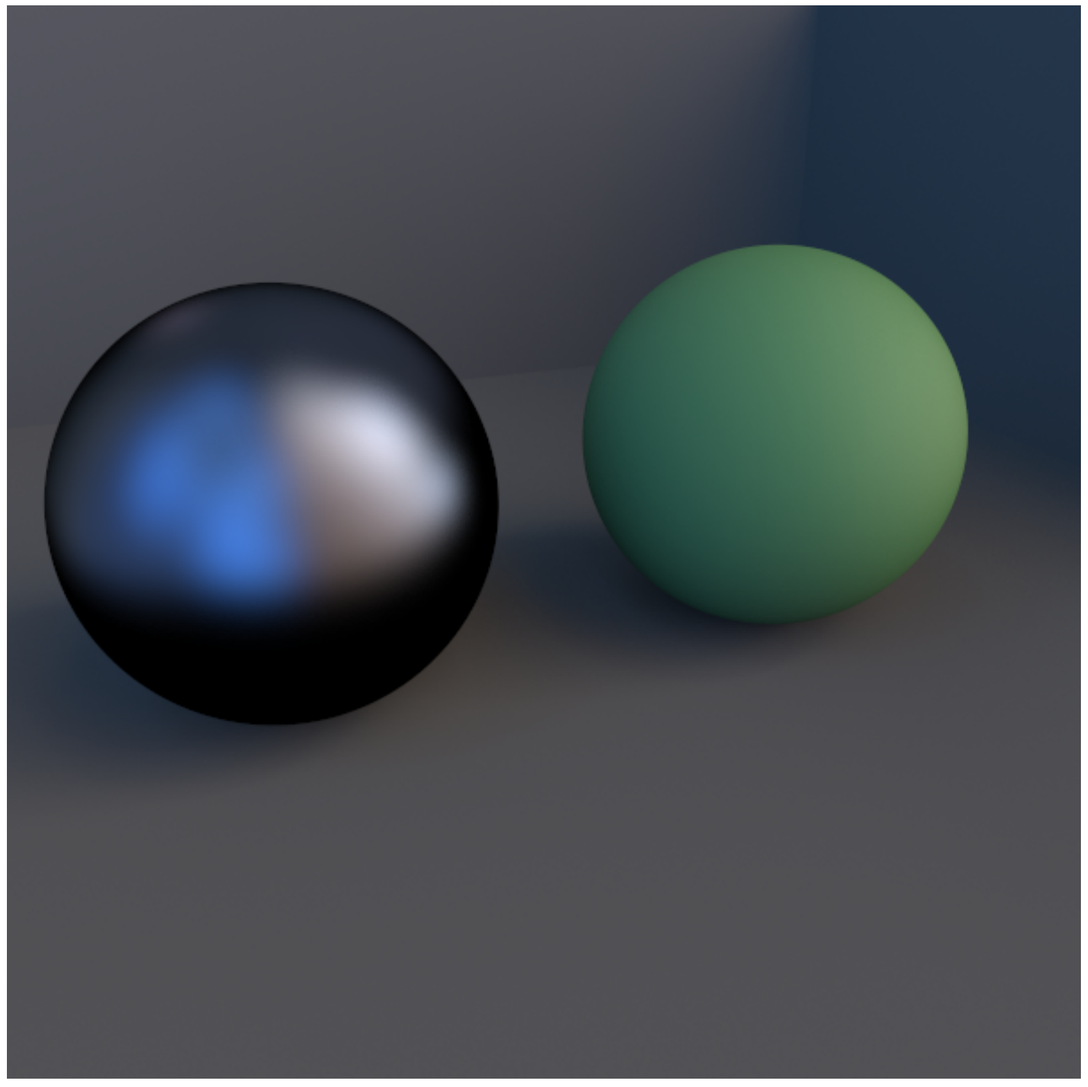
Image credit: Henrik Wann Jensen

Visual Richness from Indirect Lighting

Visual Richness from Complex Lighting



Point Light



Environment Map Lighting

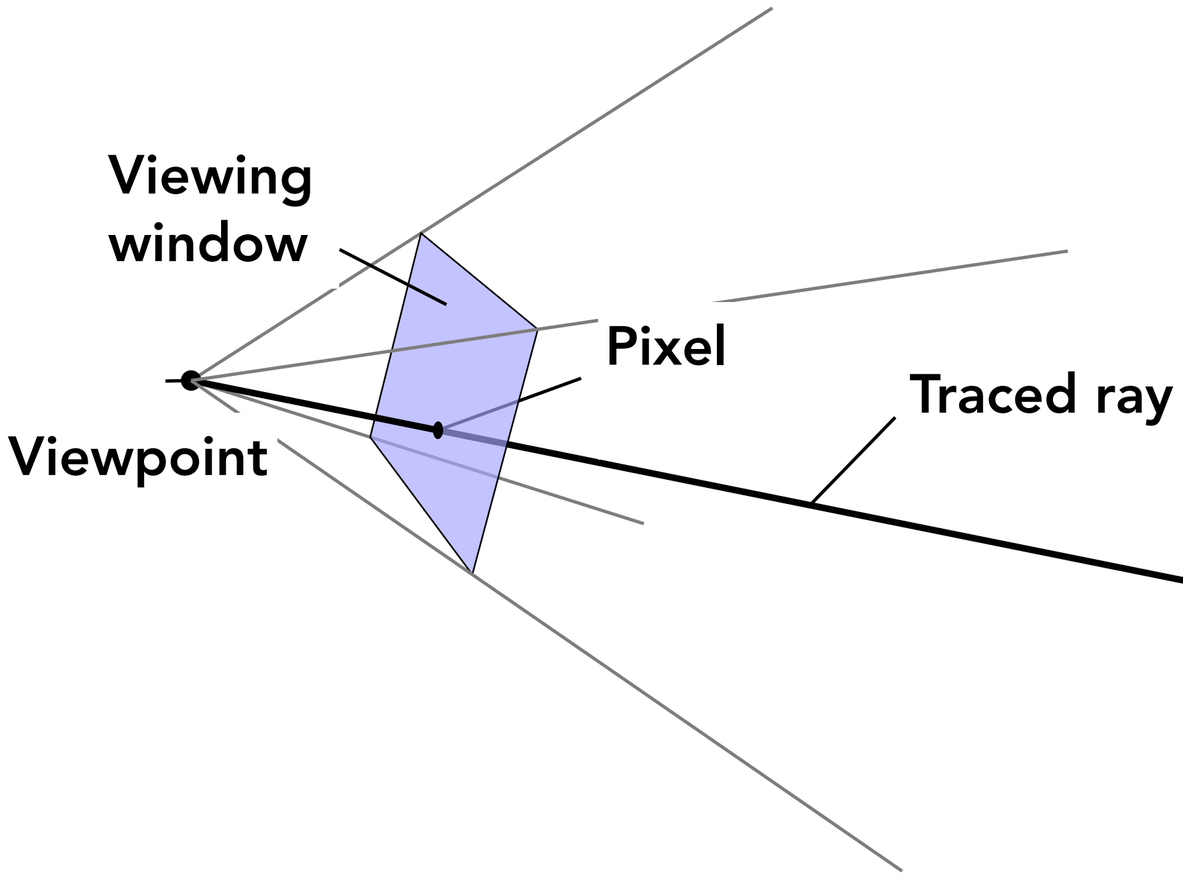
Visual Richness from Complex Materials

Credit: Bertrand Benoit. “Sweet Feast,” 2009. [Blender /VRay]

Cornell Box – Photograph vs Rendering

Photograph (CCD) vs. global illumination rendering

Ray Tracer Samples Radiance Along A Ray



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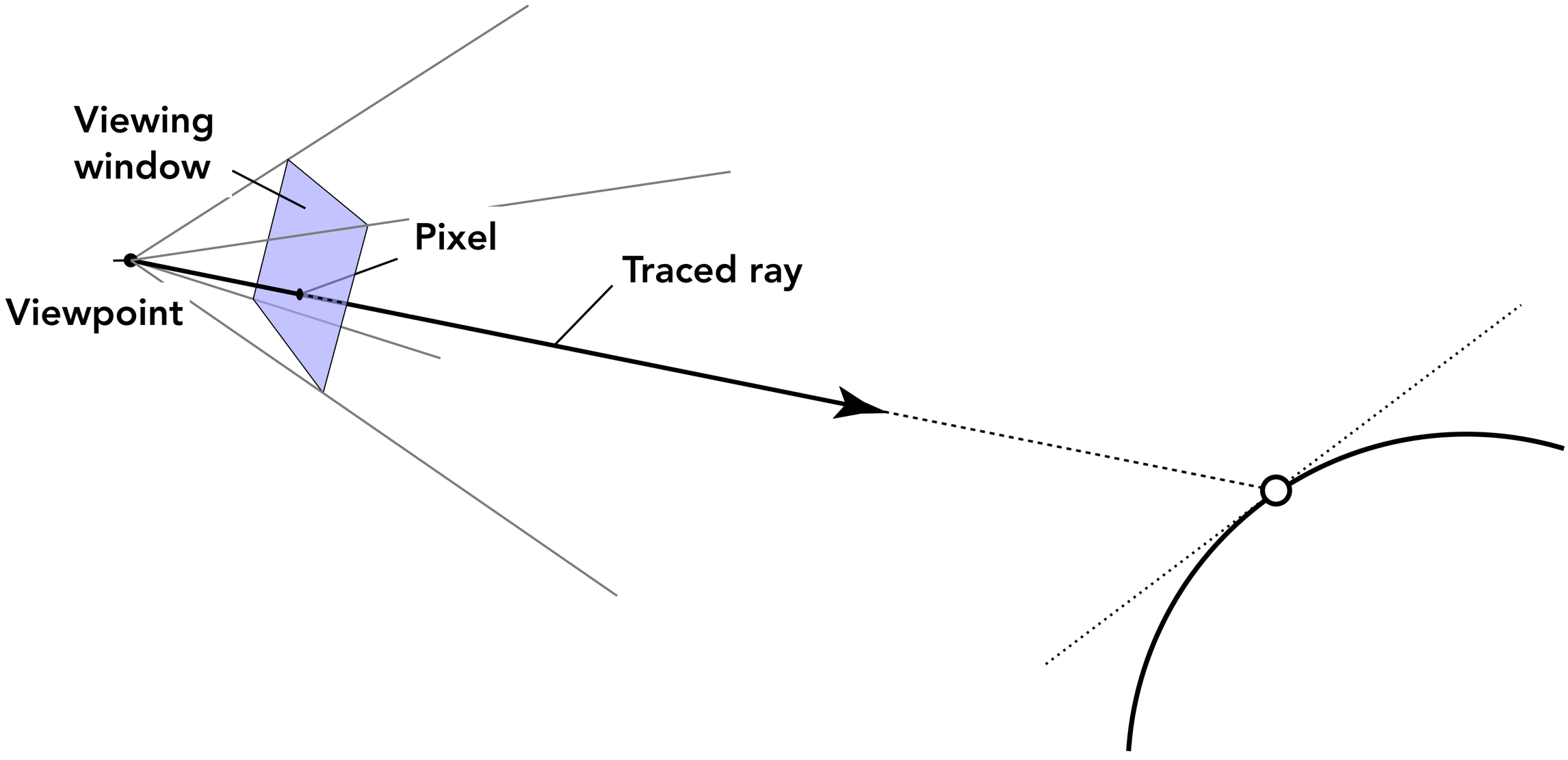

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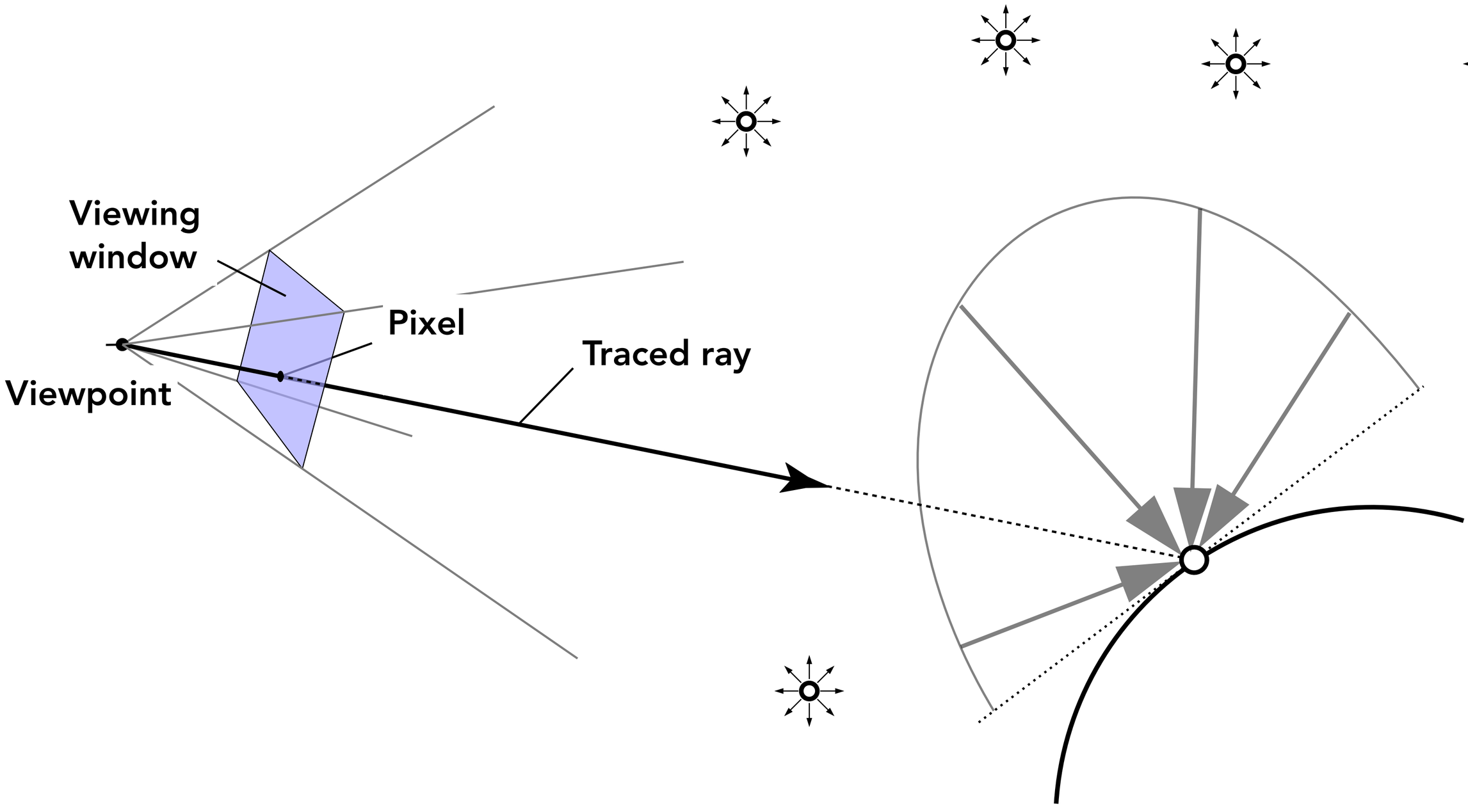
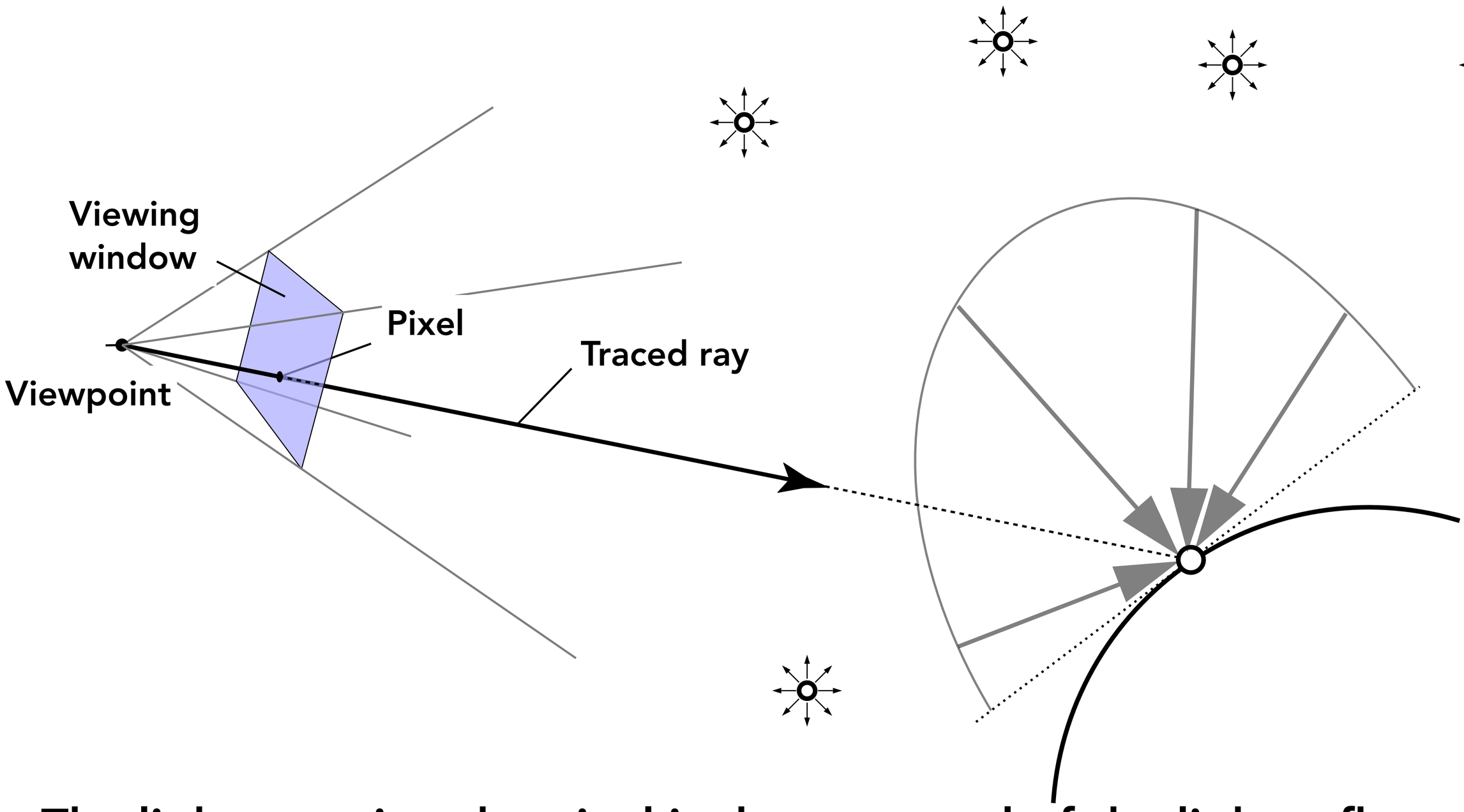


Ray Tracer Samples Radiance Along A Ray



The light entering the pixel is the sum total of the light reflected off the surface into the ray's (reverse) direction

Mini-Intro To Material Reflection

Reflection

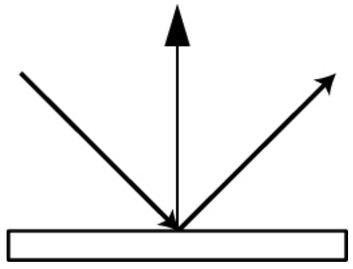
Definition: reflection is the process by which light incident on a surface interacts with the surface such that it leaves on the incident (same) side without change in frequency

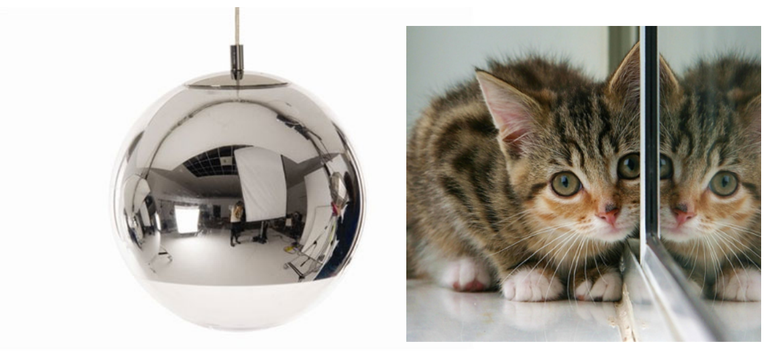
Categories of Reflection Functions

Categories of Reflection Functions

Ideal specular

- Perfect mirror reflection





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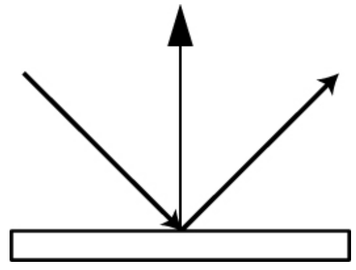
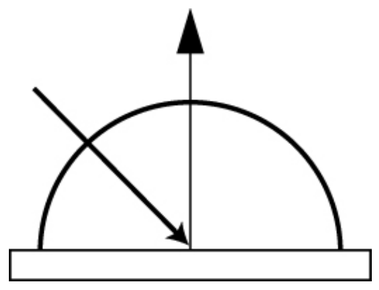


Ideal diffuse

- Equal reflection in all directions



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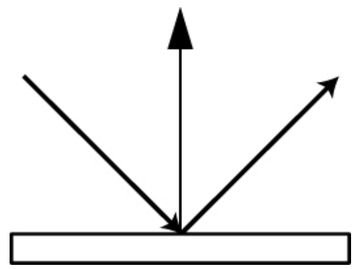

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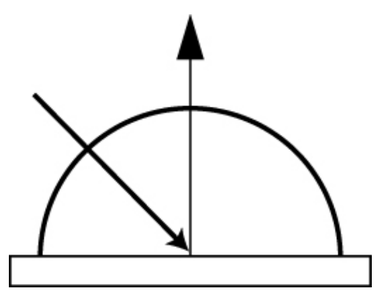
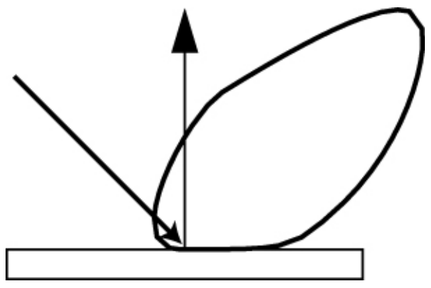


Glossy specular

- Majority of light reflected near mirror direction



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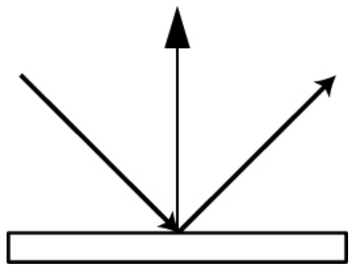

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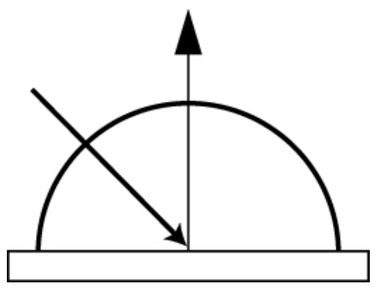

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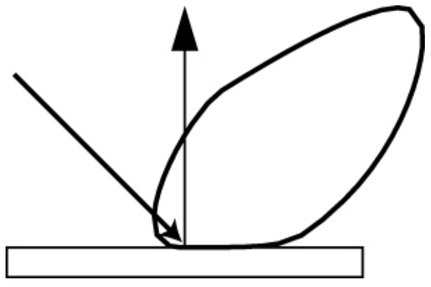
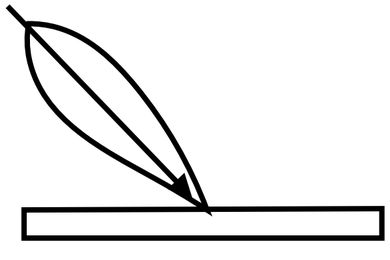
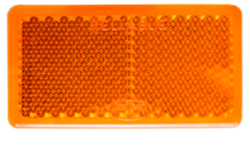


Retro-reflective

- Light reflected back towards light source





Diagrams illustrate how light from incoming direction is reflected in various outgoing directions.

Materials: Mirror

Materials: Diffuse

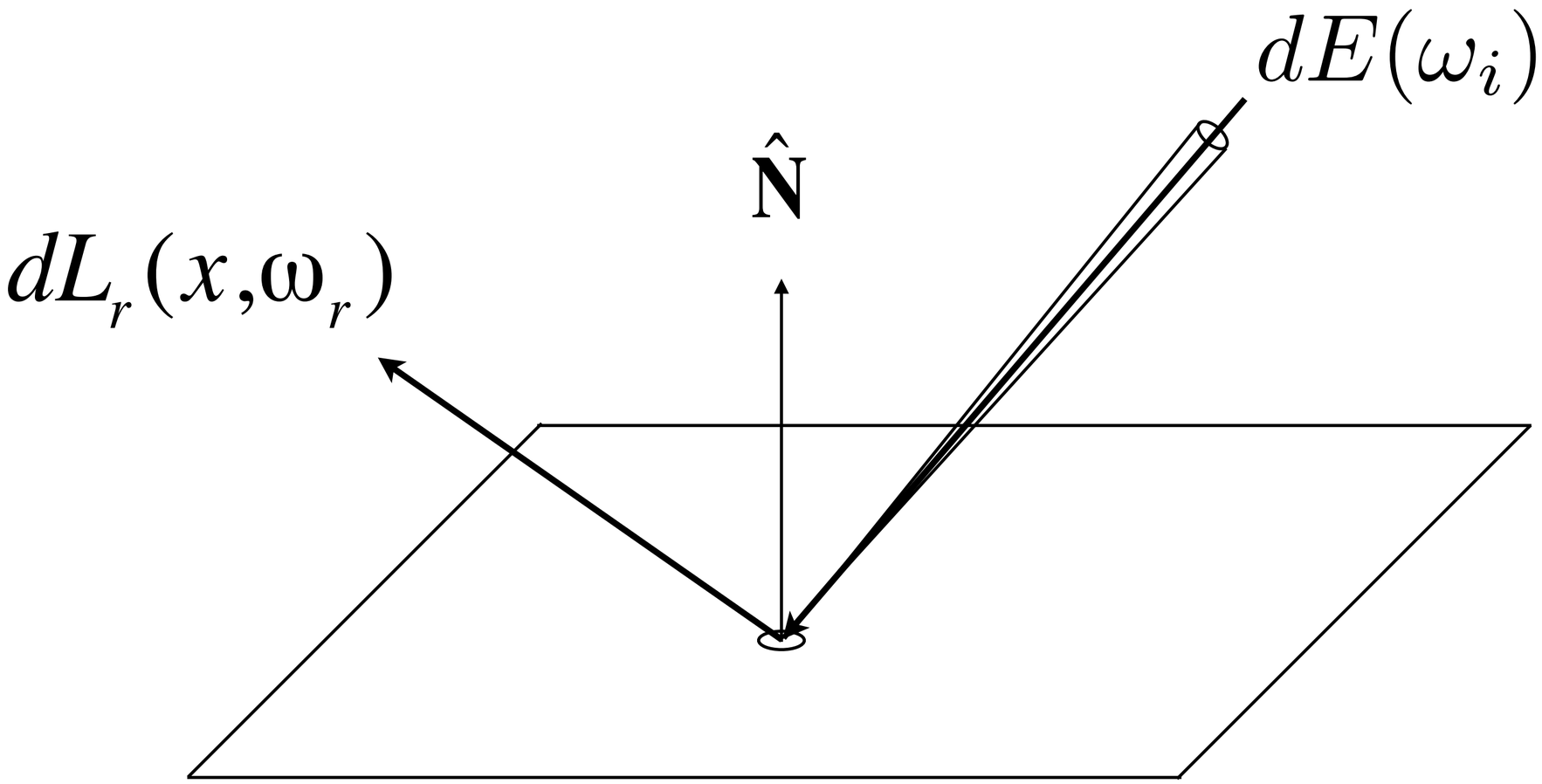
Materials: Gold

Materials: Plastic

Materials: Red Semi-Gloss Paint

Materials: Ford Mystic Lacquer Paint

Reflection at a Point

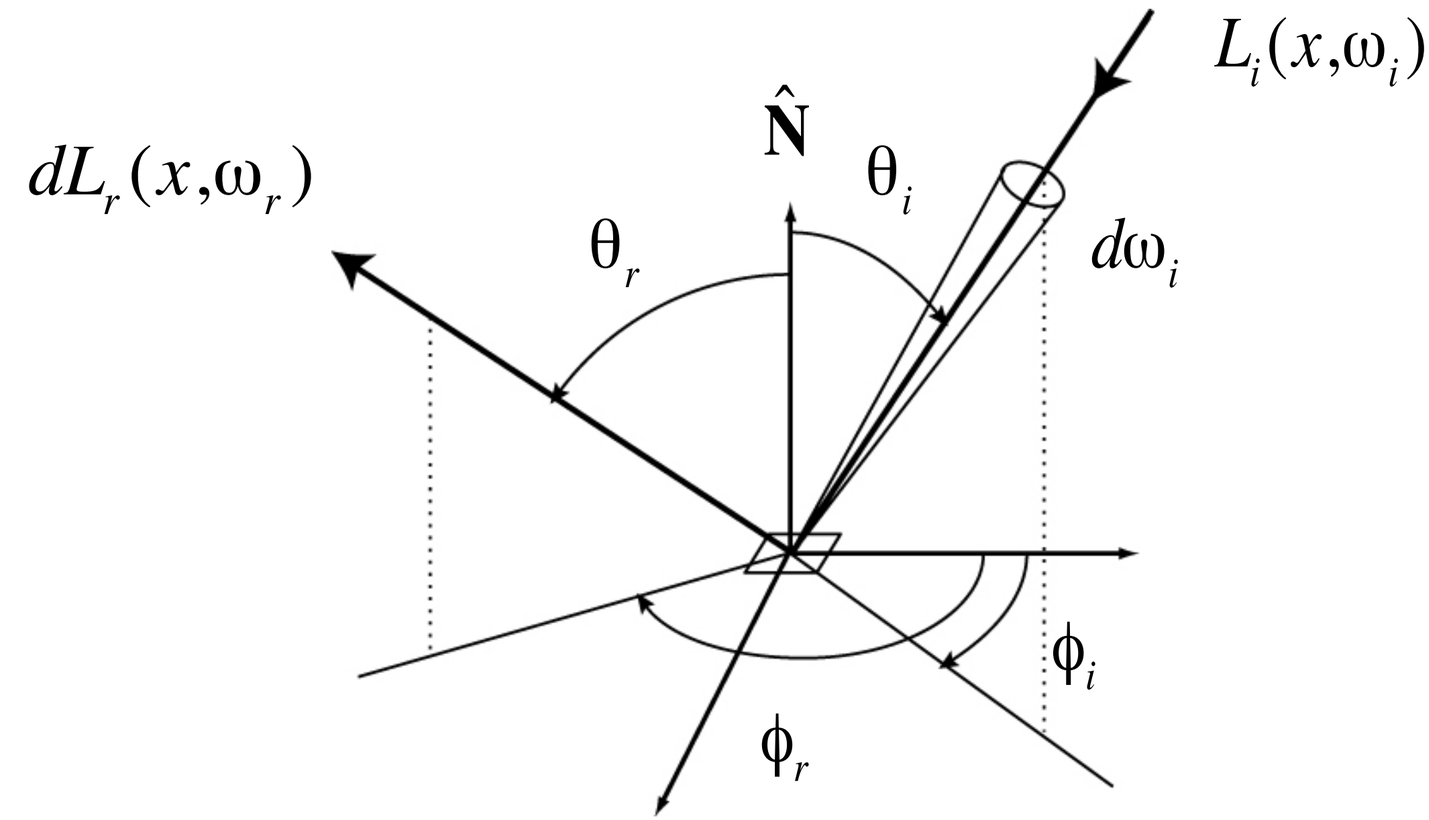


Differential irradiance incoming: $dE(\omega_i) = L(\omega_i)\, cos\, \theta_i\, d\omega_i$

Differential radiance exiting (due to $dE(\omega_i)$**)** $dL_r(\omega_r)$

BRDF

Definition: The bidirectional reflectance distribution function (BRDF) represents how much light is reflected into each outgoing direction ω_r from each incoming direction



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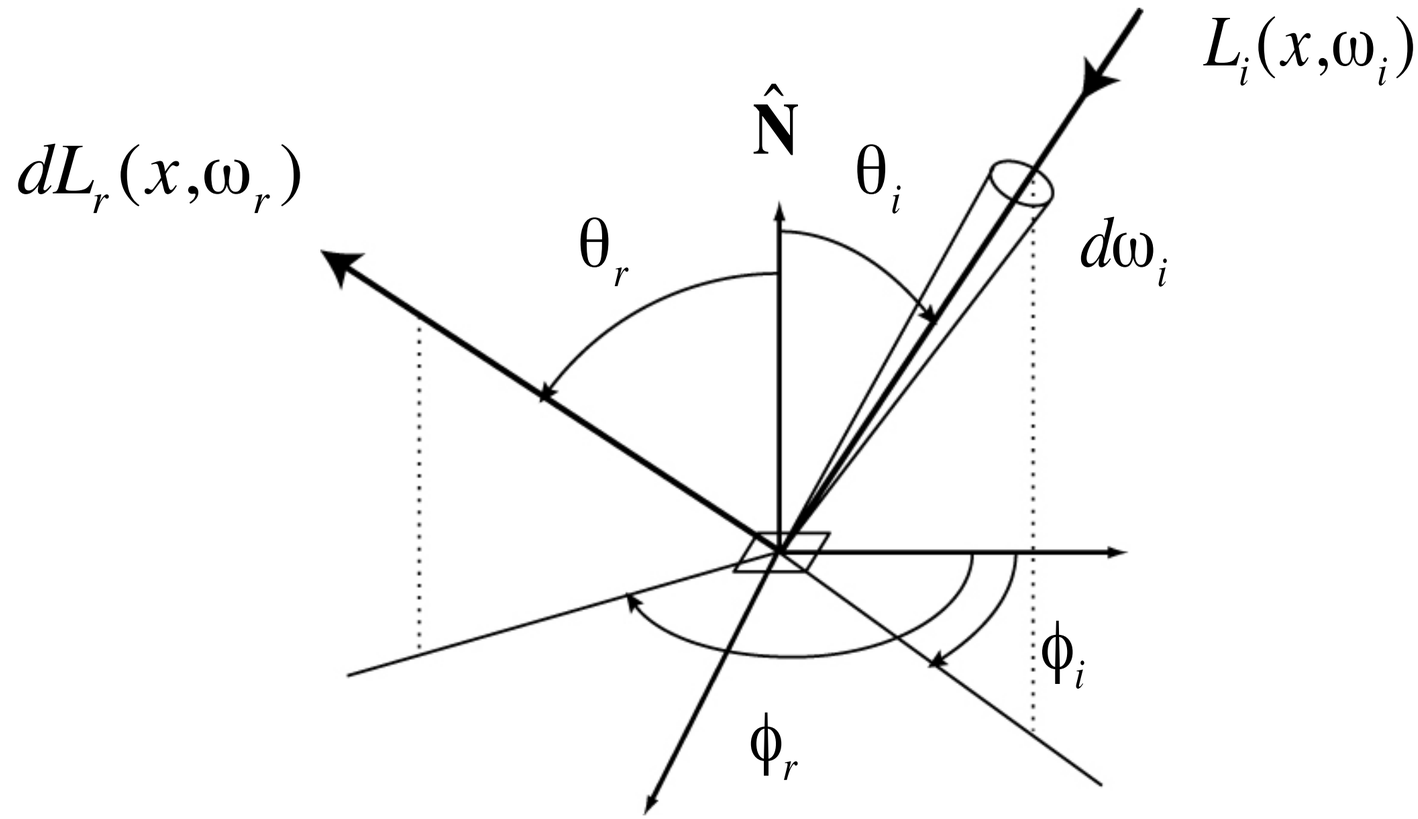

$$f_r(\omega_i \rightarrow \omega_r) = \frac{\mathrm{d}L_r(\omega_r)}{\mathrm{d}E_i(\omega_i)} = \frac{\mathrm{d}L_r(\omega_r)}{L_i(\omega_i)\cos\theta_i\,\mathrm{d}\omega_i} \quad \left[\frac{1}{\mathrm{sr}}\right]$$

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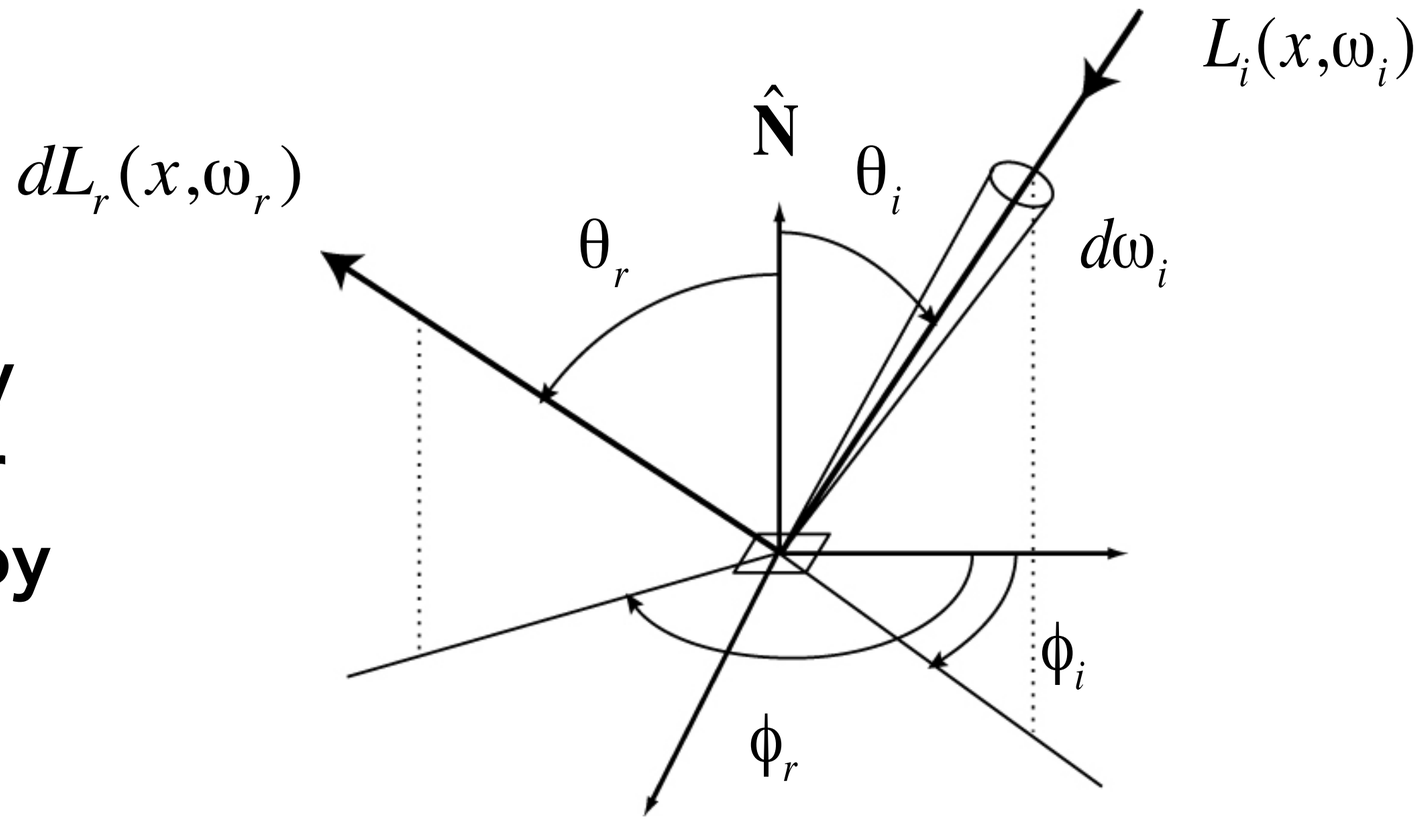
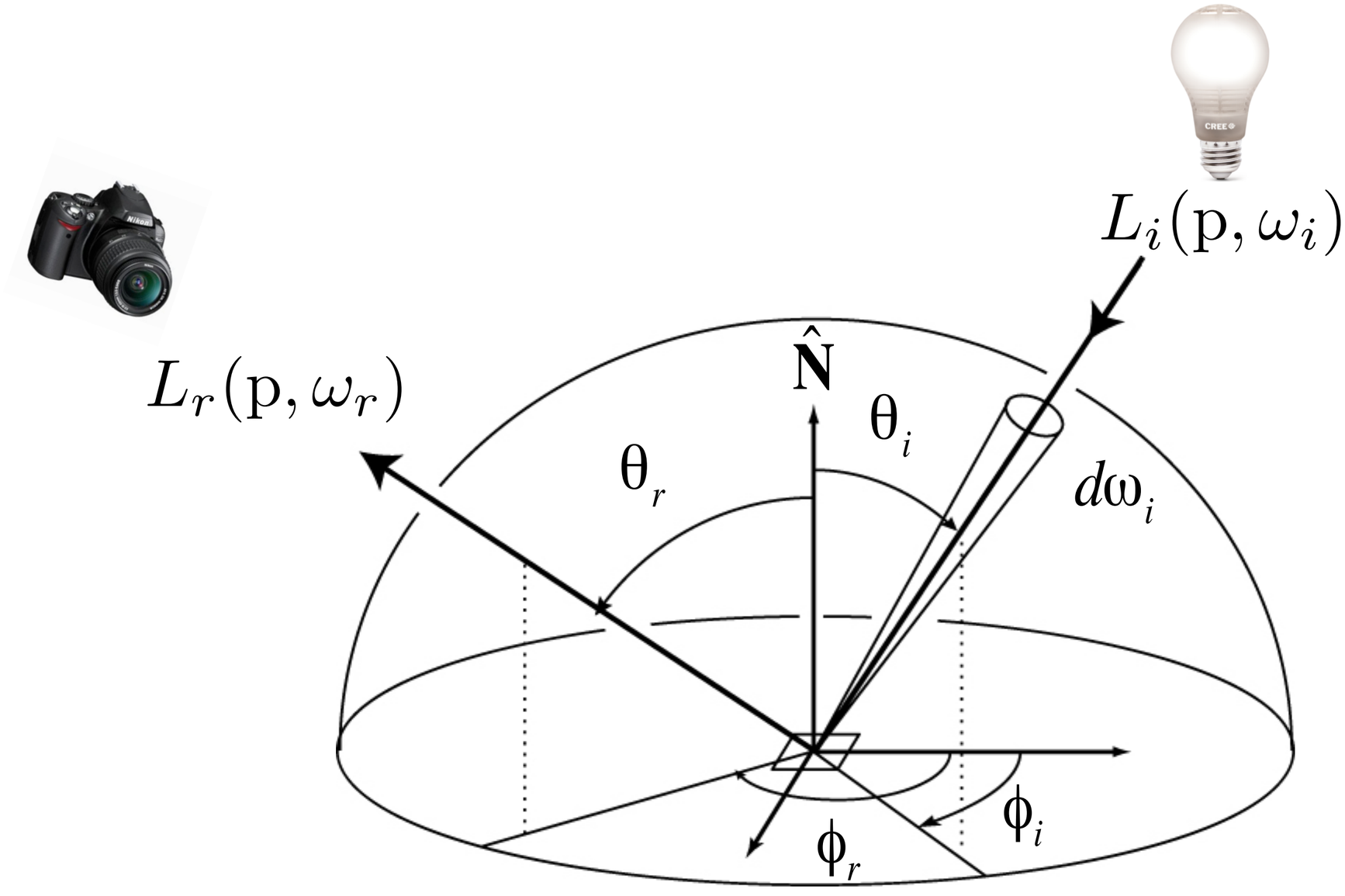


NB: ω_i points away from surface rather than into surface, by convention.

$$f_r(\omega_i \to \omega_r) = \frac{\mathrm{d}L_r(\omega_r)}{\mathrm{d}E_i(\omega_i)} = \frac{\mathrm{d}L_r(\omega_r)}{L_i(\omega_i)\cos\theta_i\,\mathrm{d}\omega_i} \quad \left[\frac{1}{\mathrm{sr}}\right]$$

The Reflection Equation



$$L_r(\mathrm{p}, \omega_r) = \int_{H^2} f_r(\mathrm{p}, \omega_i \rightarrow \omega_r)\, L_i(\mathrm{p}, \omega_i)\, \cos\theta_i \, \mathrm{d}\omega_i$$

Solving the Reflection Equation

$$L_r(\mathrm{p}, \omega_r) = \int_{H^2} f_r(\mathrm{p}, \omega_i \rightarrow \omega_r)\, L_i(\mathrm{p}, \omega_i)\, \cos\theta_i \, \mathrm{d}\omega_i$$

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Monte Carlo estimate:

- **Generate directions ω_j sampled from some distribution $p(\omega)$**
- **Choices for $p(\omega)$**
  - **Uniformly sample hemisphere**
  - **Importance sample BRDF (proportional to BRDF)**
  - **Importance sample lights (sample position on lights)**
- **Compute the estimator**

Solving the Reflection Equation

$$L_r(\mathrm{p}, \omega_r) = \int_{H^2} f_r(\mathrm{p}, \omega_i \rightarrow \omega_r)\, L_i(\mathrm{p}, \omega_i)\, \cos\theta_i \, \mathrm{d}\omega_i$$

Monte Carlo estimate:

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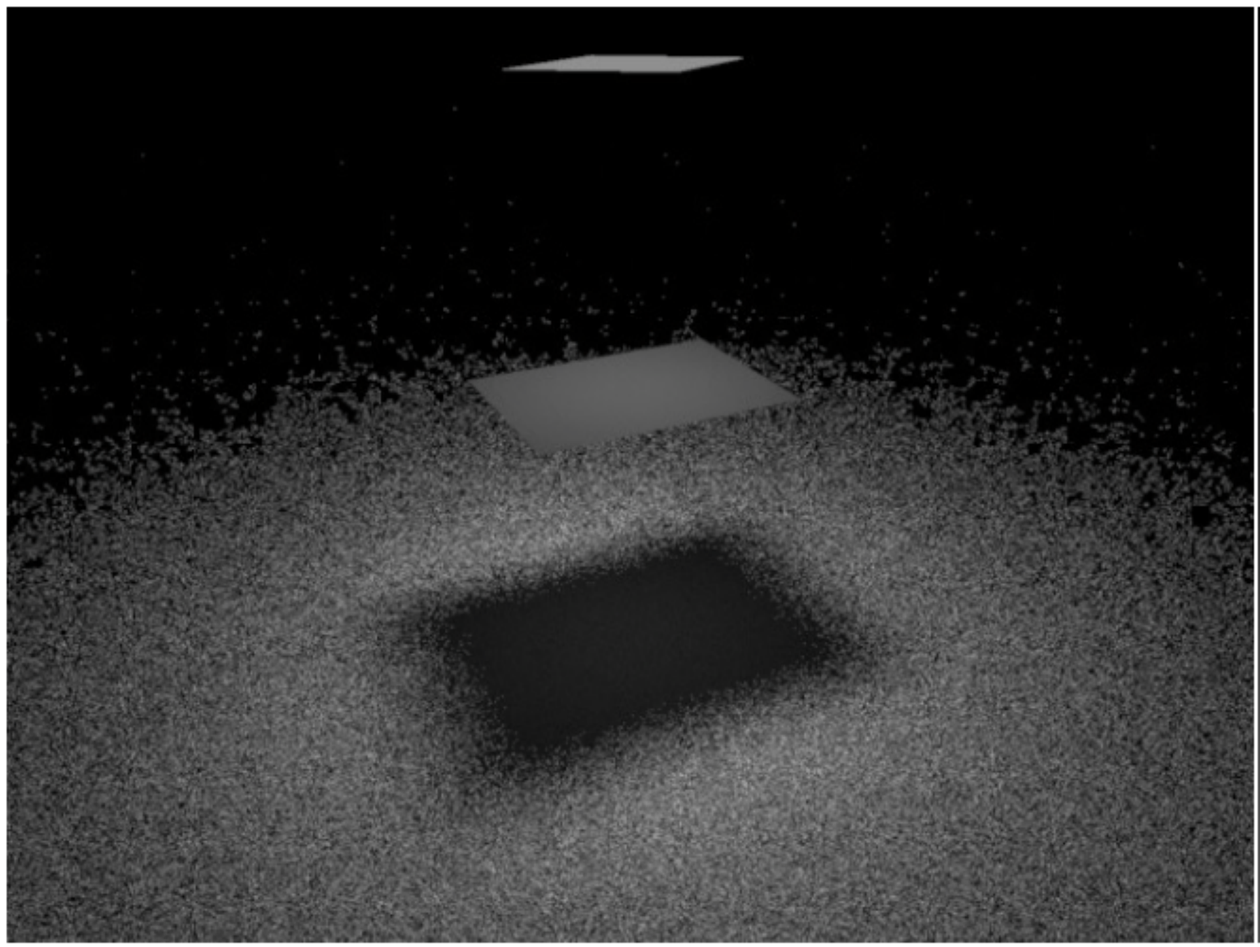
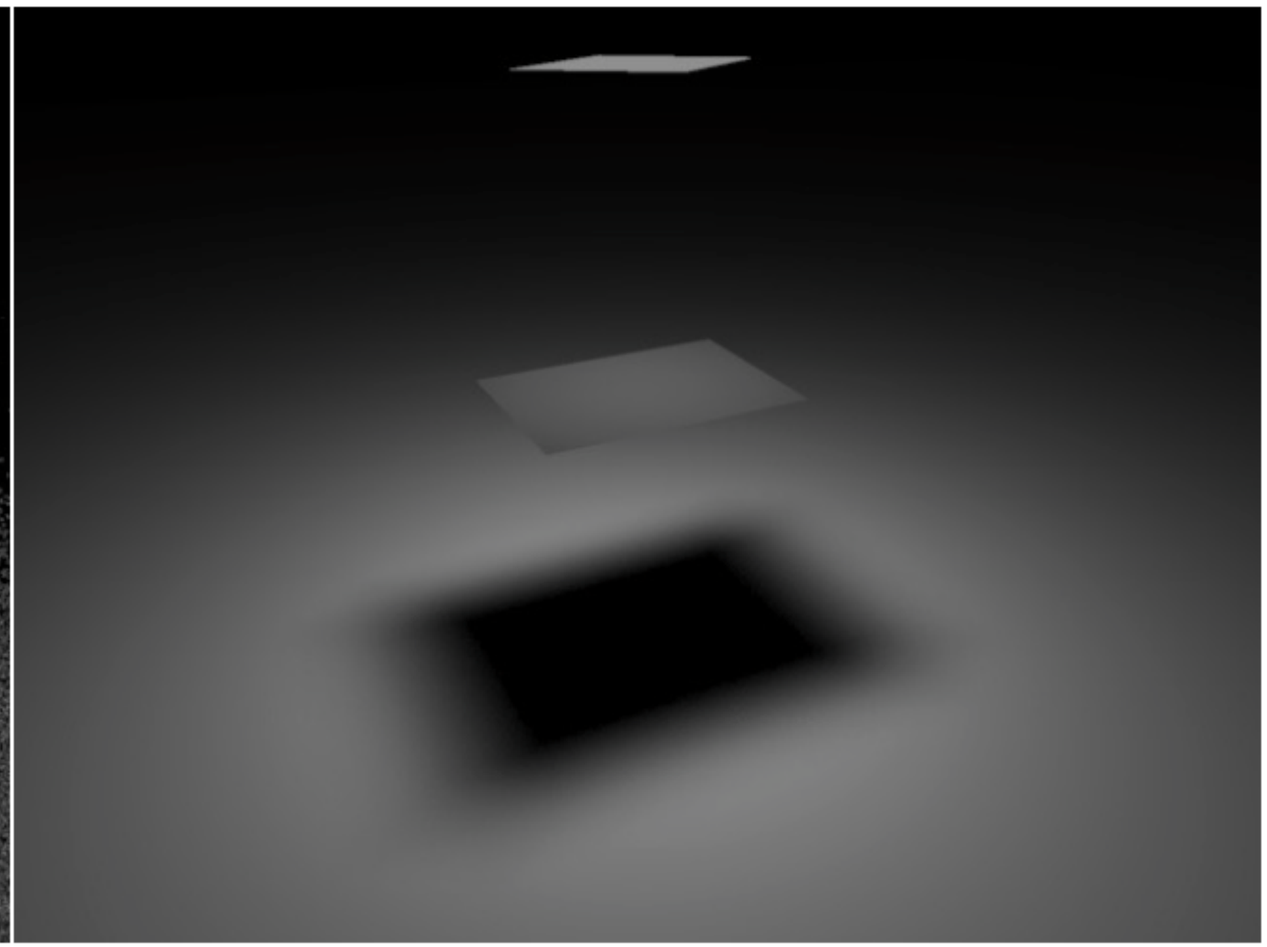
- **Choices for $p(\omega)$**
  - **Uniformly sample hemisphere**
  - **Importance sample BRDF (proportional to BRDF)**
  - **Importance sample lights (sample position on lights)**
- **Compute the estimator**

$$\frac{1}{N} \sum_{j=1}^{N} \frac{f_r(\mathrm{p}, \omega_j \rightarrow \omega_r)\, L_i(\mathrm{p}, \omega_j)\, \cos\theta_j}{p(\omega_j)}$$

Recall: Hemisphere vs Light Sampling



Sample hemisphere uniformly



Sample points on light

Direct Lighting Pseudocode (Uniform Random Sampling)

```
DirectLightingSampleUniform(p, ωo)
    ωi = hemisphere.sampleUniform();   // uniform random sampling
    pdf = 1.0 / (2 * pi);

    if (scene.shadowIntersection(p, ωi))      // Shadow ray
        return 0;
  else
    L = lights.radiance(intersect(p, ωi), –ωi);
    return L * p.brdf(ωi, ωo) * costheta / pdf;
```

Direct Lighting Pseudocode (Importance Sampling of BRDF)

```
DirectLightingSampleBRDF(p, ωo)
    ωi, pdf = p.brdf.sampleDirection(ωo);                    // Imp. Sample BRDF


    if (scene.shadowIntersection(p, ωi))      // Shadow ray
        return 0;
    else
        L = lights.radiance(intersect(p, ωi), –ωi);
        return L * p.brdf(ωi, ωo) * costheta / pdf;
```

Direct Lighting Pseudocode (Importance Sampling of Lights)

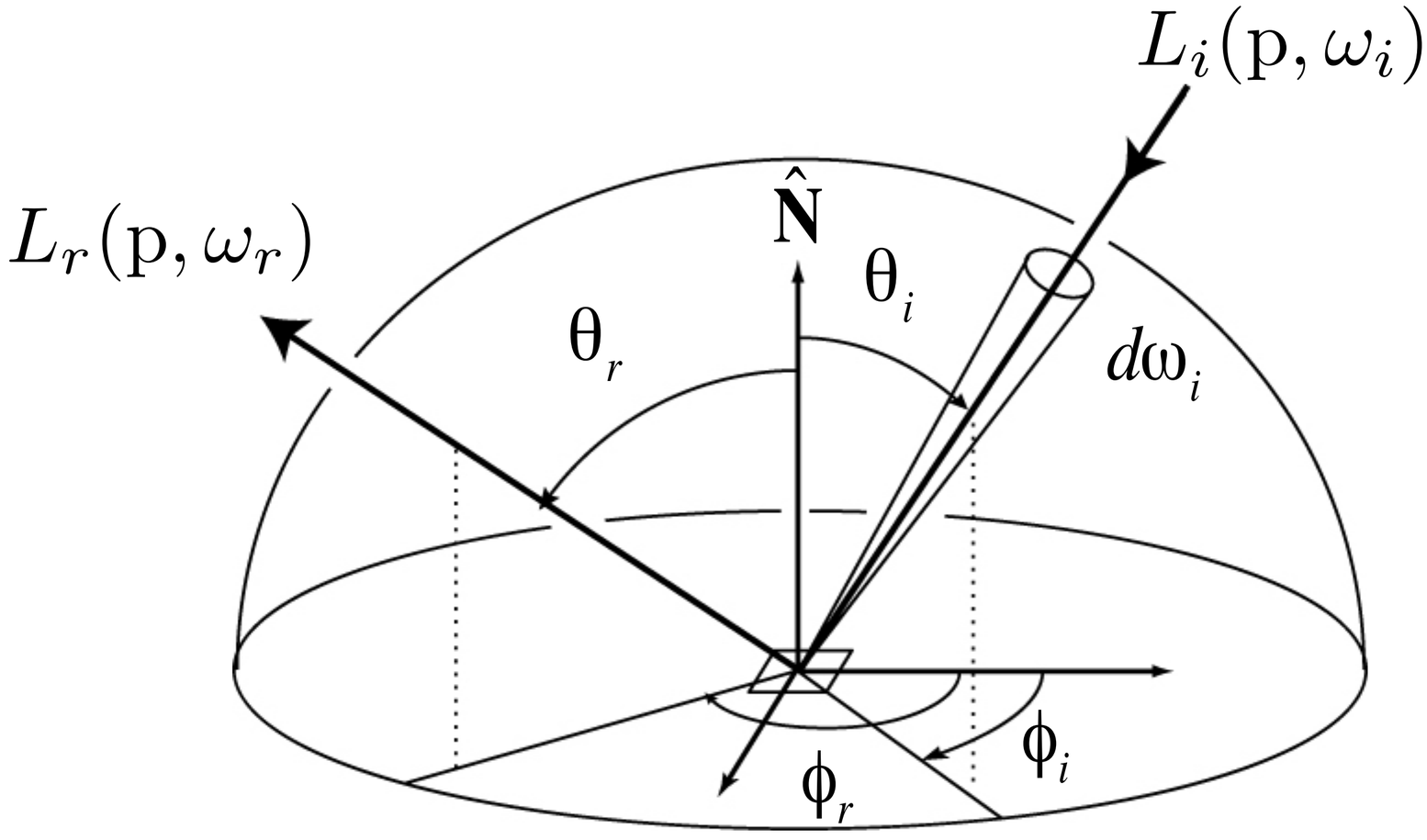
```
DirectLightingSampleLights(p, ωo)
    L, ωi, pdf = lights.sampleDirection(p);   // Imp. sampl lights


    if (scene.shadowIntersection(p, ωi))      // Shadow ray
        return 0;
    else
        return L * p.brdf(ωi, ωo) *costheta / pdf;

// Note: only one random sample over all lights.
// Assignment 3-1 asks you to, alternatively, loop over
// multiple lights and take multiple samples
```

Global Illumination:
Deriving the Rendering Equation

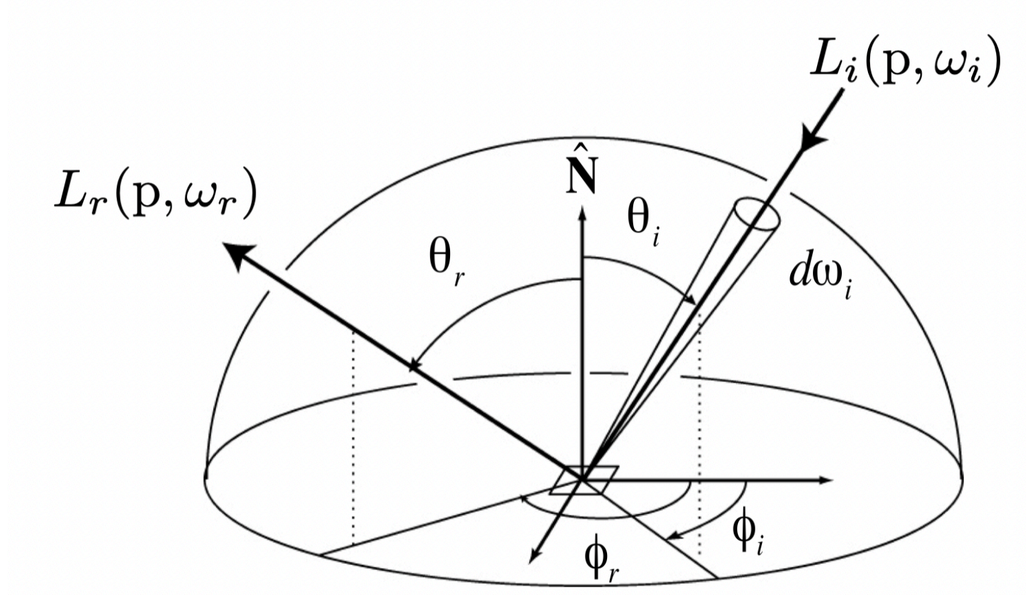
Again: Reflection Equation



$$L_r(\mathrm{p}, \omega_r) = \int_{H^2} f_r(\mathrm{p}, \omega_i \rightarrow \omega_r)\, L_i(\mathrm{p}, \omega_i)\, \cos\theta_i \, \mathrm{d}\omega_i$$

Challenge: This is Actually A Recursive Equation

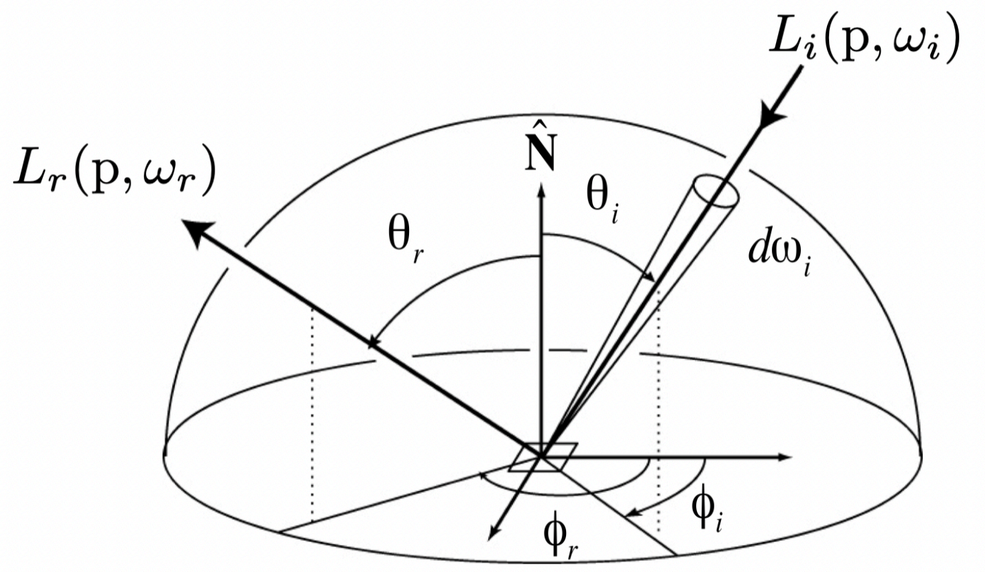
$$L_r(\mathrm{p}, \omega_r) = \int_{H^2} f_r(\mathrm{p}, \omega_i \rightarrow \omega_r)\, L_i(\mathrm{p}, \omega_i)\, \cos\theta_i \, \mathrm{d}\omega_i$$



Challenge: This is Actually A Recursive Equation

Reflected radiance depends on incoming radiance

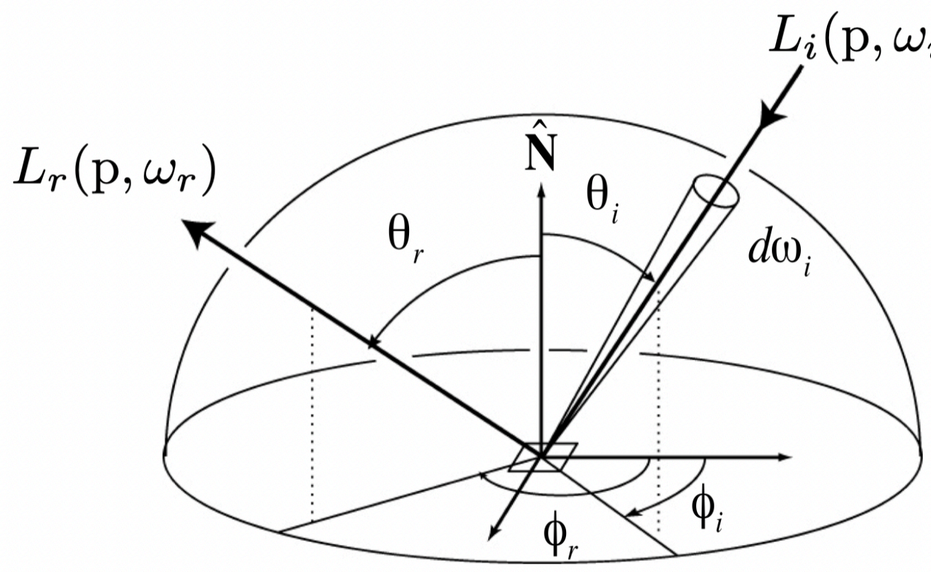
$$\boxed{L_r(\mathrm{p}, \omega_r)} = \int_{H^2} f_r(\mathrm{p}, \omega_i \rightarrow \omega_r) \boxed{L_i(\mathrm{p}, \omega_i)} \cos\theta_i \, \mathrm{d}\omega_i$$



Challenge: This is Actually A Recursive Equation

Reflected radiance depends on incoming radiance

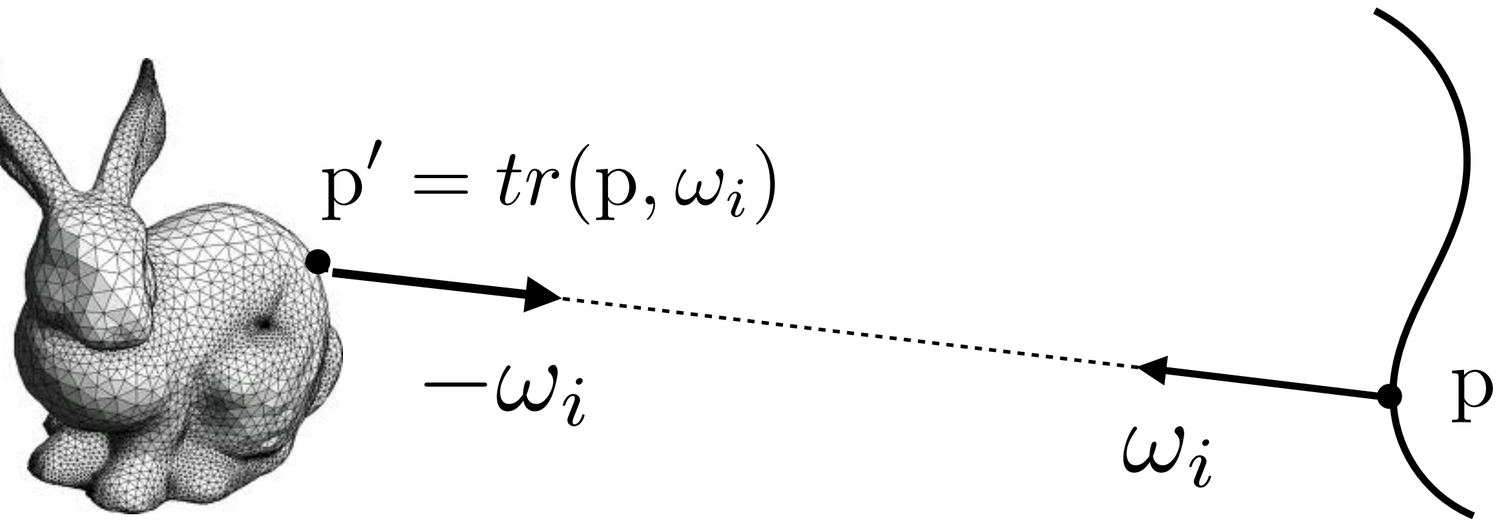
$$L_r(\mathrm{p}, \omega_r) = \int_{H^2} f_r(\mathrm{p}, \omega_i \rightarrow \omega_r) L_i(\mathrm{p}, \omega_i) \cos\theta_i \, \mathrm{d}\omega_i$$



But incoming radiance depends on reflected radiance (at another point in the scene)

Transport Function & Radiance Invariance

Definition: the Transport Function, $tr(\mathrm{p}, \omega)$, returns the first surface intersection point in the scene along ray (p, ω)



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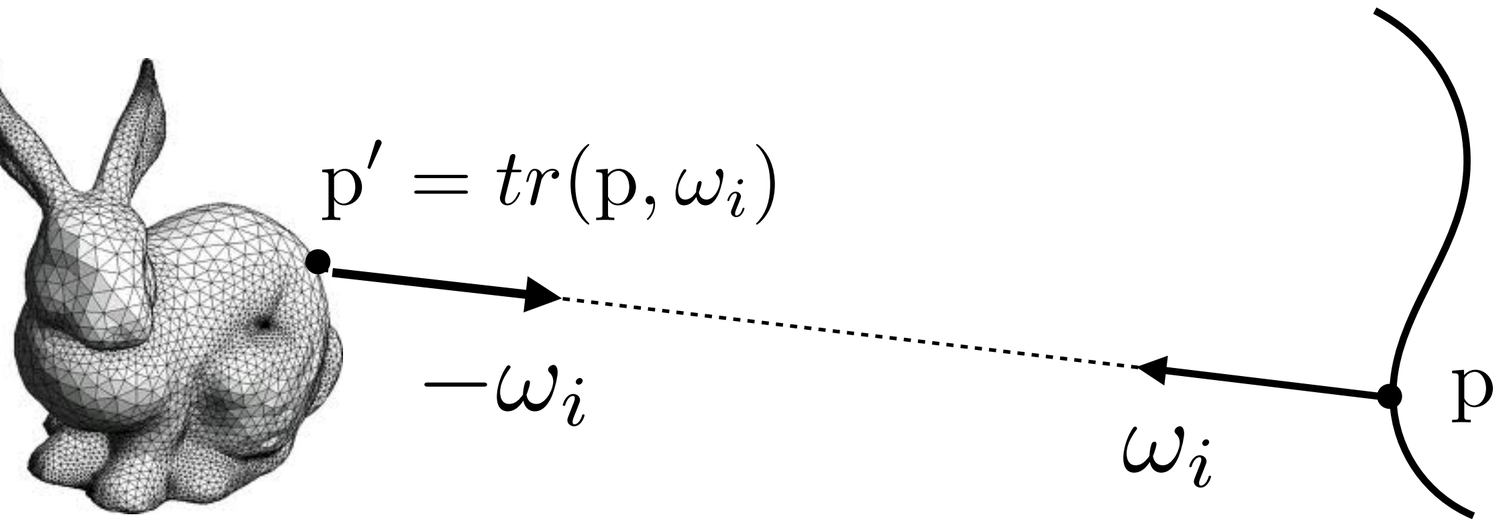
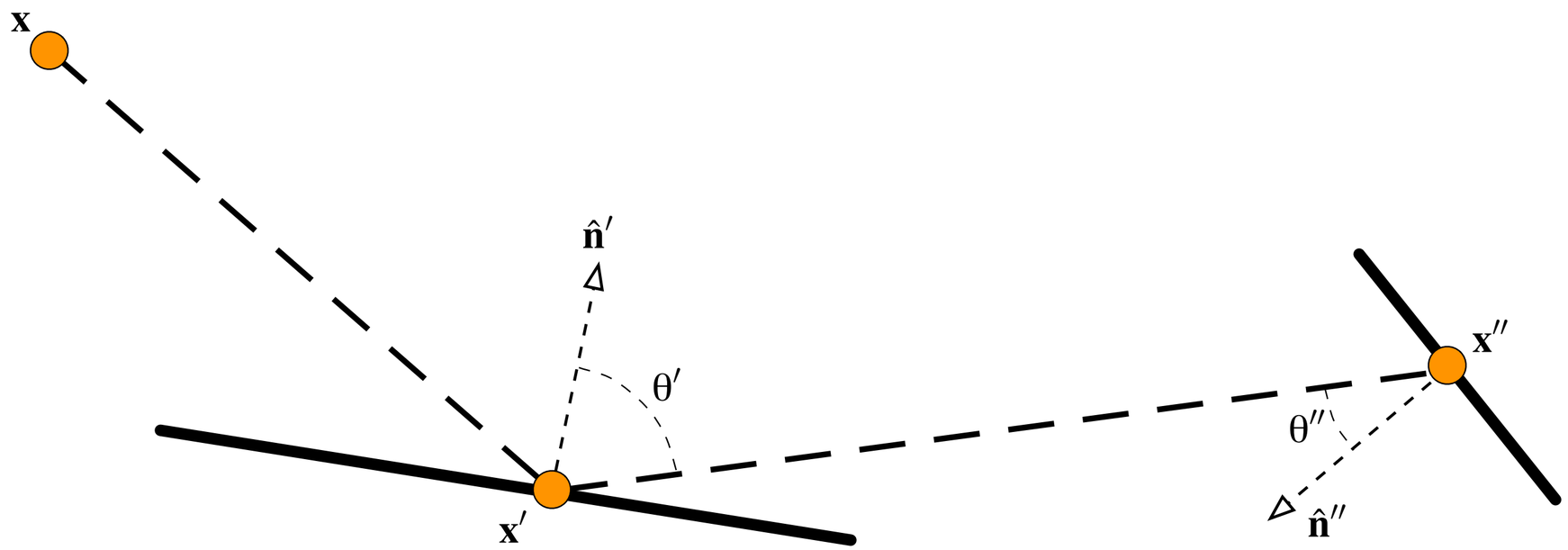


Radiance invariance along rays: $L_o(tr(\mathrm{p}, \omega_i), -\omega_i) = L_i(\mathrm{p}, \omega_i)$

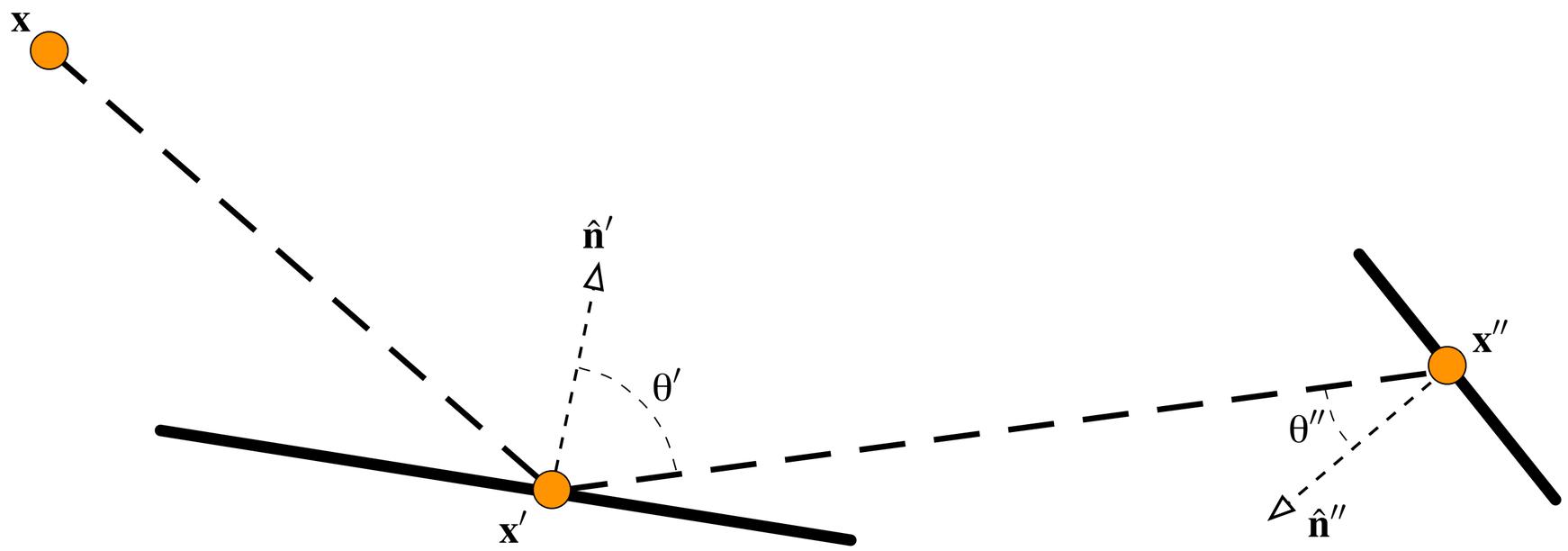
"Radiance arriving at p from direction ω_i is equal to the radiance leaving p' in direction $-\omega_i$"

The Rendering Equation



The light shining on x from x' is equal to:
- the emitted light from x' toward x, plus
- for each bit of surface in the scene, how much light shines from that bit onto x' and is reflected toward x, scaled appropriately

The Rendering Equation



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$$L_s(\mathbf{x},\mathbf{x}') = \delta(\mathbf{x},\mathbf{x}')\left[E(\mathbf{x},\mathbf{x}') + \int_S \rho_{x'}(\mathbf{x},\mathbf{x}'')L_s(\mathbf{x}',\mathbf{x}'')\frac{\cos(\theta')\cos(\theta'')}{||\mathbf{x}'-\mathbf{x}''||^2}d\mathbf{x}''\right]$$

The Rendering Equation

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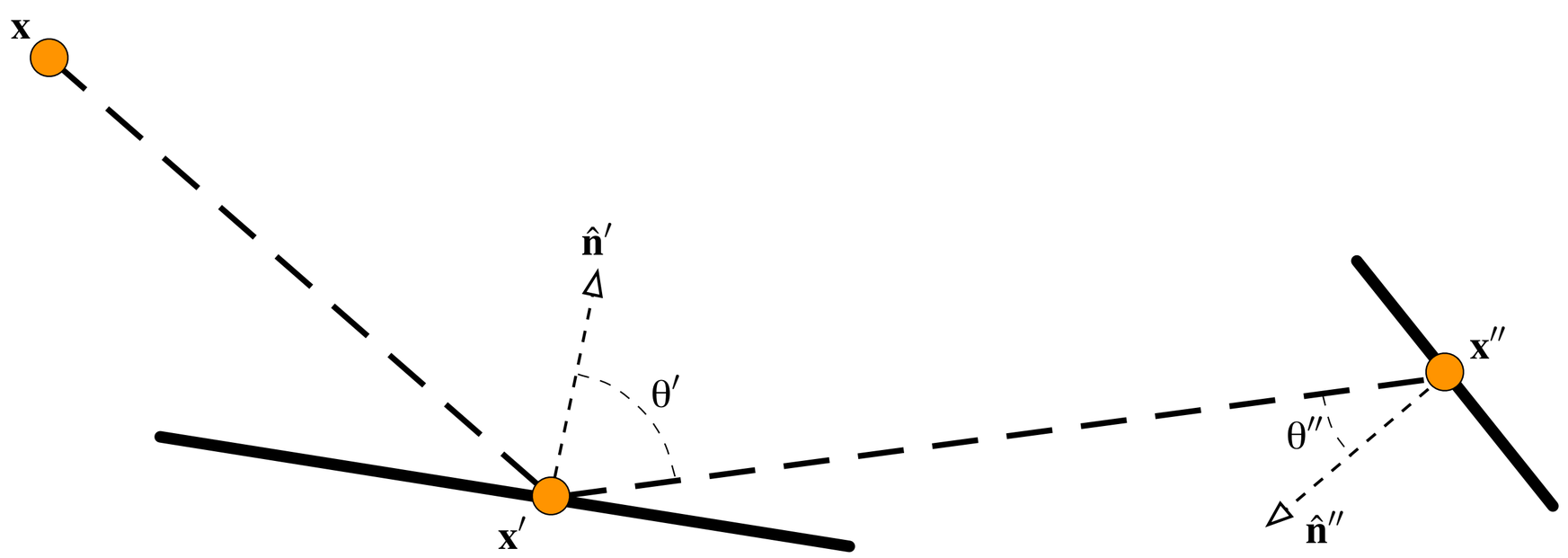
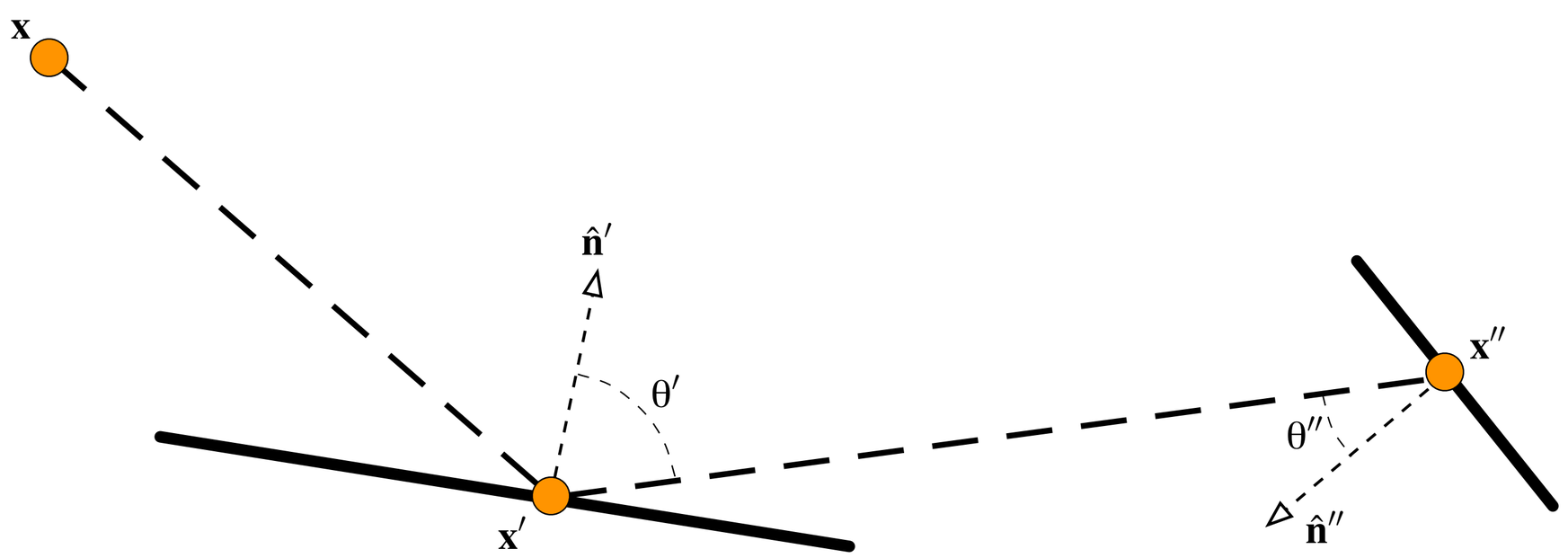


The Rendering Equation

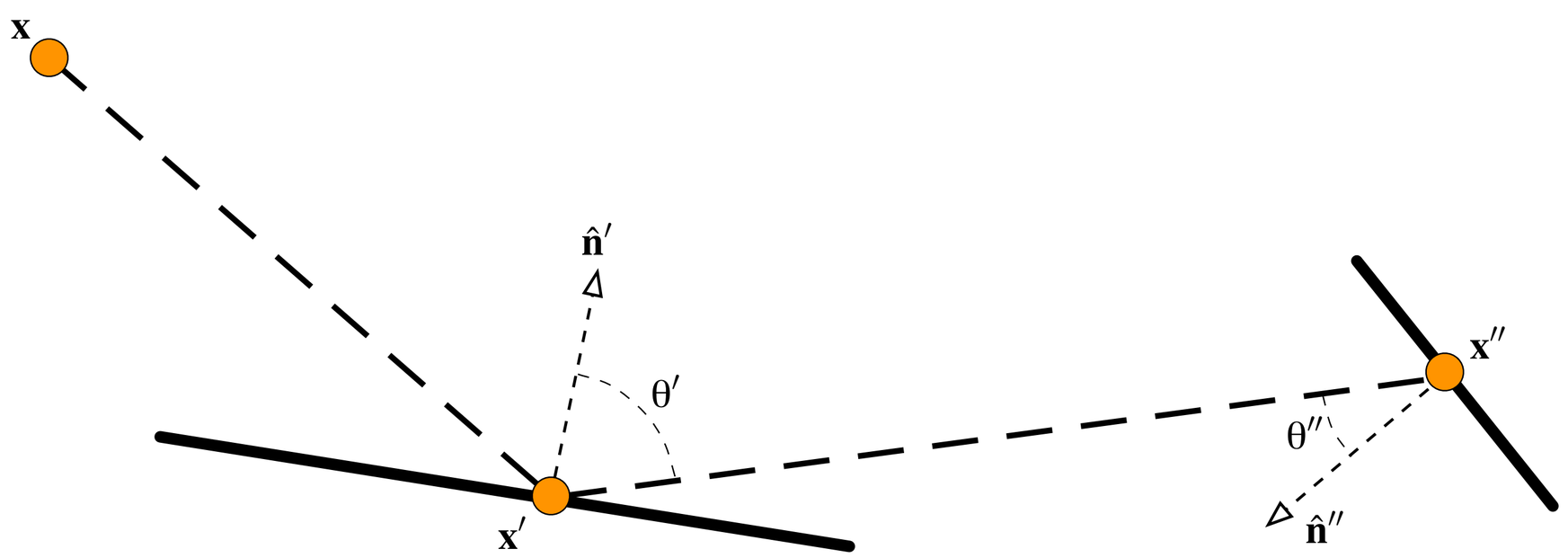
$$L_s(\mathbf{x},\mathbf{x}') = \delta(\mathbf{x},\mathbf{x}')\left[E(\mathbf{x},\mathbf{x}') + \int_S \rho_{x'}(\mathbf{x},\mathbf{x}'')L_s(\mathbf{x}',\mathbf{x}'')\frac{\cos(\theta')\cos(\theta'')}{||\mathbf{x}'-\mathbf{x}''||^2}\mathrm{d}\mathbf{x}''\right]$$

Light energy hitting x from x’



The Rendering Equation

$$L_s(\mathbf{x},\mathbf{x}') = \delta(\mathbf{x},\mathbf{x}')\left[E(\mathbf{x},\mathbf{x}') + \int_S \rho_{x'}(\mathbf{x},\mathbf{x}'')L_s(\mathbf{x}',\mathbf{x}'')\frac{\cos(\theta')\cos(\theta'')}{||\mathbf{x}'-\mathbf{x}''||^2}\mathrm{d}\mathbf{x}''\right]$$



The Rendering Equation

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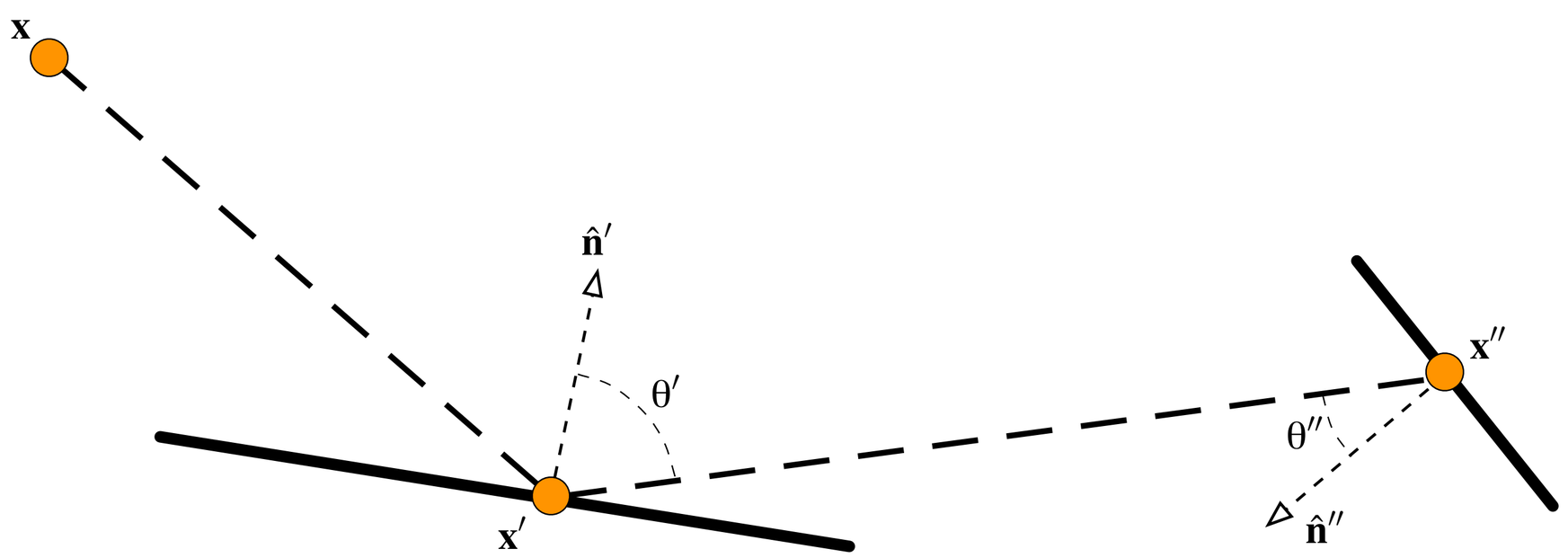
Can x see x' ?



The Rendering Equation

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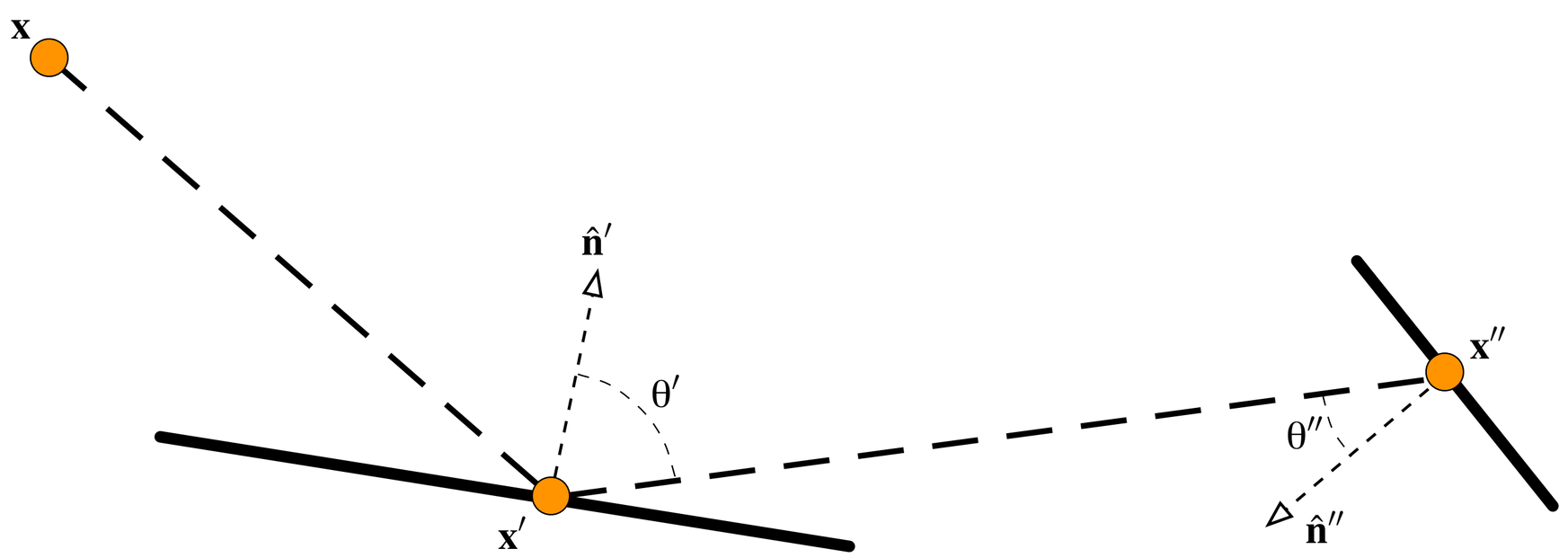

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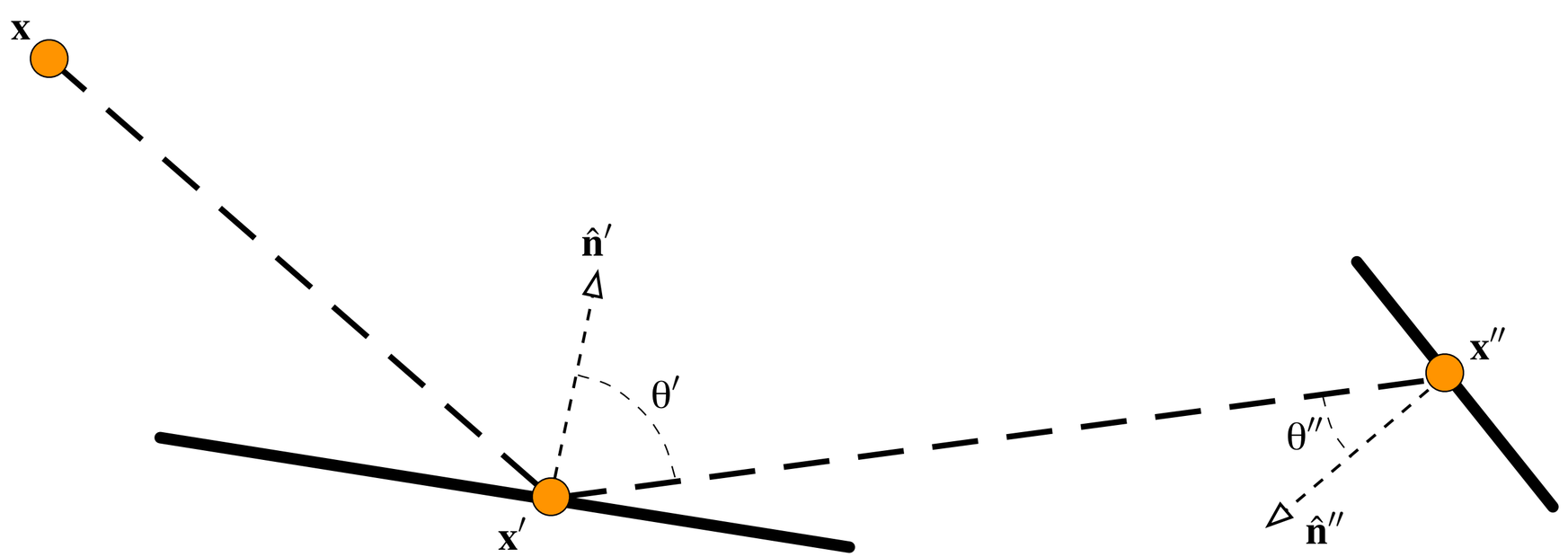
Light emitted from x' toward x



The Rendering Equation

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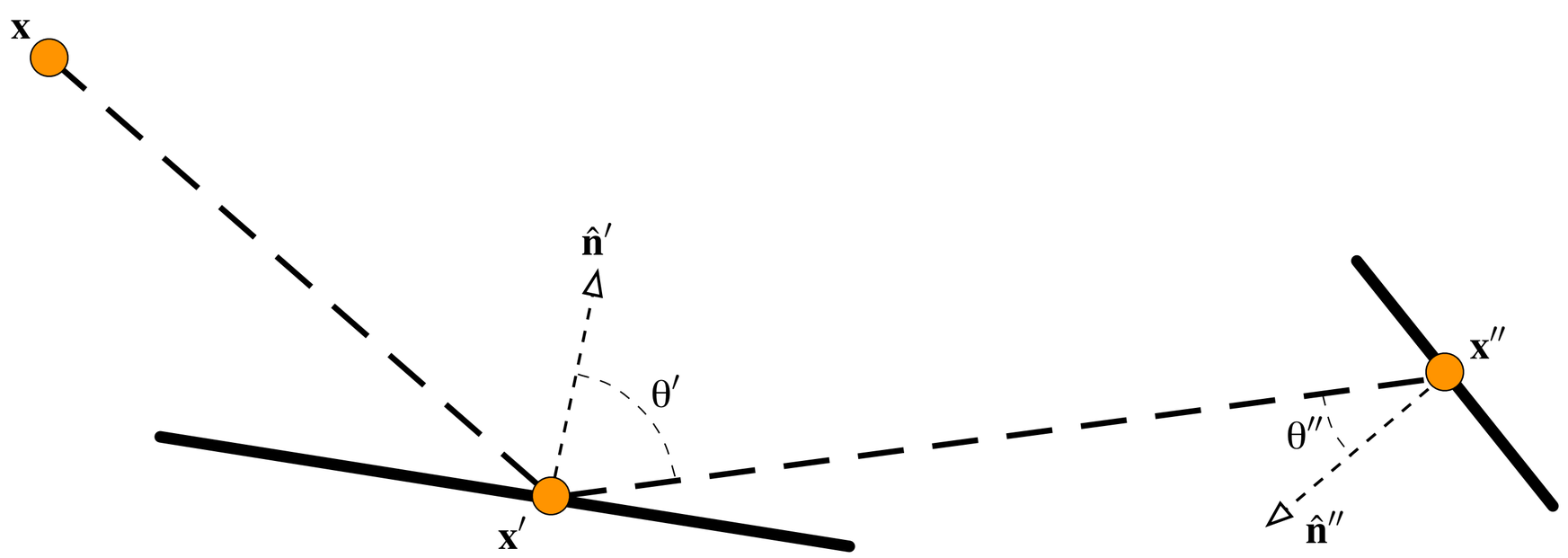
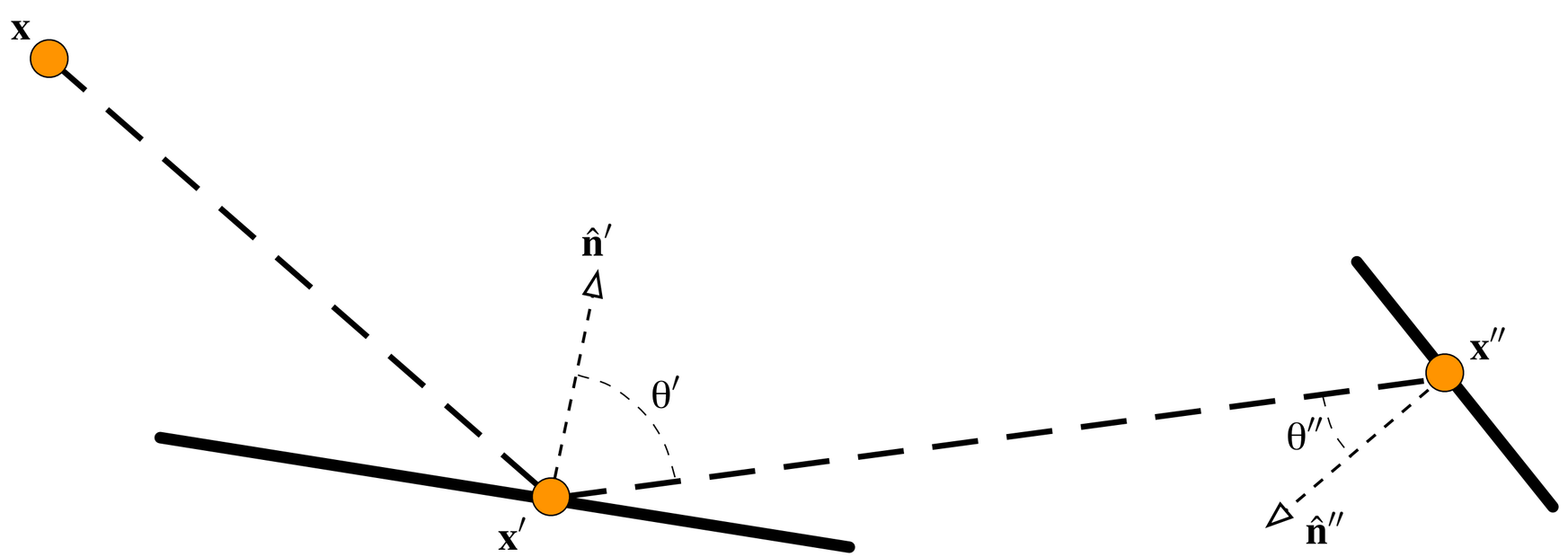


The Rendering Equation

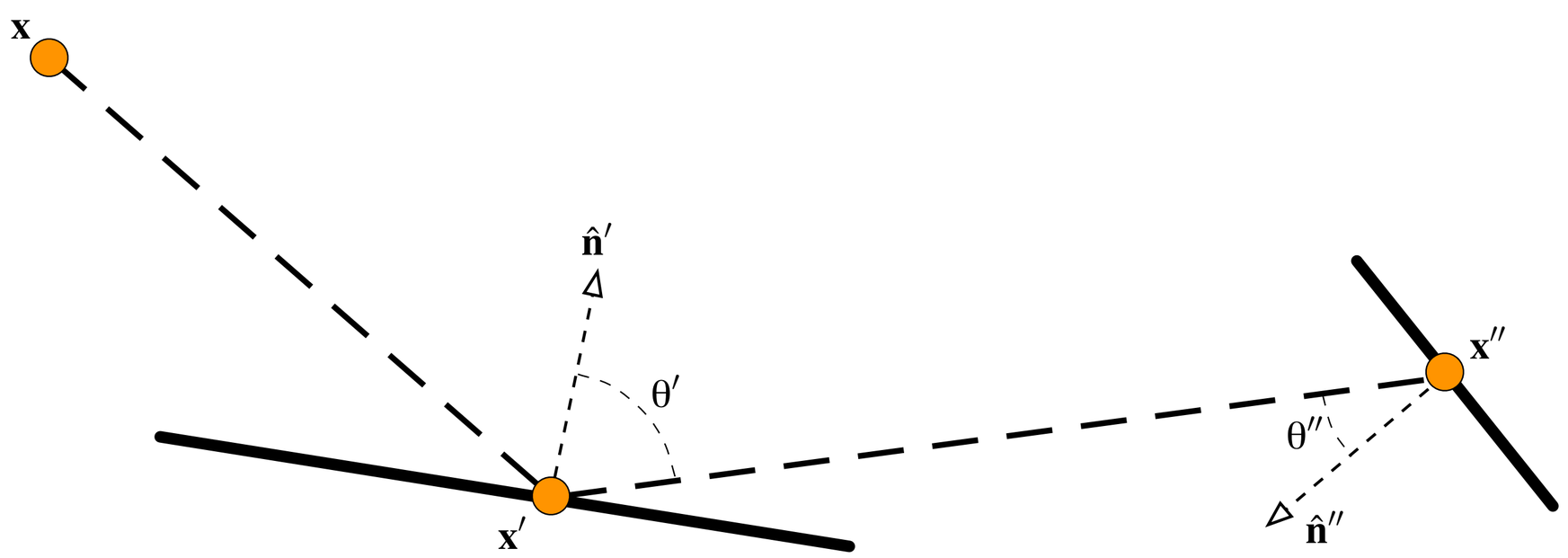
$$L_s(\mathbf{x},\mathbf{x}') = \delta(\mathbf{x},\mathbf{x}')\left[E(\mathbf{x},\mathbf{x}') + \int_S \rho_{x'}(\mathbf{x},\mathbf{x}'')L_s(\mathbf{x}',\mathbf{x}'')\frac{\cos(\theta')\cos(\theta'')}{||\mathbf{x}'-\mathbf{x}''||^2}\mathrm{d}\mathbf{x}''\right]$$

sum over every bit of surface in the scene



The Rendering Equation

$$L_s(\mathbf{x},\mathbf{x}') = \delta(\mathbf{x},\mathbf{x}')\left[E(\mathbf{x},\mathbf{x}') + \int_S \rho_{x'}(\mathbf{x},\mathbf{x}'')L_s(\mathbf{x}',\mathbf{x}'')\frac{\cos(\theta')\cos(\theta'')}{||\mathbf{x}'-\mathbf{x}''||^2}\mathrm{d}\mathbf{x}''\right]$$



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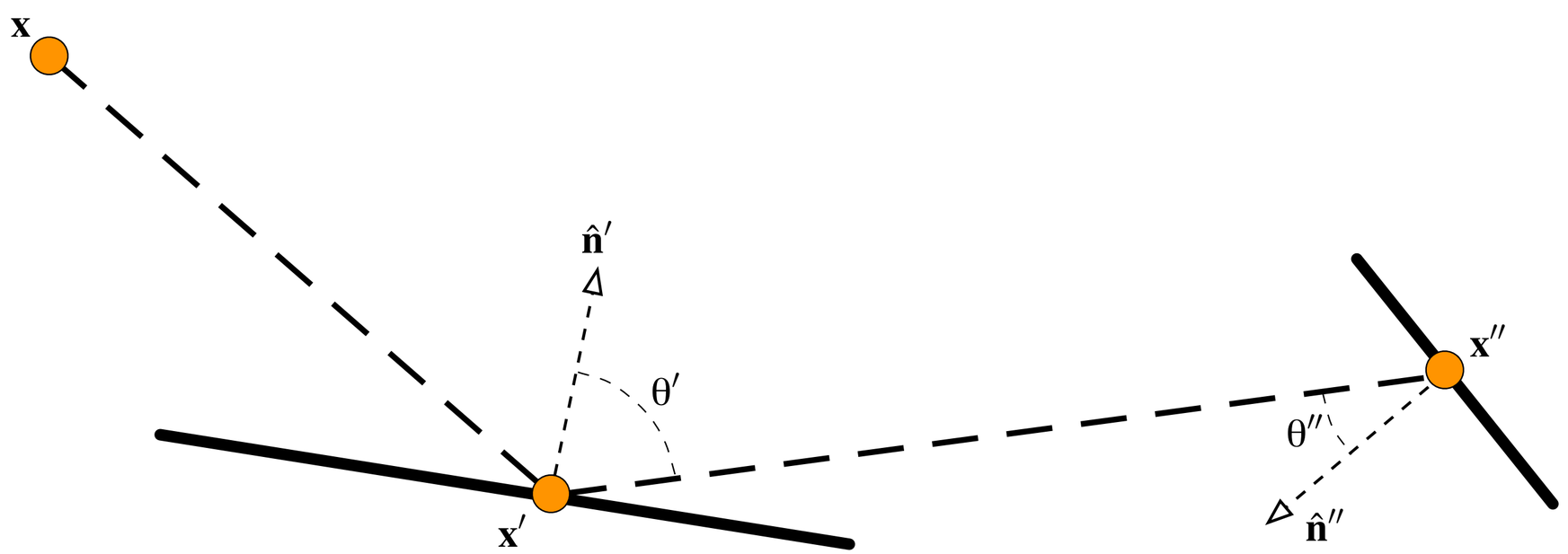
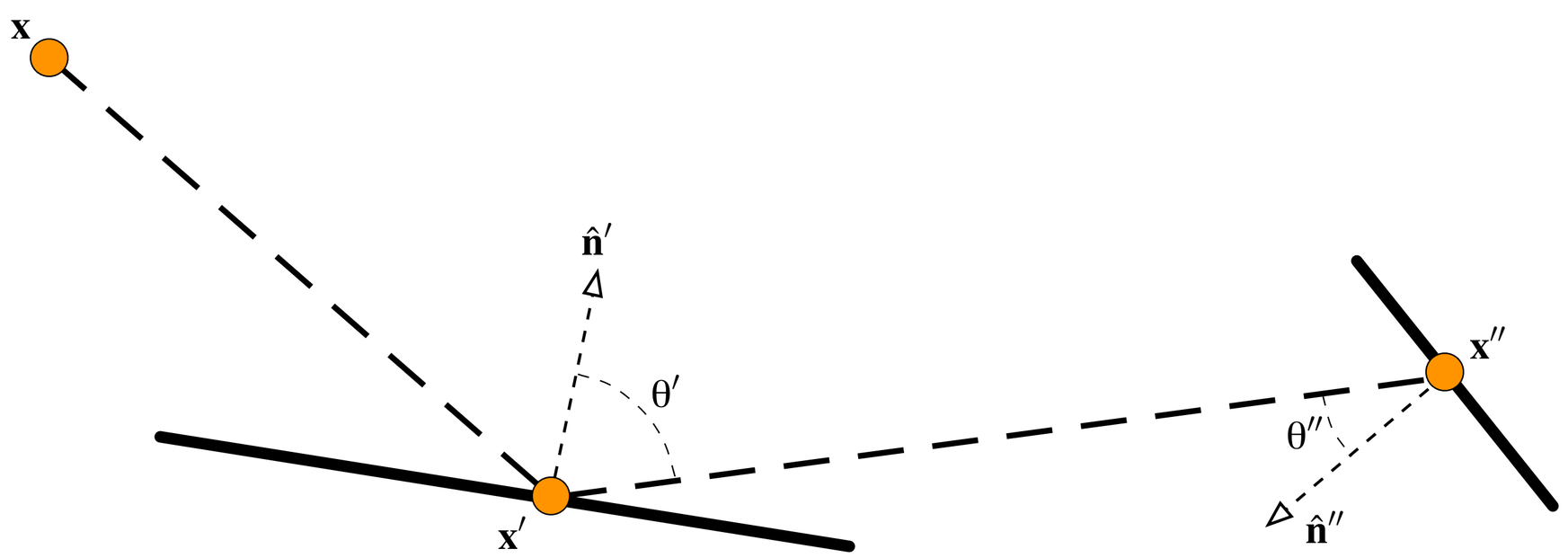
Light emitted from x'' toward x'



The Rendering Equation

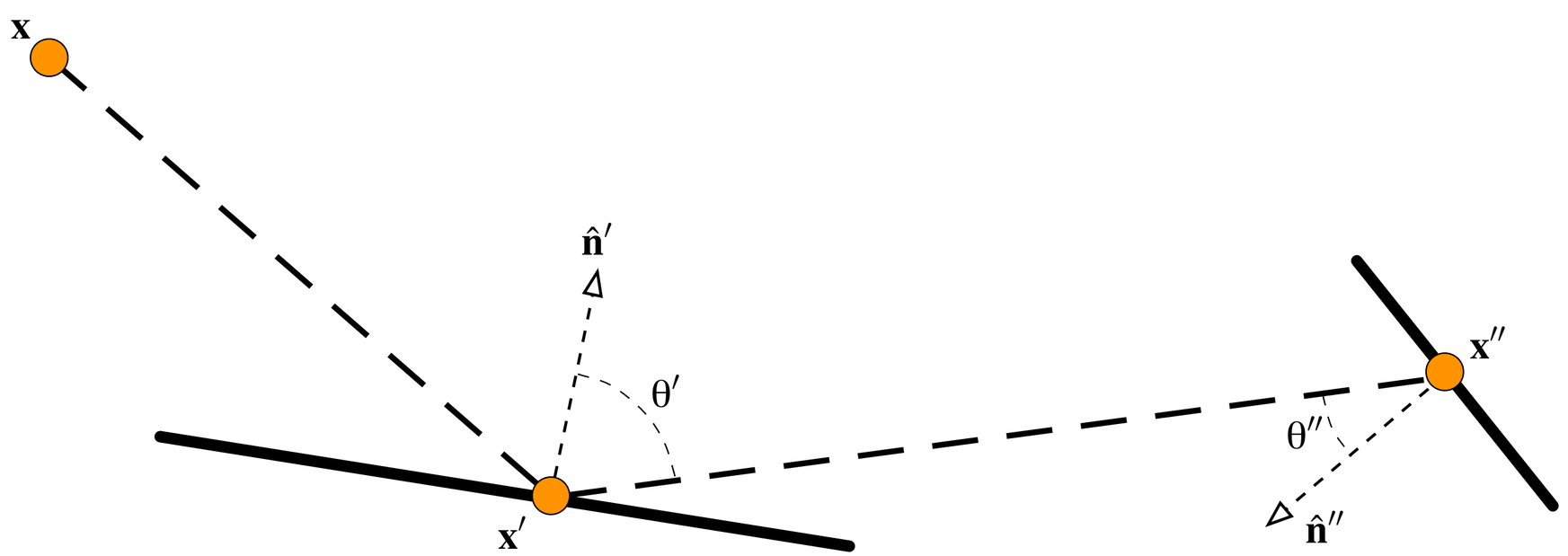
$$L_s(\mathbf{x},\mathbf{x}') = \delta(\mathbf{x},\mathbf{x}')\left[E(\mathbf{x},\mathbf{x}') + \int_S \rho_{x'}(\mathbf{x},\mathbf{x}'')L_s(\mathbf{x}',\mathbf{x}'')\frac{\cos(\theta')\cos(\theta'')}{||\mathbf{x}'-\mathbf{x}''||^2}\mathrm{d}\mathbf{x}''\right]$$



The Rendering Equation

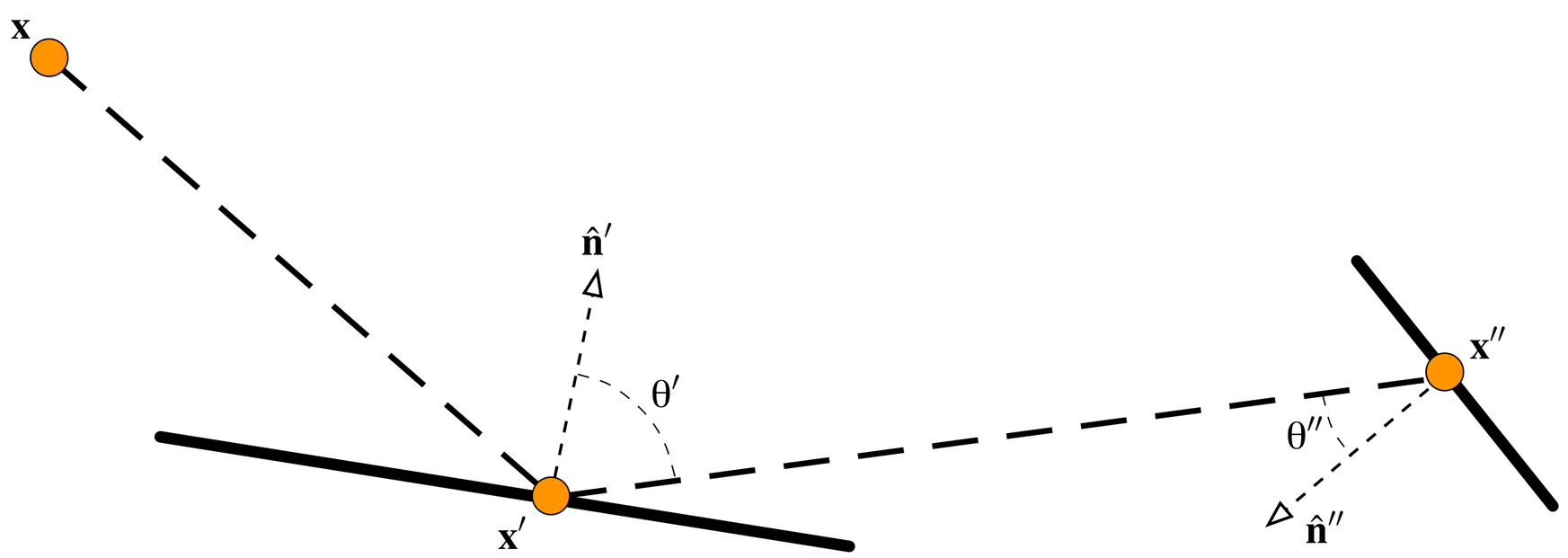
$$L_s(\mathbf{x},\mathbf{x}') = \delta(\mathbf{x},\mathbf{x}')\left[E(\mathbf{x},\mathbf{x}') + \int_S \rho_{x'}(\mathbf{x},\mathbf{x}'')L_s(\mathbf{x}',\mathbf{x}'')\frac{\cos(\theta')\cos(\theta'')}{||\mathbf{x}'-\mathbf{x}''||^2}\mathrm{d}\mathbf{x}''\right]$$

scaled down by the BRDF of x'



The Rendering Equation

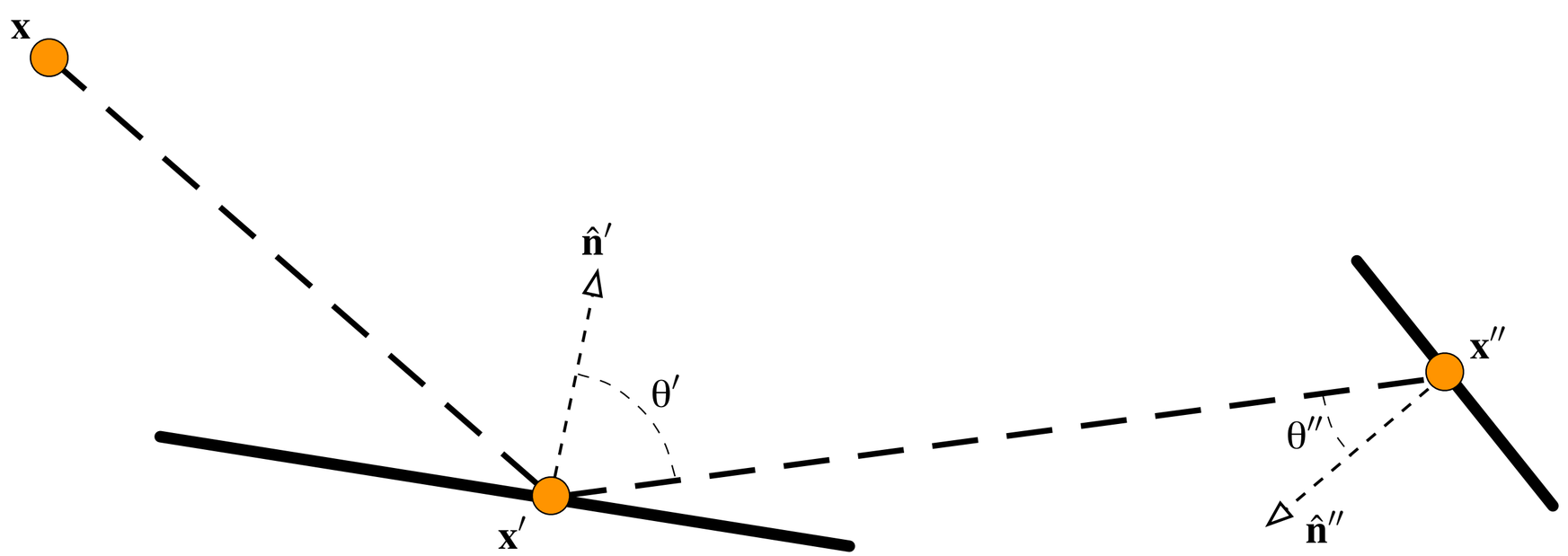
$$L_s(\mathbf{x},\mathbf{x}') = \delta(\mathbf{x},\mathbf{x}')\left[E(\mathbf{x},\mathbf{x}') + \int_S \rho_{x'}(\mathbf{x},\mathbf{x}'')L_s(\mathbf{x}',\mathbf{x}'')\frac{\cos(\theta')\cos(\theta'')}{||\mathbf{x}'-\mathbf{x}''||^2}\mathrm{d}\mathbf{x}''\right]$$



The Rendering Equation

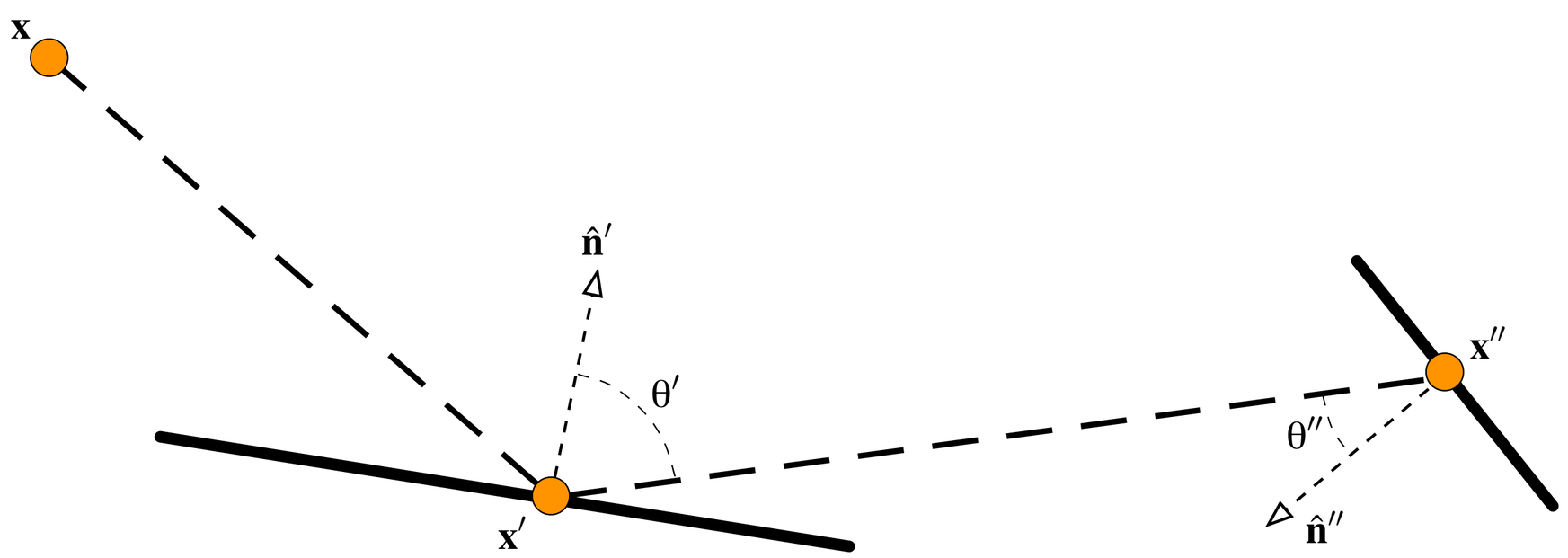
$$L_s(\mathbf{x},\mathbf{x}') = \delta(\mathbf{x},\mathbf{x}')\left[E(\mathbf{x},\mathbf{x}') + \int_S \rho_{x'}(\mathbf{x},\mathbf{x}'')L_s(\mathbf{x}',\mathbf{x}'')\frac{\cos(\theta')\cos(\theta'')}{||\mathbf{x}'-\mathbf{x}''||^2}d\mathbf{x}''\right]$$

scaled down by distance and relative orientation ("form factor")



The Rendering Equation

$$L_s(\mathbf{x},\mathbf{x}') = \delta(\mathbf{x},\mathbf{x}')\left[E(\mathbf{x},\mathbf{x}') + \int_S \rho_{x'}(\mathbf{x},\mathbf{x}'')L_s(\mathbf{x}',\mathbf{x}'')\frac{\cos(\theta')\cos(\theta'')}{||\mathbf{x}'-\mathbf{x}''||^2}\mathrm{d}\mathbf{x}''\right]$$



The Rendering Equation

Re-write the reflection equation:

$$L_o(\mathrm{p}, \omega_o) = L_e(\mathrm{p}, \omega_o) + \int_{H^2} f_r(\mathrm{p}, \omega_i \rightarrow \omega_o)\, L_i(\mathrm{p}, \omega_i)\, \cos\theta_i \, \mathrm{d}\omega_i$$

Using the transport function: $L_i(\mathrm{p}, \omega_i) = L_o(tr(\mathrm{p}, \omega_i), -\omega_i)$

The Rendering Equation

$$L_o(\mathrm{p}, \omega_o) = L_e(\mathrm{p}, \omega_o) + \int_{H^2} f_r(\mathrm{p}, \omega_i \rightarrow \omega_o)\, L_o(tr(\mathrm{p}, \omega_i), -\omega_i)\, \cos\theta_i \, \mathrm{d}\omega_i$$

Note: recursion is now explicit

How to solve?

Radiosity

Assume all materials are perfectly Lambertian (diffuse only, no specularities)

- Removes all dependance on directions
- Reduces dimensionality of lightfield
- Allows a FEM solution (break up into chunks)

Can also relax assumption slightly...

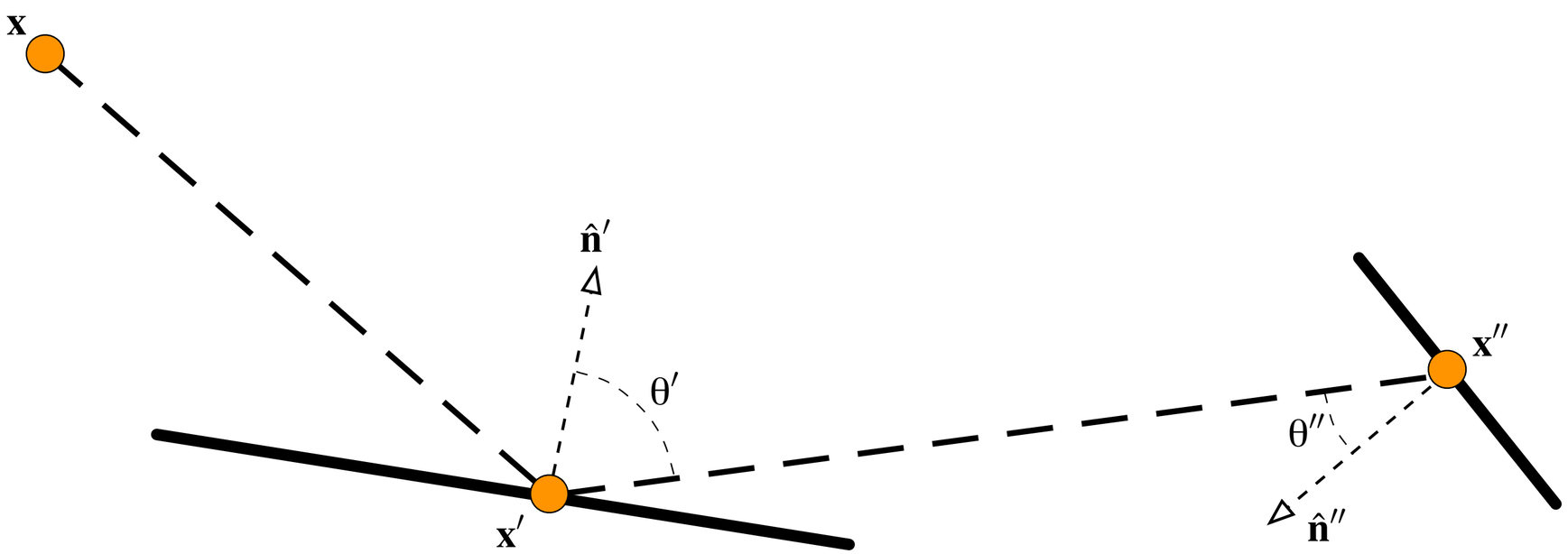
Early radiosity

from Hanrahan 2000

Assume Lambertian

$$L_s(\mathbf{x},\mathbf{x}') = \delta(\mathbf{x},\mathbf{x}')\left[E(\mathbf{x},\mathbf{x}') + \int_S \rho_{x'}(\mathbf{x},\mathbf{x}'')L_s(\mathbf{x}',\mathbf{x}'')\frac{\cos(\theta')\cos(\theta'')}{||\mathbf{x}'-\mathbf{x}''||^2}\mathrm{d}\mathbf{x}''\right]$$

$$L_s(\mathbf{x},\mathbf{x}') = \delta(\mathbf{x},\mathbf{x}')\left[E_{x'} + \int_S \rho_{x'}L_s(\mathbf{x}',\mathbf{x}'')\frac{\cos(\theta')\cos(\theta'')}{||\mathbf{x}'-\mathbf{x}''||^2}\mathrm{d}\mathbf{x}''\right]$$

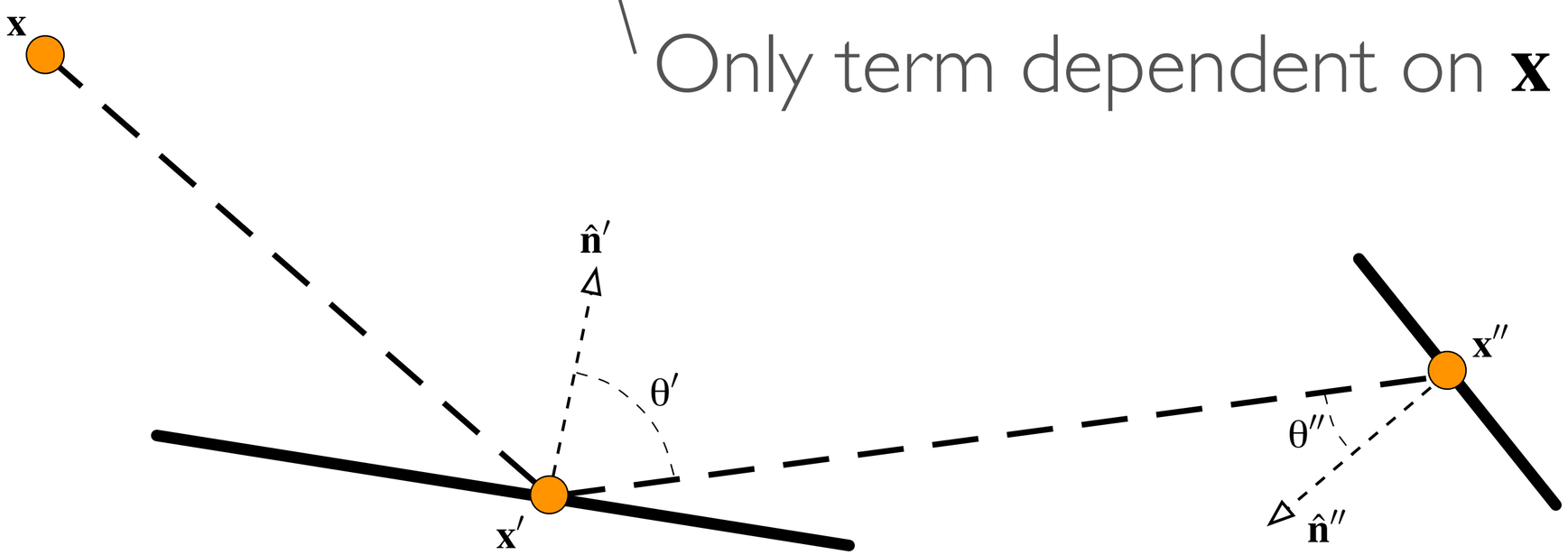


Assume Lambertian

$$L_s(\mathbf{x},\mathbf{x}') = \delta(\mathbf{x},\mathbf{x}')\left[E(\mathbf{x},\mathbf{x}') + \int_S \rho_{x'}(\mathbf{x},\mathbf{x}'')L_s(\mathbf{x}',\mathbf{x}'')\frac{\cos(\theta')\cos(\theta'')}{||\mathbf{x}'-\mathbf{x}''||^2}\mathrm{d}\mathbf{x}''\right]$$

$$L_s(\mathbf{x},\mathbf{x}') = \underline{\delta(\mathbf{x},\mathbf{x}')}\left[E_{x'} + \int_S \rho_{x'}L_s(\mathbf{x}',\mathbf{x}'')\frac{\cos(\theta')\cos(\theta'')}{||\mathbf{x}'-\mathbf{x}''||^2}\mathrm{d}\mathbf{x}''\right]$$

Only term dependent on $\mathbf{x}$

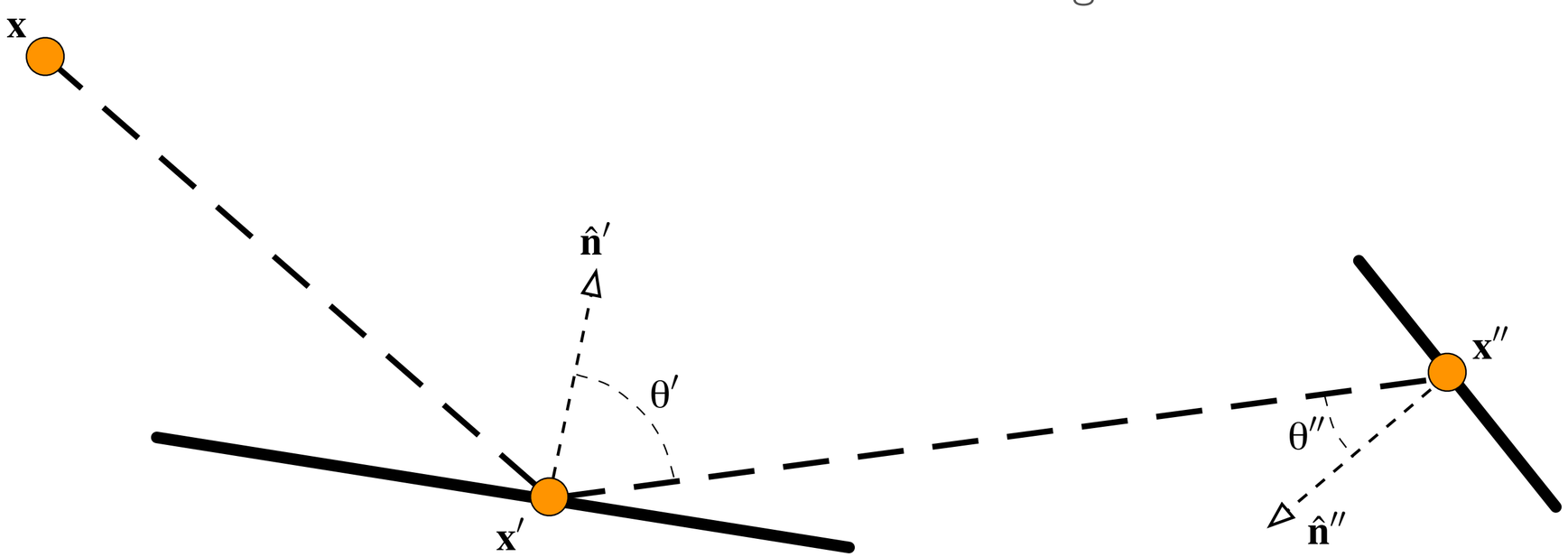


Rewrite in Terms of Radiosity

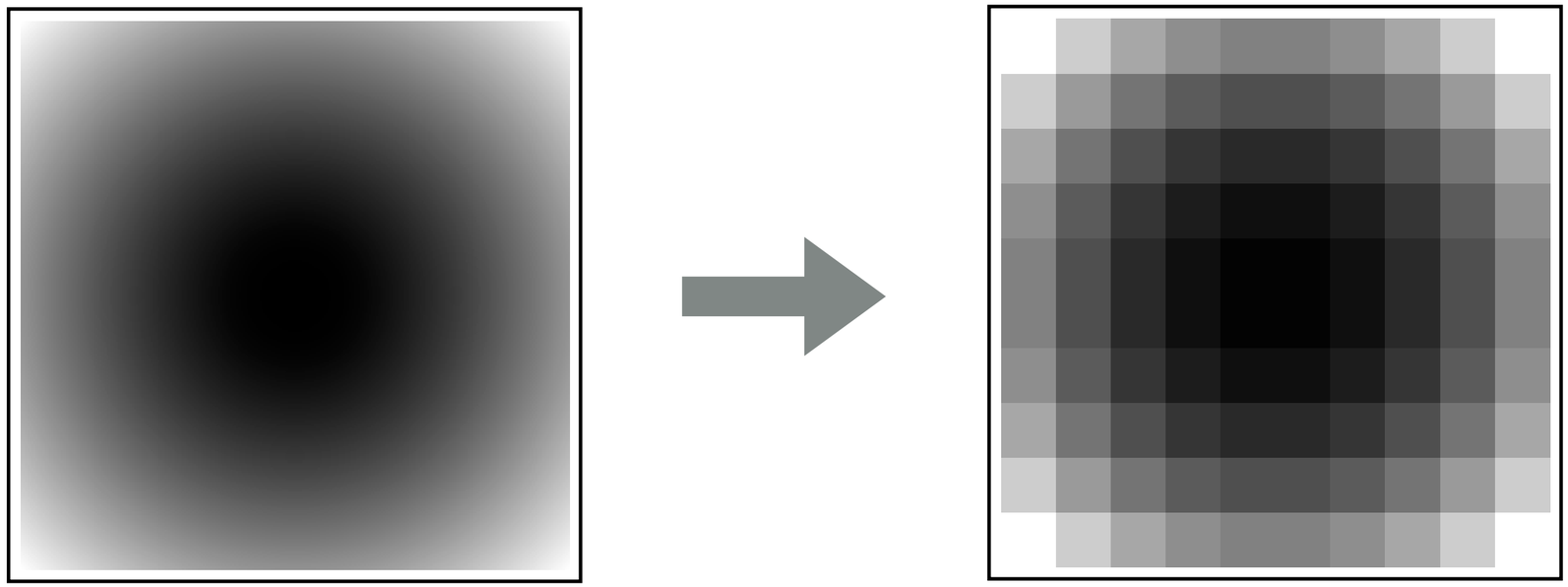
$$L_s(\mathbf{x},\mathbf{x}') = \delta(\mathbf{x},\mathbf{x}')\left[E_{x'} + \int_S \rho_{x'} L_s(\mathbf{x}',\mathbf{x}'')\frac{\cos(\theta')\cos(\theta'')}{||\mathbf{x}'-\mathbf{x}''||^2}\mathrm{d}\mathbf{x}''\right]$$

$$H_{x'} = E_{x'} + \rho_{x'}\int_S \delta(\mathbf{x}',\mathbf{x}'')\frac{H_{x''}}{2\pi}\frac{\cos(\theta')\cos(\theta'')}{||\mathbf{x}'-\mathbf{x}''||^2}\mathrm{d}\mathbf{x}''$$

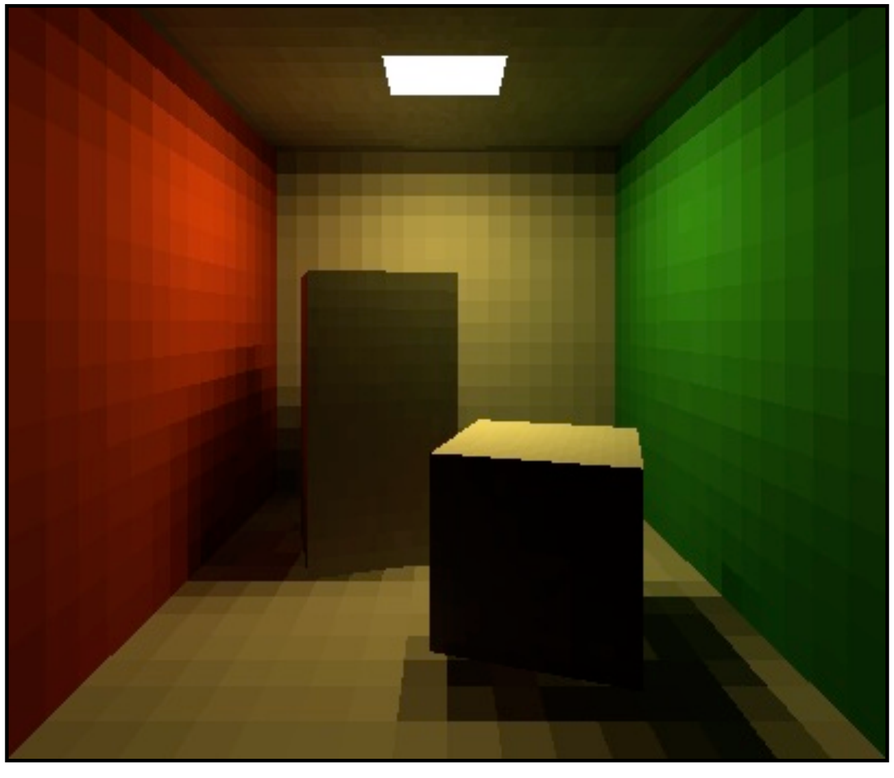
Note: we changed E from radiance to radiosity.



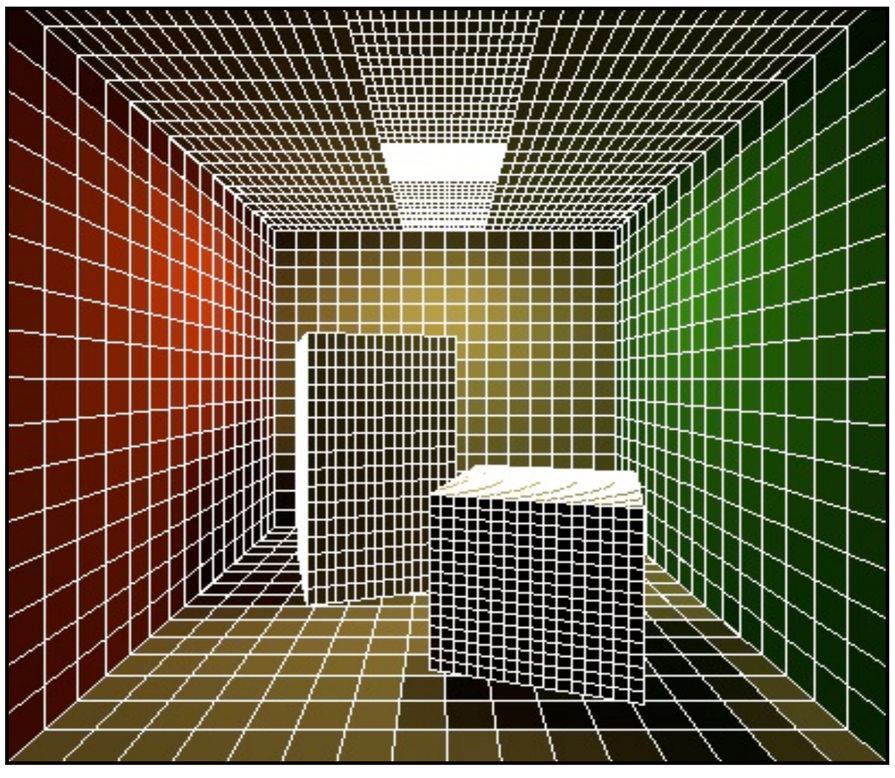
Discretize into Patches



Piece-wise
constant patches



Example mesh for Cornell Box
by Mark Schmelzenbach



Discretize into Patches

The Candlestick Theater,
Mark Mack Architects.

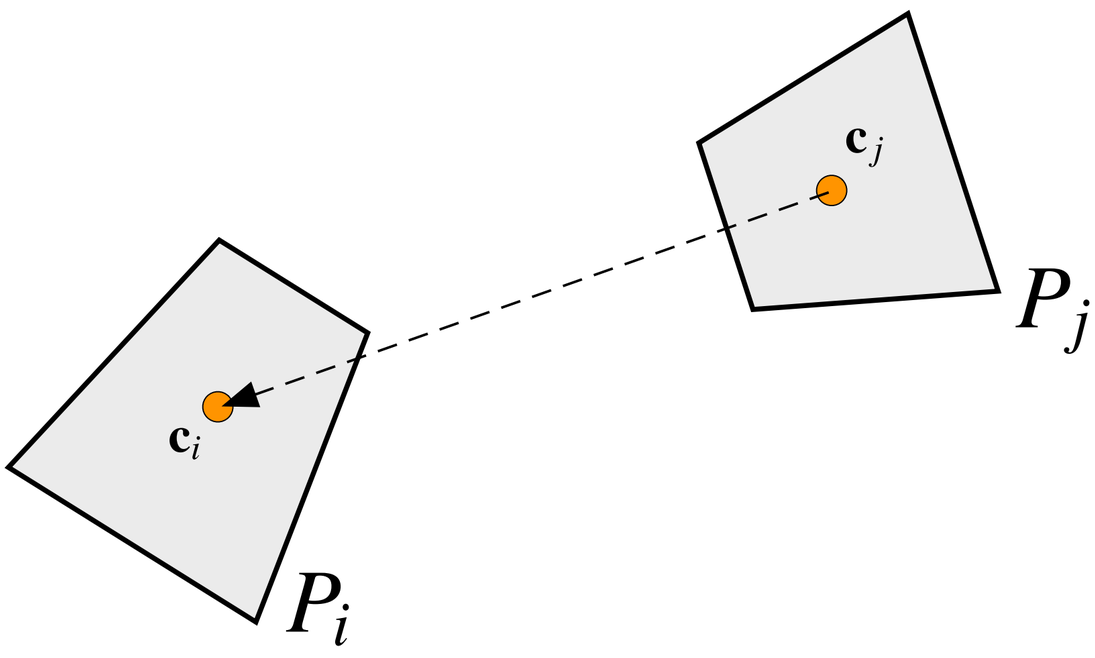
Discretize into Patches

The Candlestick Theater,
Mark Mack Architects.

Rewrite in Terms of Patches

$$H_{x'} = E_{x'} + \rho_{x'} \int_S \delta(\mathbf{x}', \mathbf{x}'') \frac{H_{x''} \cos(\theta') \cos(\theta'')}{||\mathbf{x}' - \mathbf{x}''||^2} d\mathbf{x}''$$

$$H_i = E_i + \rho_i \sum_j H_j \int_{S_j} \delta_{ij} \frac{\cos(\theta_i) \cos(\theta_j)}{2\pi ||\mathbf{c}_i - \mathbf{x}||^2} d\mathbf{x}$$

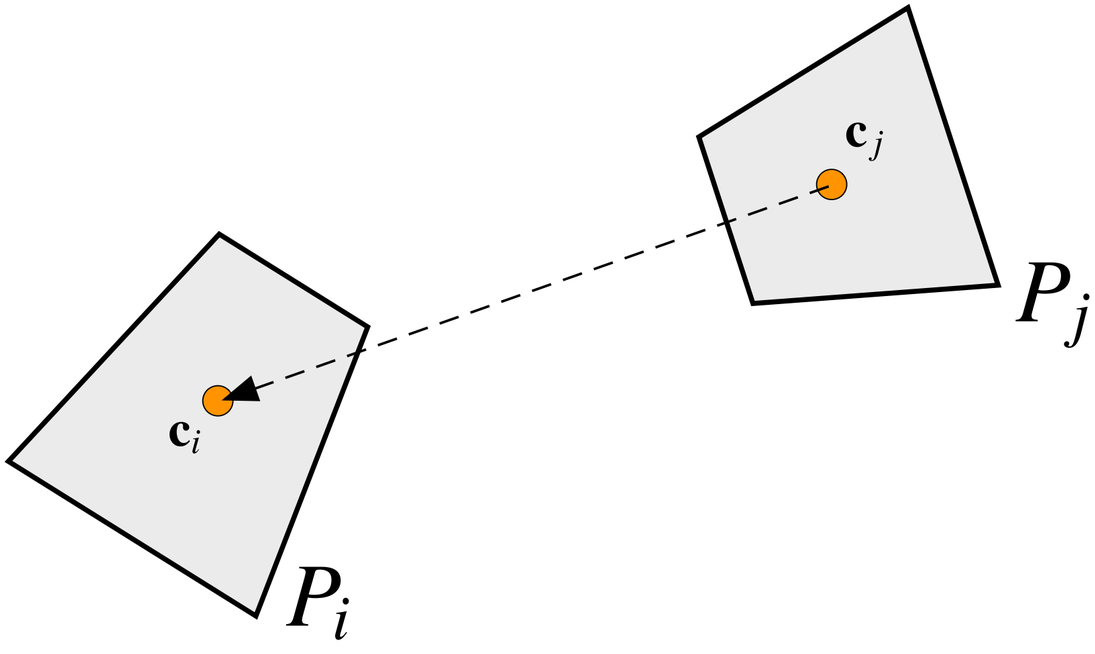


Rewrite in Terms of Patches

$$H_{x'} = E_{x'} + \rho_{x'} \int_S \delta(\mathbf{x}', \mathbf{x}'') \frac{H_{x''} \cos(\theta') \cos(\theta'')}{2\pi \, ||\mathbf{x}' - \mathbf{x}''||^2} d\mathbf{x}''$$

$$H_i = E_i + \rho_i \sum_j H_j \int_{S_j} \delta_{ij} \frac{\cos(\theta_i) \cos(\theta_j)}{2\pi ||\mathbf{c}_i - \mathbf{x}||^2} d\mathbf{x}$$

Form factor from ***j*** to ***i***, F_{ij}

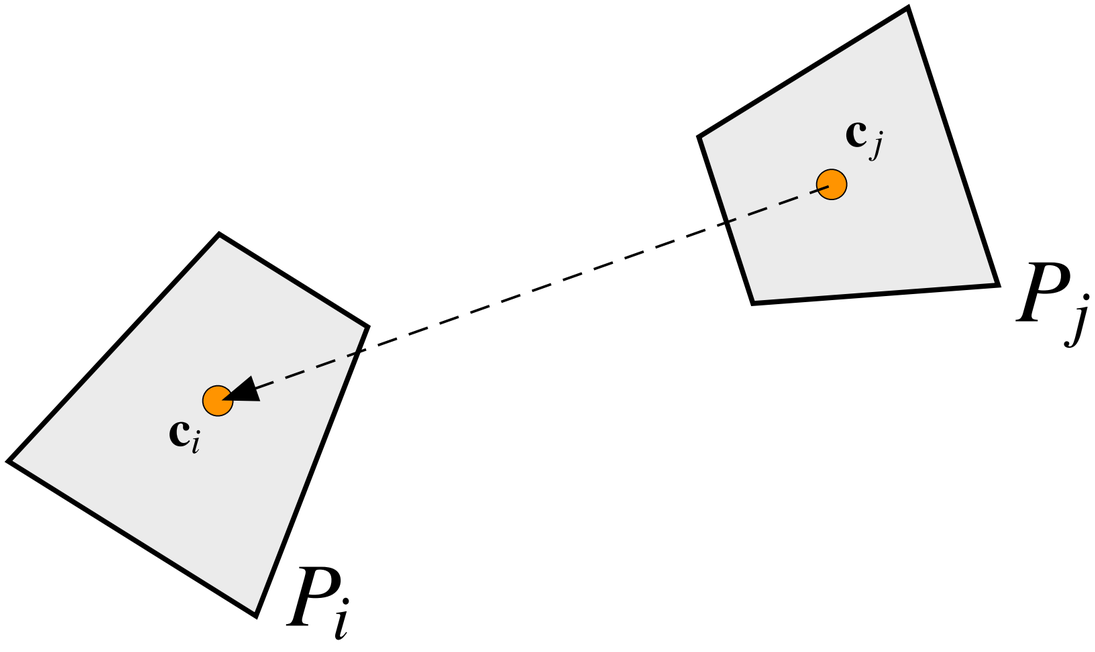


Rewrite in Terms of Patches

$$H_{x'} = E_{x'} + \rho_{x'} \int_S \delta(\mathbf{x}', \mathbf{x}'') \frac{H_{x''} \cos(\theta') \cos(\theta'')}{2\pi} \frac{}{||\mathbf{x}' - \mathbf{x}''||^2} d\mathbf{x}''$$

$$H_i = E_i + \rho_i \sum_j H_j \int_{S_j} \delta_{ij} \frac{\cos(\theta_i) \cos(\theta_j)}{2\pi ||\mathbf{c}_i - \mathbf{x}||^2} d\mathbf{x}$$

Form factor from ***j*** to ***i***, F_{ij}



Example of a rough approximation:

$$F_{ij} \approx \delta_{ij} \frac{\cos(\theta_i) \cos(\theta_j)}{2\pi ||\mathbf{c}_i - \mathbf{c}_j||^2} A_j$$

Radiosity Method

E_i ρ_i

F_{ij}

$$H_i = E_i + \rho_i \sum_j H_j F_{ij}$$

$$\mathbf{h} = \mathbf{e} + \mathbf{A}\mathbf{h}$$

$$(\mathbf{I} - \mathbf{A})\mathbf{h} = \mathbf{e}$$

Radiosity Method

Given the light emitted and surface properties

First compute F_{ij}, form factors between patches

Then solve a linear system to balance energy between all patches

Comments:

- The system is very large
- **It is also sparse (why?)**
- **Should be solved with an iterative method**
  - e.g.: Jacobi or Gauss-Seidel
- **Solution is view independent**

Progressive Radiosity

If magnitude of eigenvalues of $\mathbf{A}<1$

$$(\mathbf{I}-\mathbf{A})^{-1}=\mathbf{I}+\mathbf{A}+\mathbf{A}^2+\mathbf{A}^3+\cdots$$

- True for form-factor matrices

- Use Gauss-Seidel-like iteration but reorder by priority

$$\mathbf{h}^{k+1}=\mathbf{h}^k+\mathbf{u}^{k+1}$$

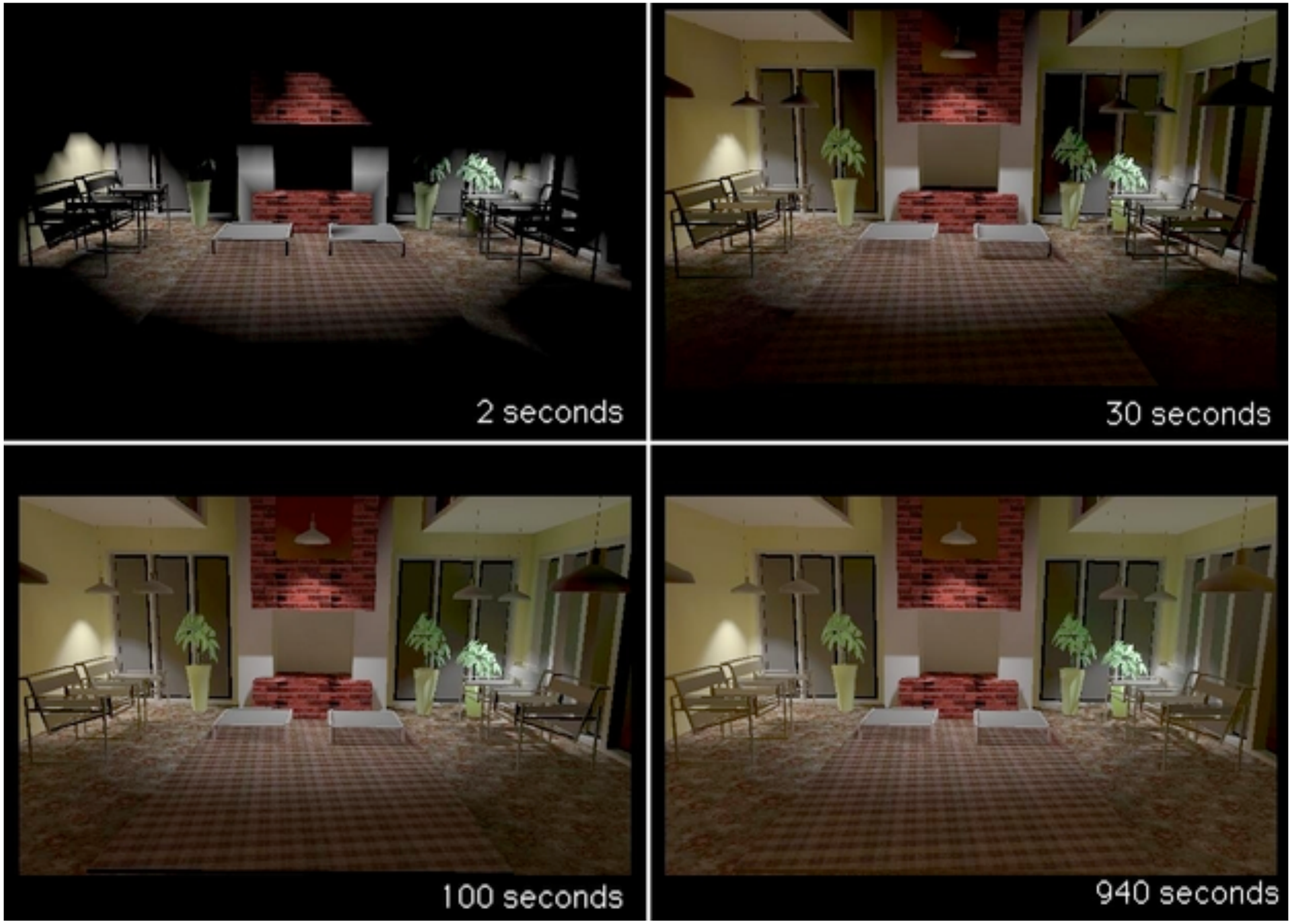
$$\mathbf{u}^{k+1}=\mathbf{A}\,\mathbf{u}^k$$

$$\mathbf{h}^0=0 \quad \mathbf{u}^0=\mathbf{e}$$

Idea: let important sources of light energy emit first, maybe don't even bother with dark things

Southwell Relaxation

Progressive Radiosity

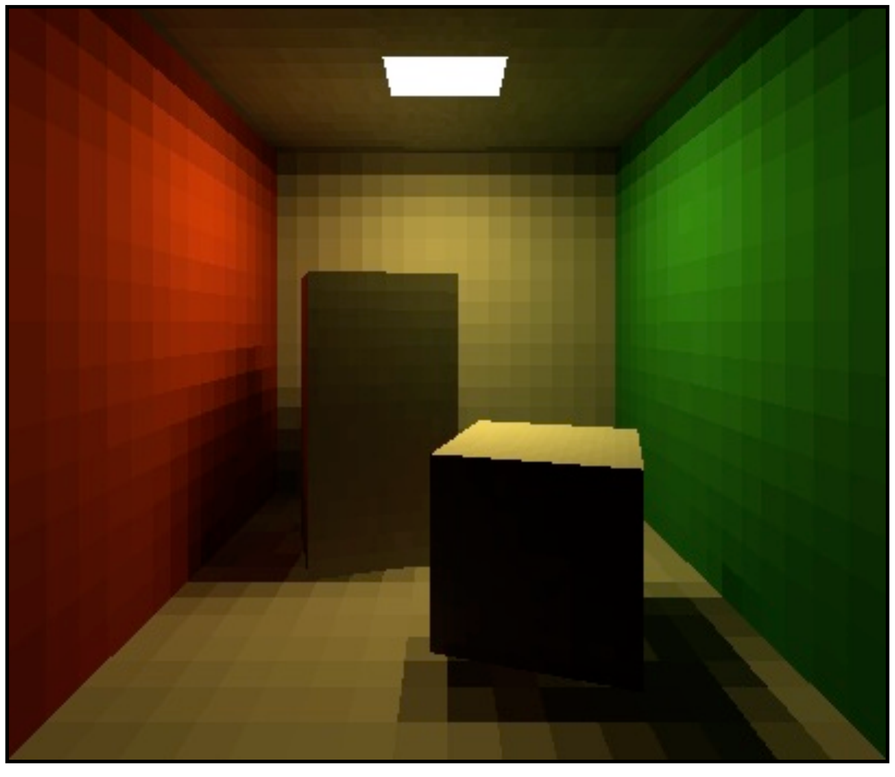


From dissertation "Efficient and predictive realistic image synthesis"
by Karol Myszkowski

Touchup

Each patch will have a constant color

- Smooth solution (*e.g.* average to vertices)



Example mesh for Cornell Box
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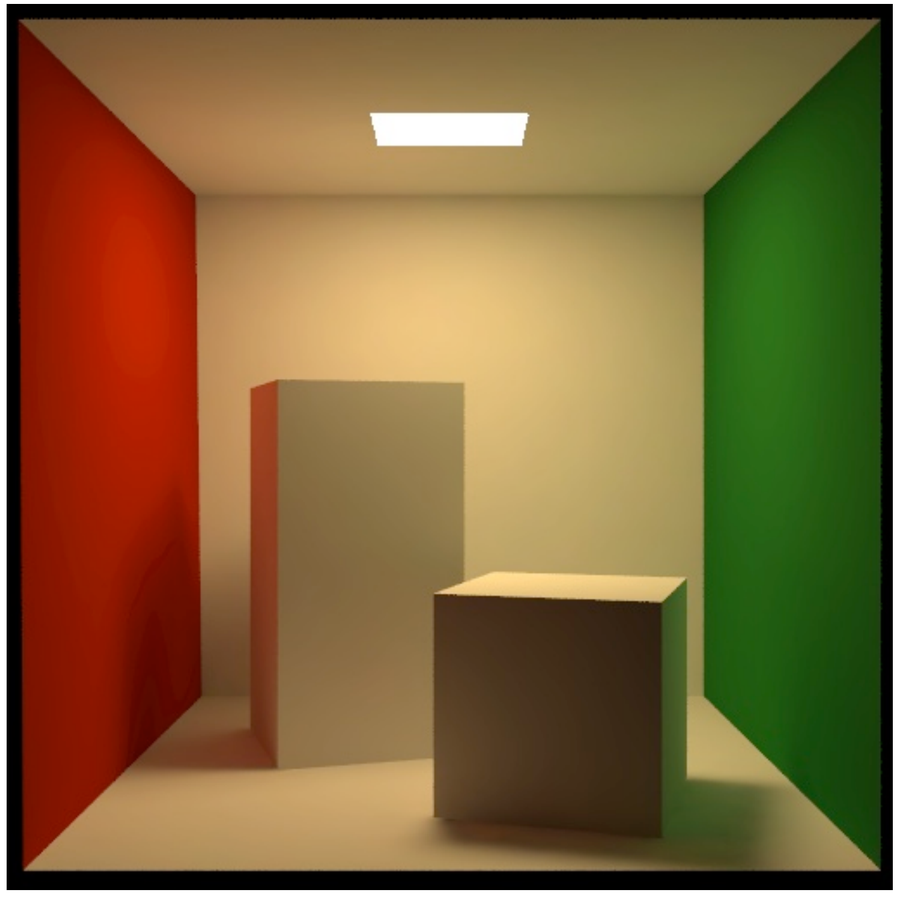


Does not match but you get the idea...

Other Things

Each patch will have a constant color

- Smooth solution (*e.g.* average to vertices)

No specular reflection

- **Add Phong specular term or raytraced specular reflection**

Grid artifacts

- **Be clever with grid...**

White Light by Paul Cocksedge

White Light by Paul Cocksedge

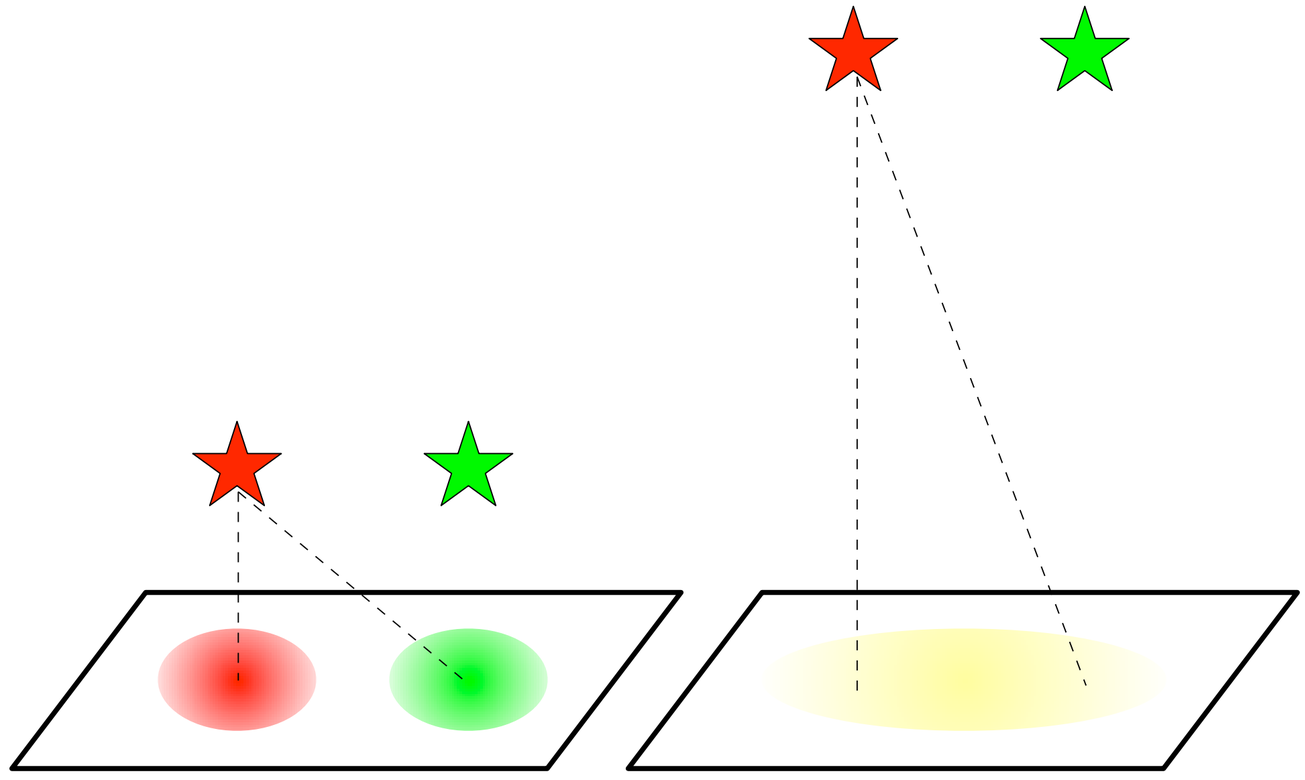
Hierarchical Radiosity

Light smoothes with distance

- Compare $1/h^2$ with $1/(h^2+d^2)$ as h gets large

Group patches into hierarchy

- Far interactions use lower-res form factors



Hierarchical Radiosity

Light smoothes with distance

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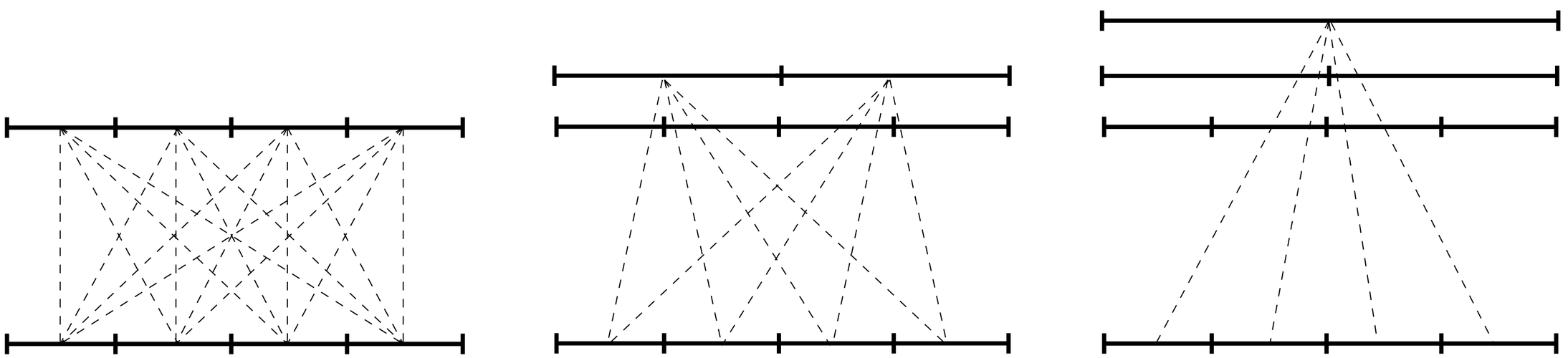


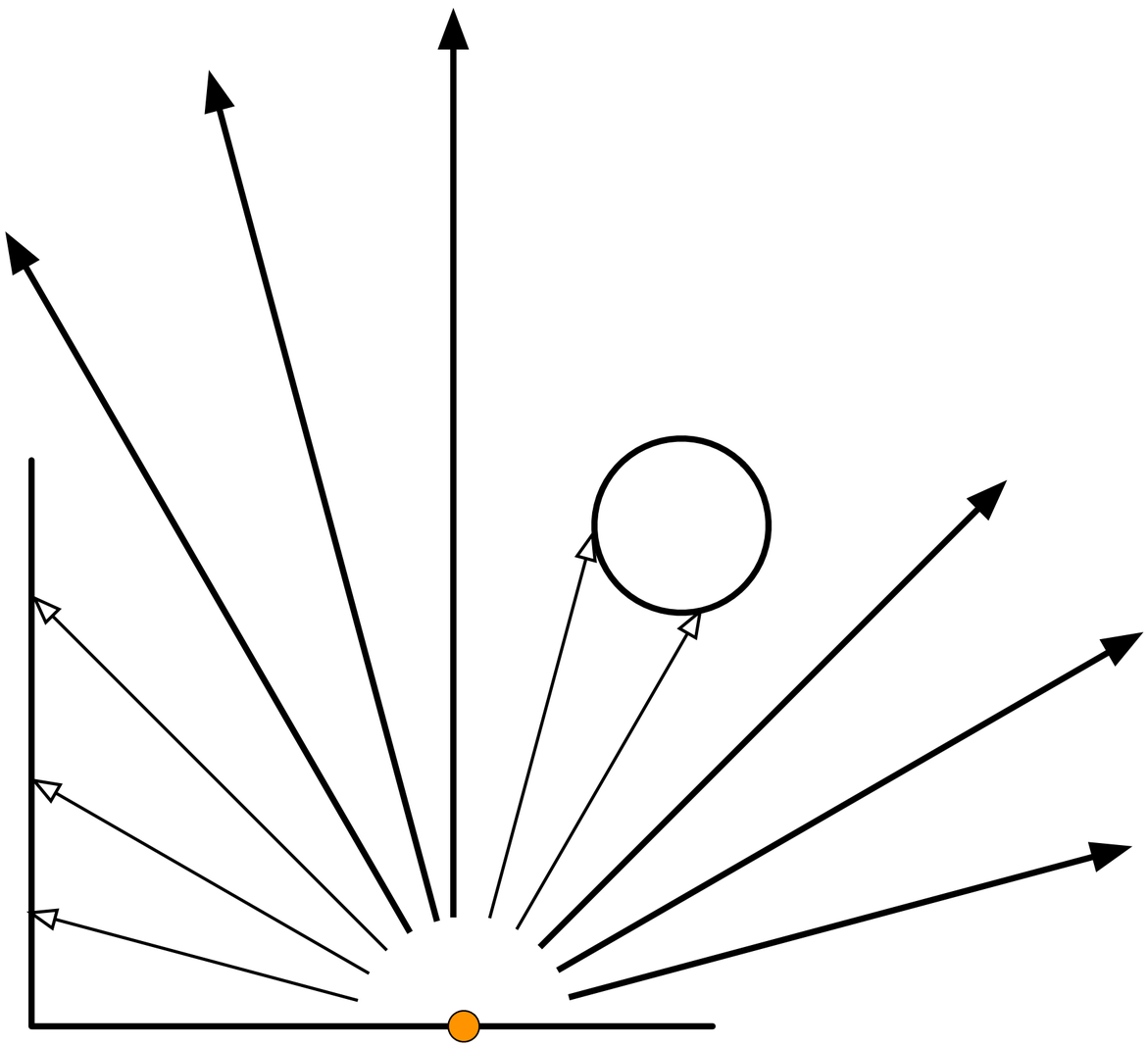
Ambient Occlusion

A “hack” to create more realistic ambient illumination cheaply

Assume light from everywhere is partially blocked by local objects

- At a point on the surface cast rays at random
- Ambient term is proportional to percent of rays that hit nothing
- Weight average by cosine of angle with normal
- Take into account how far before occluded

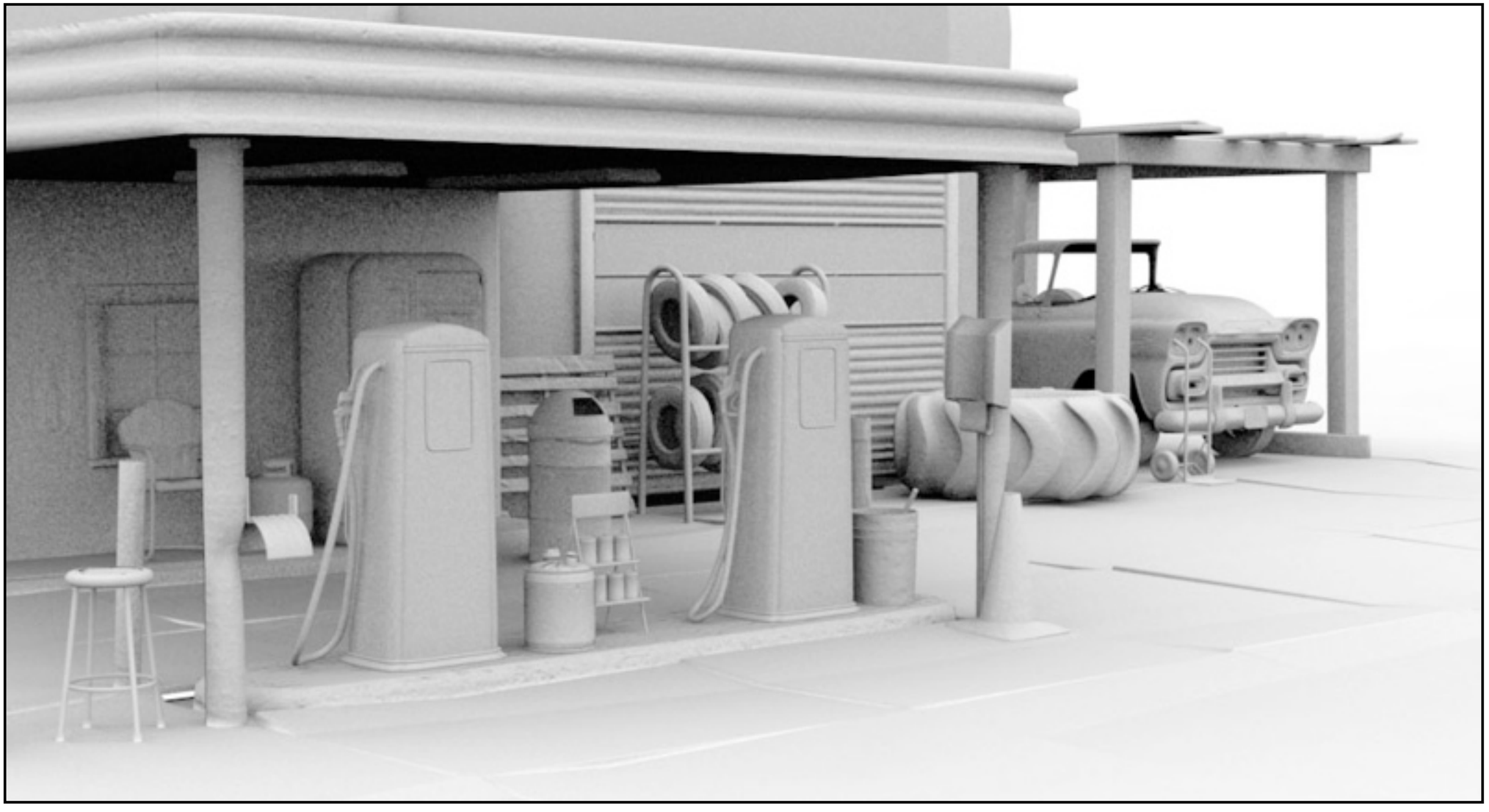
Ambient Occlusion



Ambient Occlusion

Diffuse Only Ambient Occlusion Combined

Ambient Occlusion



nVidia Gelato Demo Image

Light Transport Operators

Operators Are Higher-Order Functions

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Functions:

$$f, g : (x, \omega) \to \mathbb{R}$$

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Operators are higher-order functions:

$$P : ((x, \omega) \to \mathbb{R}) \to ((x, \omega) \to \mathbb{R})$$
$$P(f) = g$$

- **Take a function and transform it into another function**

Linear Operators

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- Linear operators act on functions like matrices act on vectors

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Linear Operators

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$$h(x) = (L(f))(x)$$

- **They are linear in that:**

$$L(af + bg) = aL(f) + bL(g)$$

- **Examples of linear operators:**

$$H(f)(x) = \int h(x, x')\, f(x')\, \mathrm{d}x'$$

$$D(f)(x) = \frac{\delta f}{\delta x}(x)$$

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  (all surface points & outgoing directions) $L_e(\mathrm{p}, \omega)$

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$$R(g)(\mathrm{p}, \omega_o) \equiv \int_{H^2} f_r(\mathrm{p}, \omega_i \rightarrow \omega_o)\, g(\mathrm{p}, \omega_i)\, \cos\theta_i \, \mathrm{d}\omega_i$$

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- **Transport operator:**

$$T(f)(\mathrm{p}, \omega_o) \equiv f(tr(\mathrm{p}, \omega), -\omega)$$

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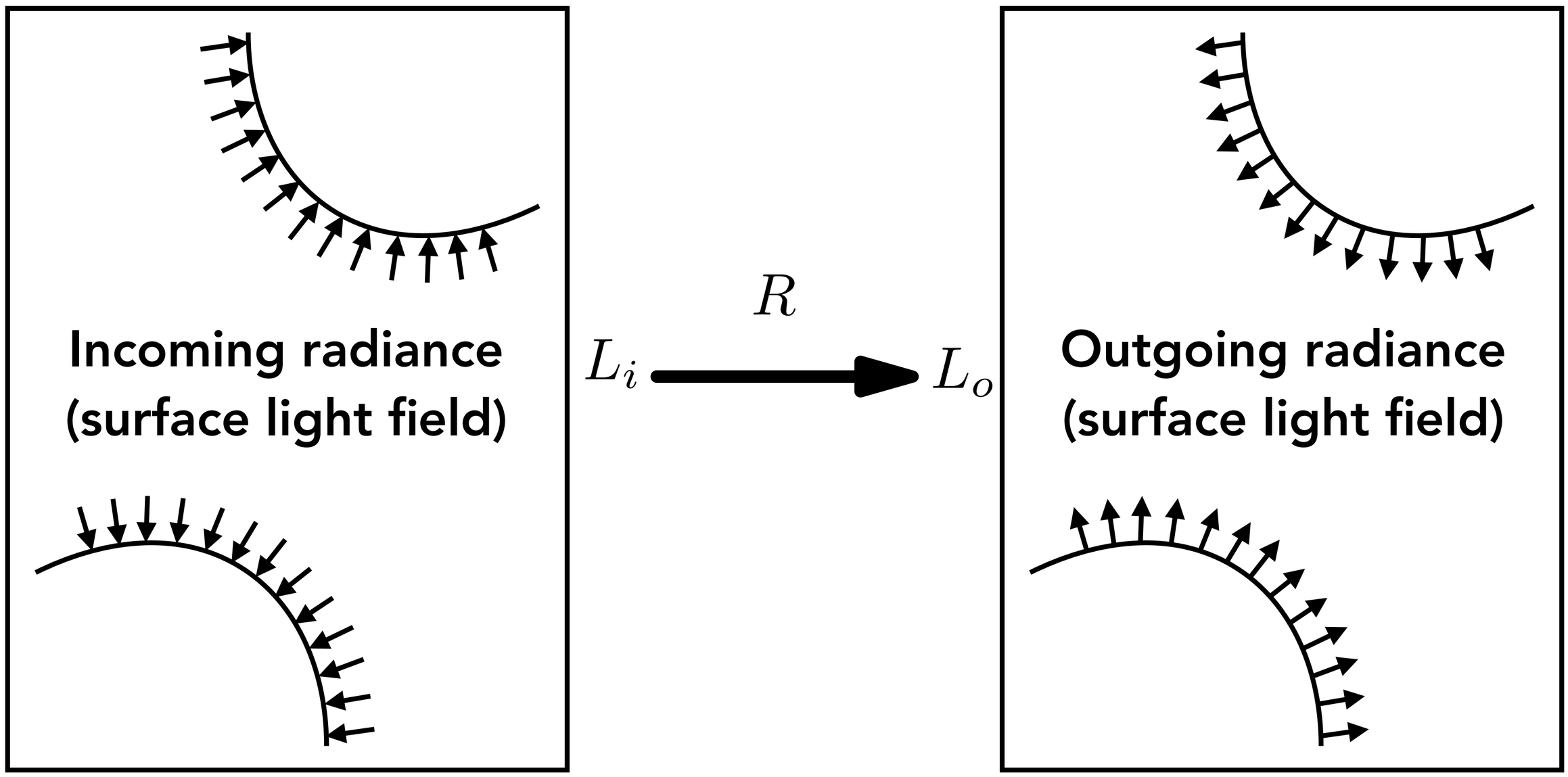
Reflection Operator



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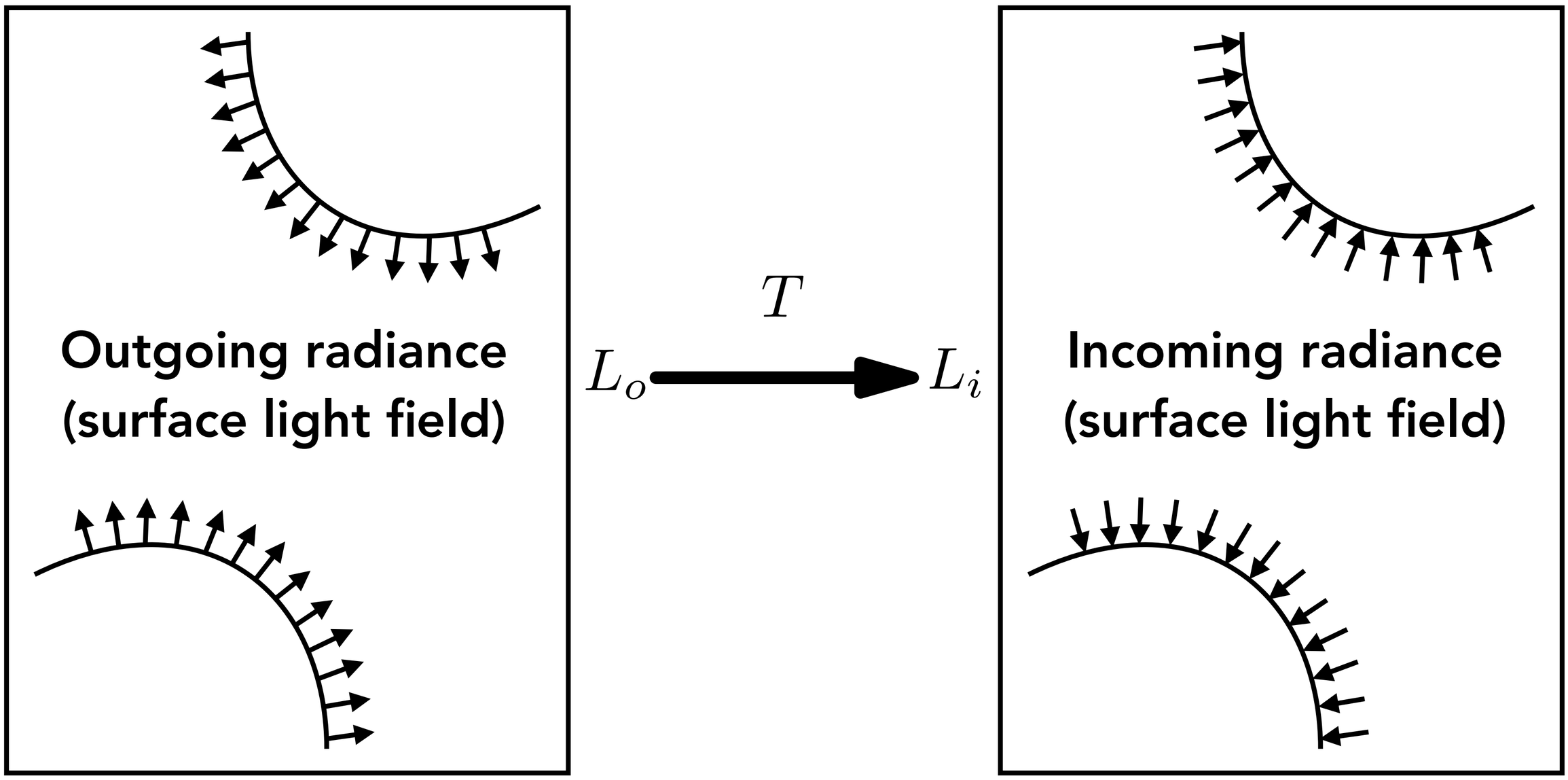
Transport Operator



$$T(f)(\mathrm{p}, \omega_o) \equiv f(tr(\mathrm{p}, \omega_o), -\omega_o)$$

$$T(L_o) = L_i$$

Rendering Equation in Operator Notation

$$L_o(\mathrm{p}, \omega_o) = L_e(\mathrm{p}, \omega_o) + \int_{H^2} f_r(\mathrm{p}, \omega_i \to \omega_o)\, L_o(tr(\mathrm{p}, \omega_i), -\omega_i)\, \cos\theta_i \, \mathrm{d}\omega_i$$

$$L_o = L_e + (R \circ T)(L_o)$$

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Define full one-bounce light transport operator: $K = R \circ T$

$$\boxed{L_o = L_e + K(L_o)}$$

Solving the Rendering Equation

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- Rendering equation:

$$L = L_e + K(L)$$

$$(I - K)(L) = L_e$$

***L* is outgoing reflected**

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***L* is outgoing reflected**

$$(I - K)(L) = L_e$$

- **Solution desired:**

$$L = (I - K)^{-1}(L_e)$$

- **How to solve?**

Solution Intuition

For scalar functions, recall:

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots$$

$$\text{converges for } \; -1 < x < 1$$

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where $||K|| < 1$ means that the "energy" of the radiance function decreases after applying K. This is intuitively true for valid scene models based on energy dissipation (though not trivial to prove, see Veach & Guibas).

Formal Solution

Neumann series:

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Check:

$$\begin{aligned}
&(I - K) \circ (I - K)^{-1} \\
&\quad = (I - K) \circ (I + K + K^2 + K^3 + \cdots) \\
&\quad = (I + K + K^2 + \cdots) - (K + K^2 + \cdots) \\
&\quad = I
\end{aligned}$$

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&\quad = I
\end{aligned}$$

Again, energy dissipation makes it possible to show that the series converges.

Rendering Equation Solution

$$
\begin{aligned}
L &= (I - K)^{-1}(L_e) \\
&= (I + K + K^2 + K^3 + \cdots)(L_e) \\
&= L_e + K(L_e) + K^2(L_e) + K^3(L_e) + \cdots
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Emitted **1-bounce** **2-bounce** **3-bounce**

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Intuitive: Sum of successive bounces of light

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Emitted **1-bounce** **2-bounce** **3-bounce**

Intuitive: Sum of successive bounces of light

This calculates the steady-state surface light field over the scene.

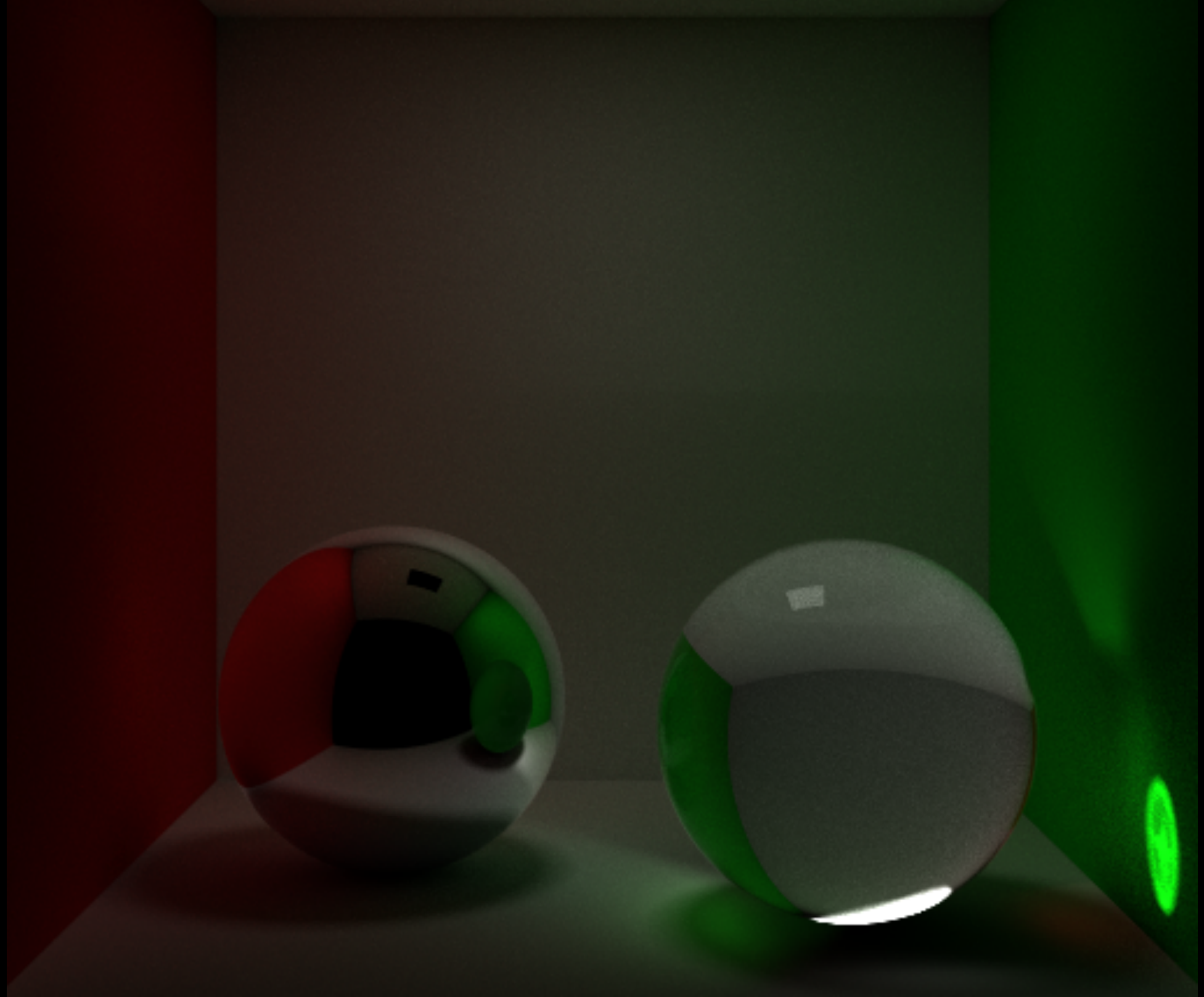
L_e

$K(L_e)$

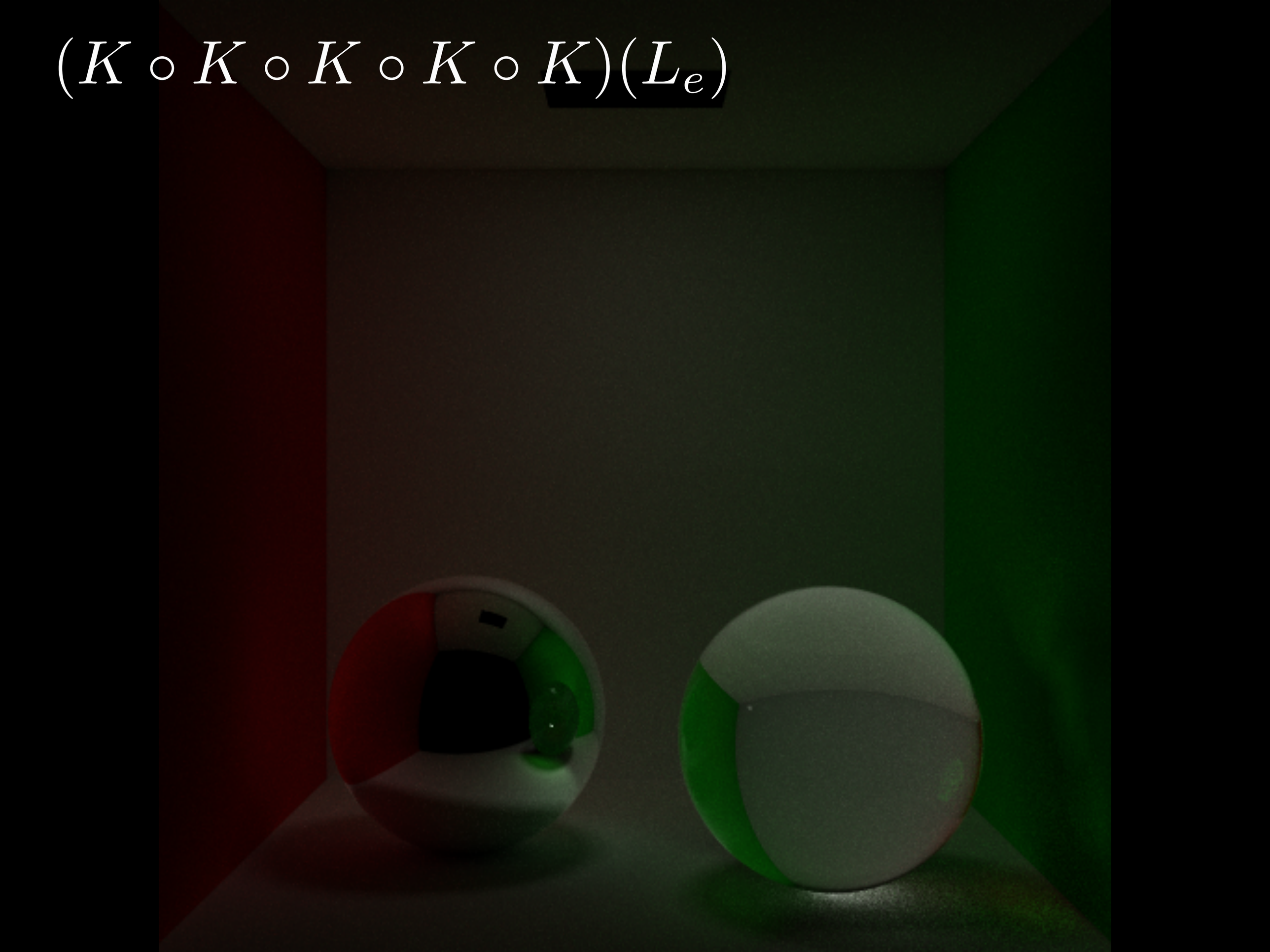
$(K \circ K)(L_e)$

$(K \circ K \circ K)(L_e)$

$$(K \circ K \circ K \circ K)(L_e)$$



$$(K \circ K \circ K \circ K \circ K)(L_e)$$



$$(K \circ K \circ K \circ K \circ K \circ K)(L_e)$$

$$\sum_{i=0}^{0} K^i(L_e)$$

$$\sum_{i=0}^{1} K^i(L_e)$$

$$\sum_{i=0}^{2} K^i(L_e)$$

$$\sum_{i=0}^{3} K^i(L_e)$$

$$\sum_{i=0}^{4} K^i(L_e)$$

$$\sum_{i=0}^{5} K^i(L_e)$$

$$\sum_{i=0}^{6} K^i(L_e)$$

Direct illumination

One-bounce global illumination

Two-bounce global illumination

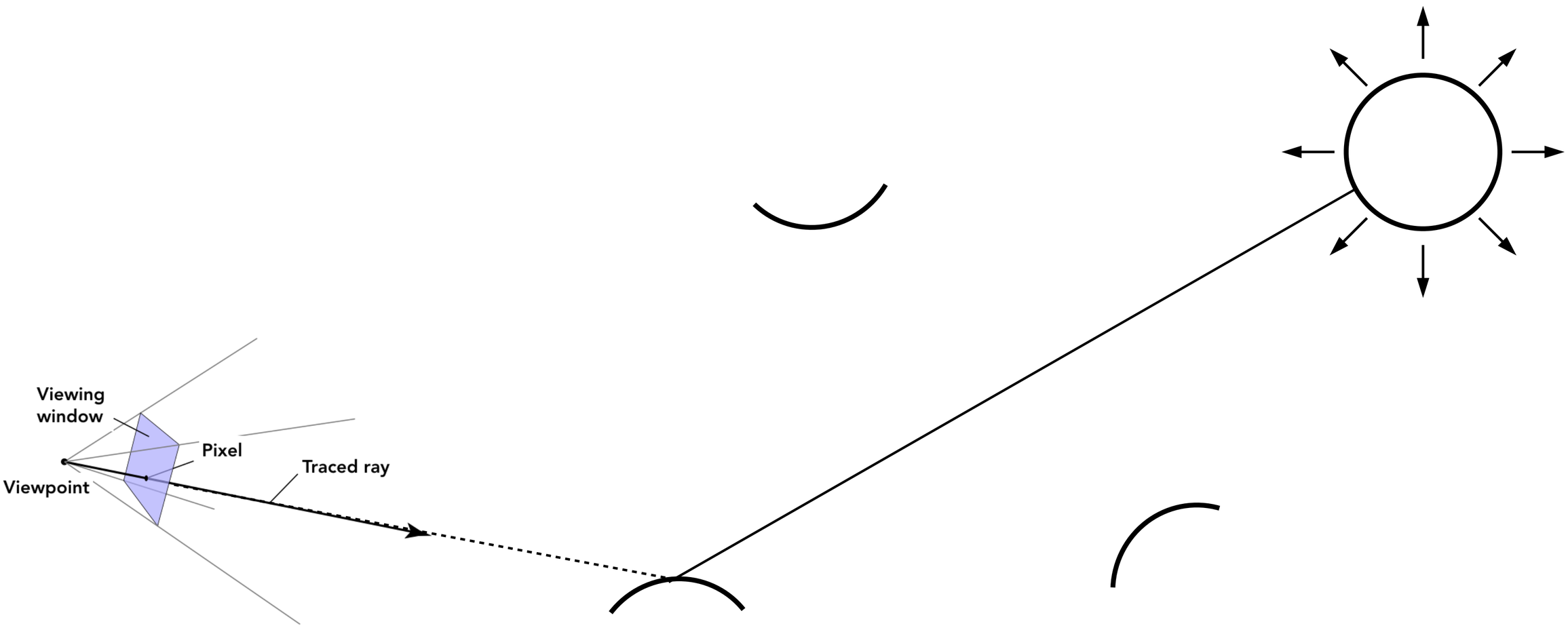
Four-bounce global illumination

Eight-bounce global illumination

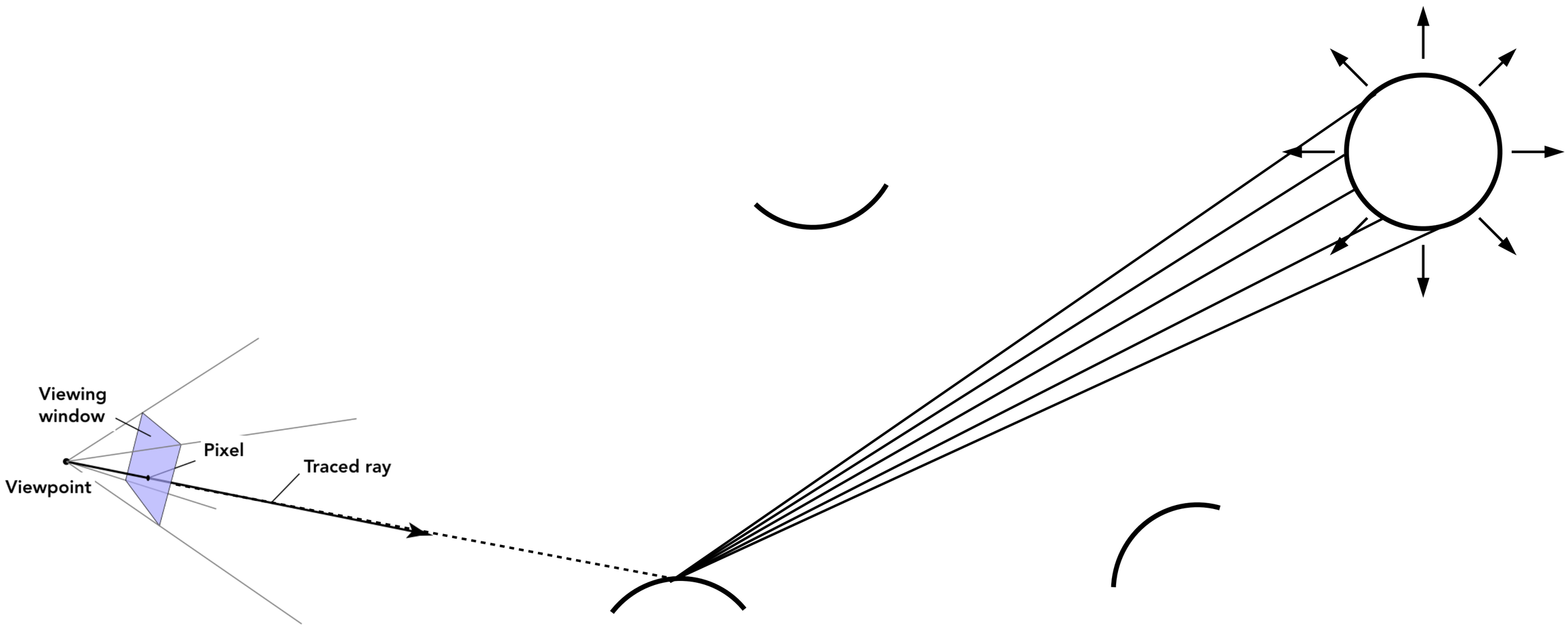
Sixteen-bounce global illumination

Light Paths

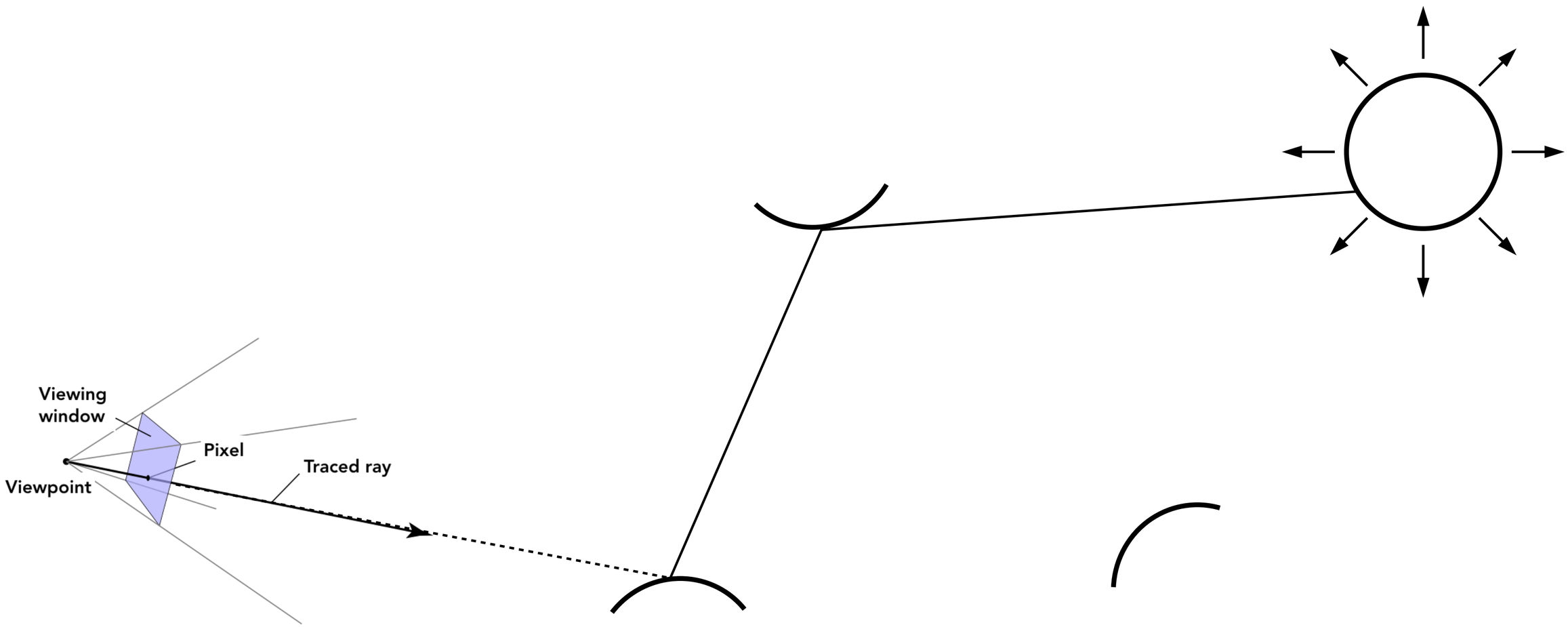
1-Bounce Path Connecting Ray to Light



1-Bounce Paths Connecting Ray to Light



2-Bounce Path Connecting Ray to Light



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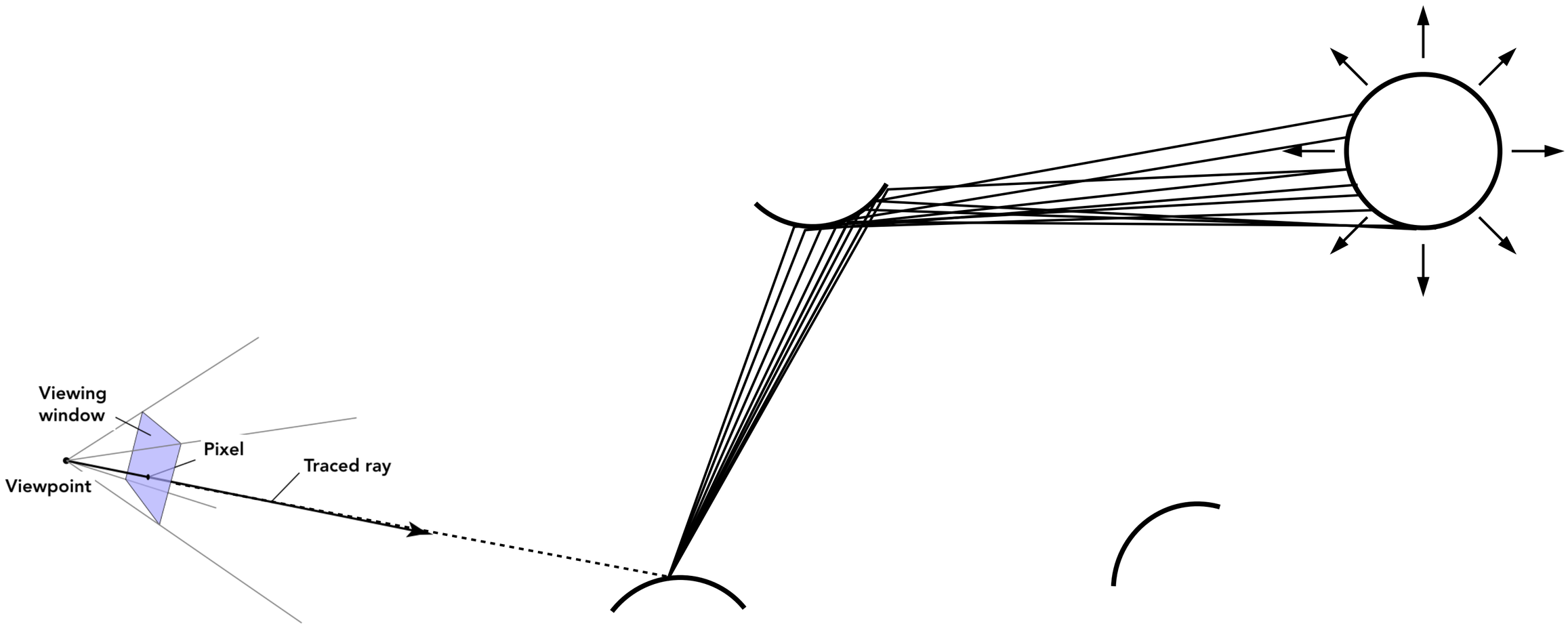
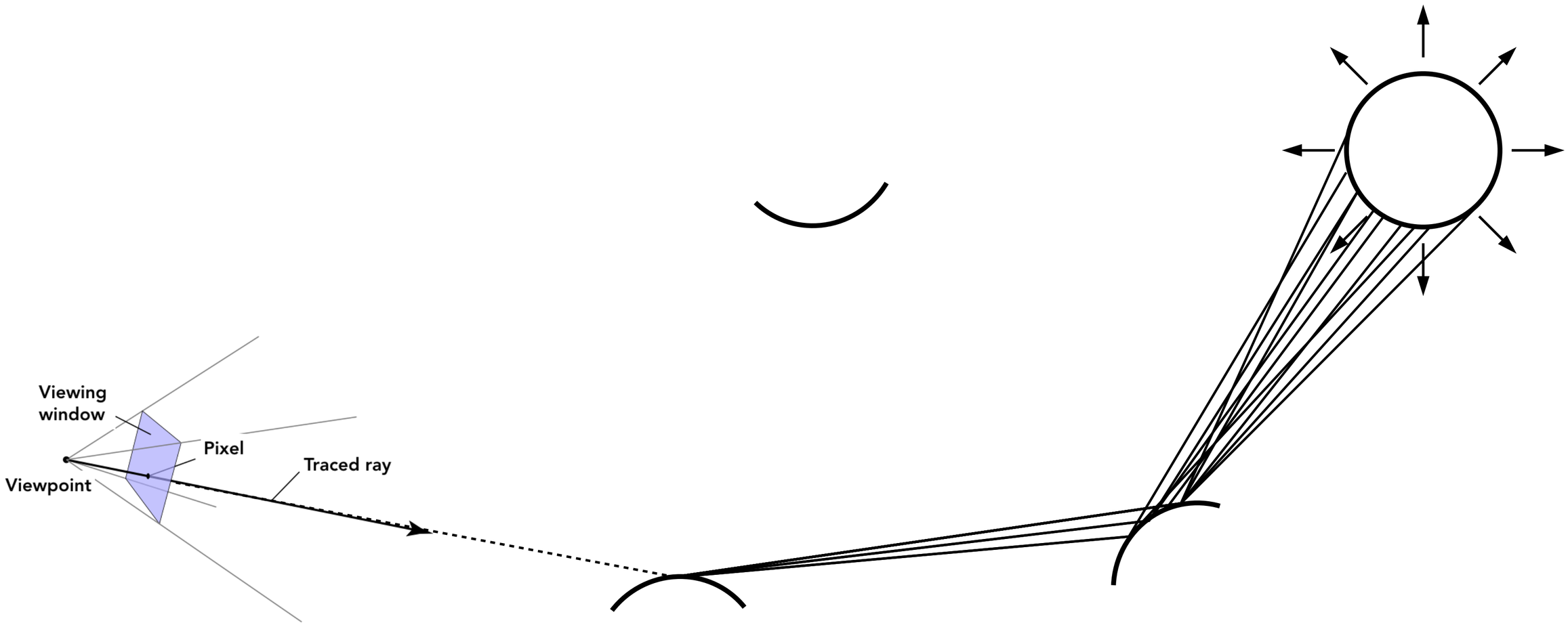


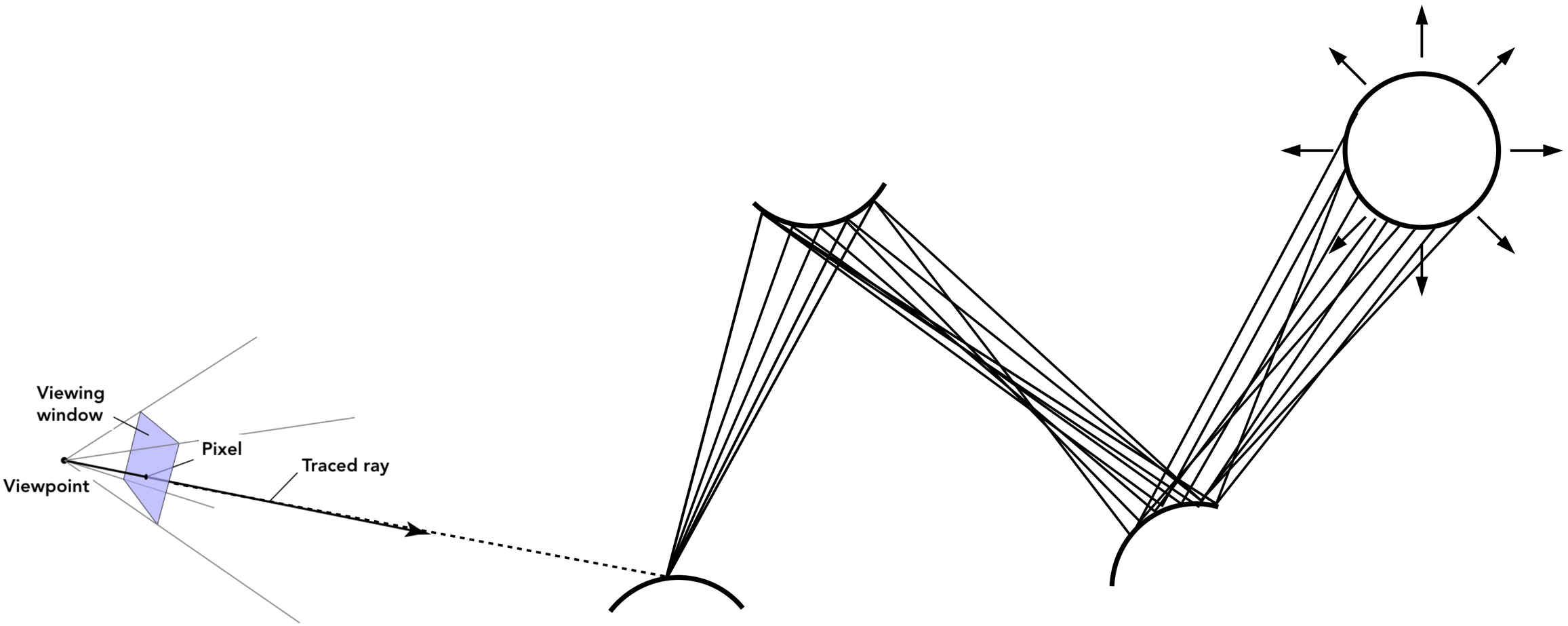
Camera

Light

2-Bounce Paths Connecting Ray to Light



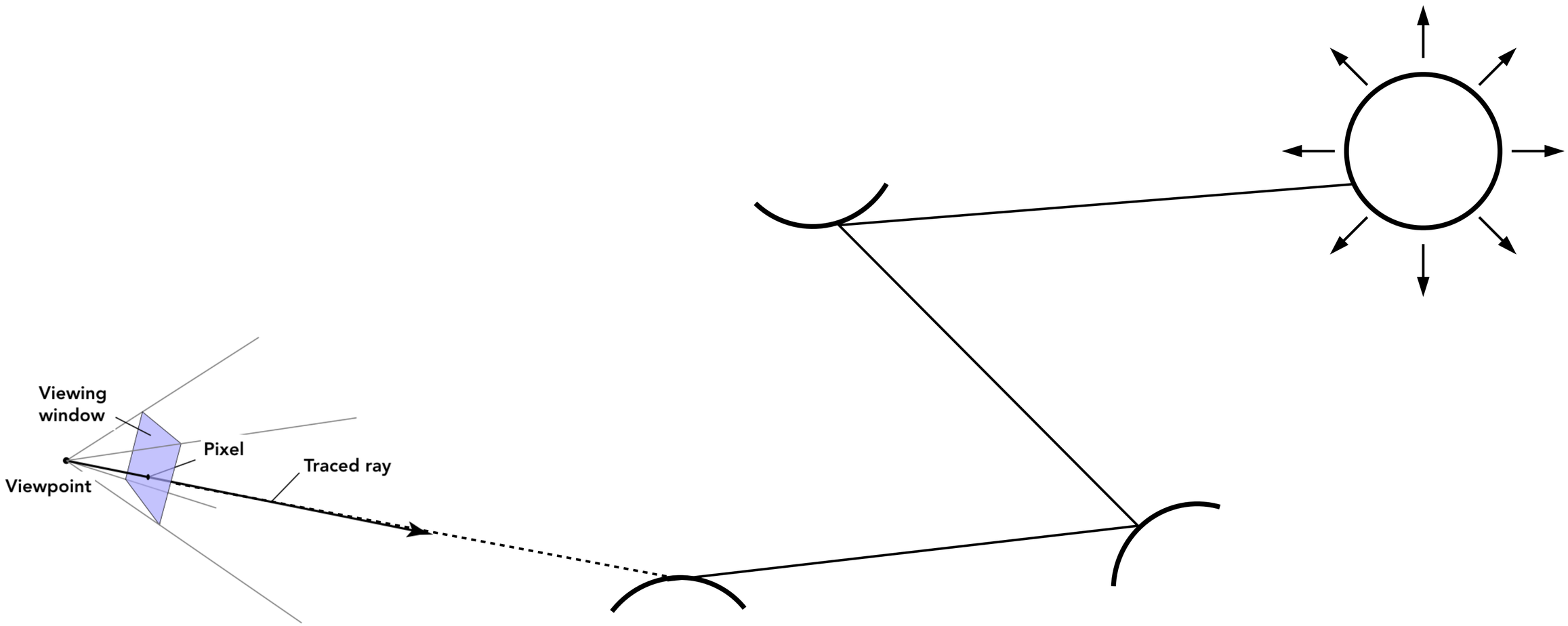
3-Bounce Paths Connecting Ray to Light



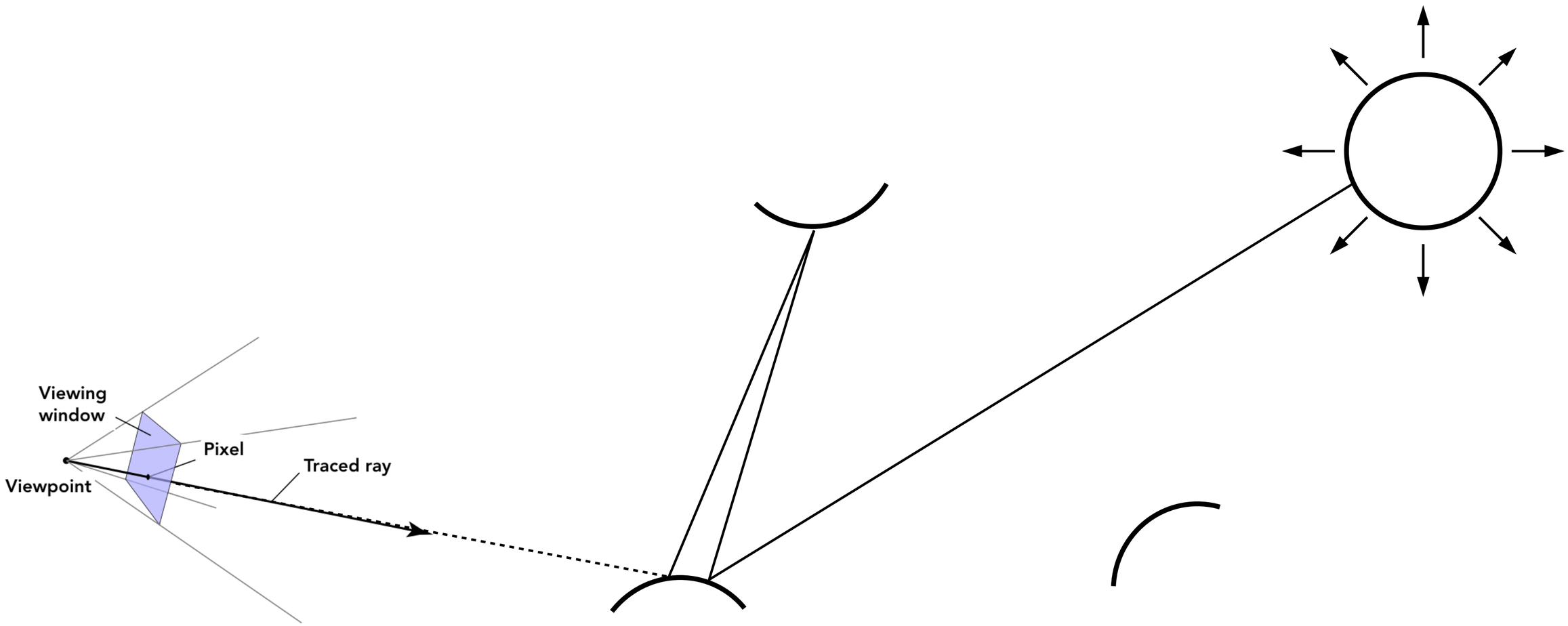
Camera

Light

3-Bounce Path Connecting Ray to Light



3-Bounce Path Connecting Ray to Light



Camera

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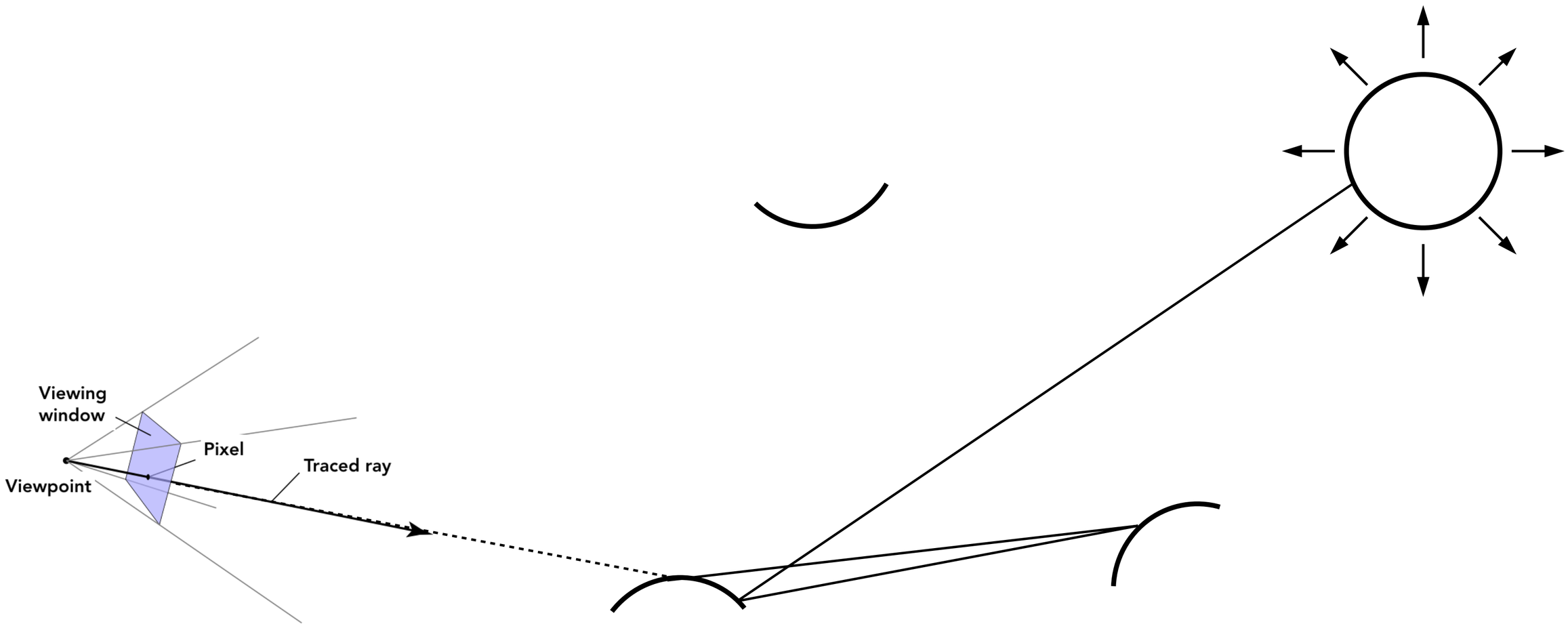

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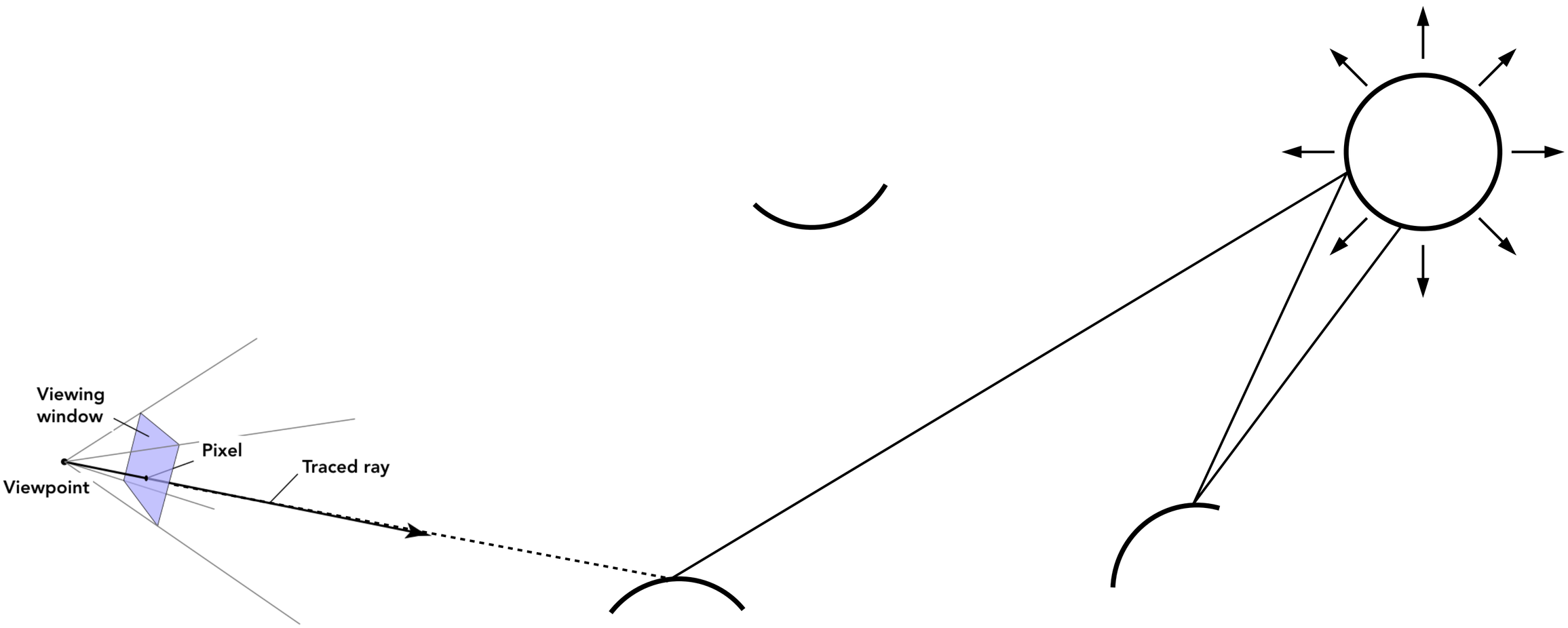

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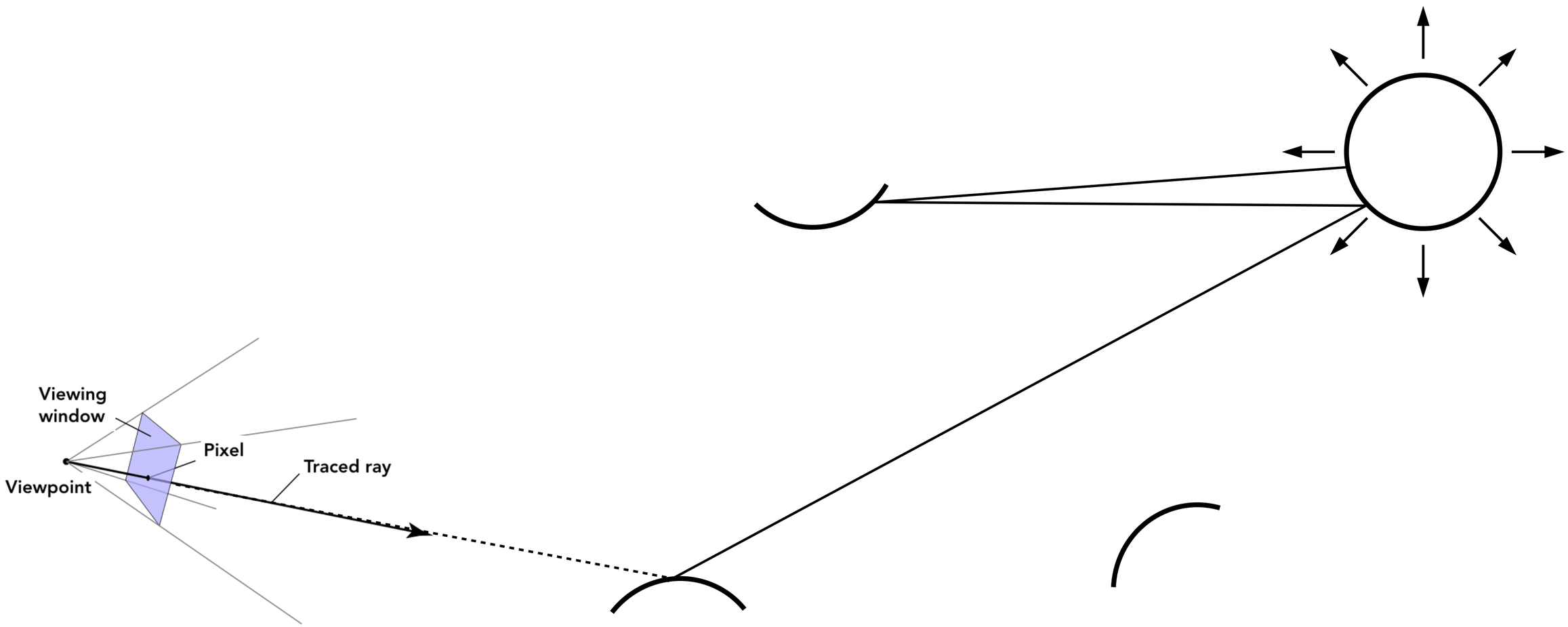

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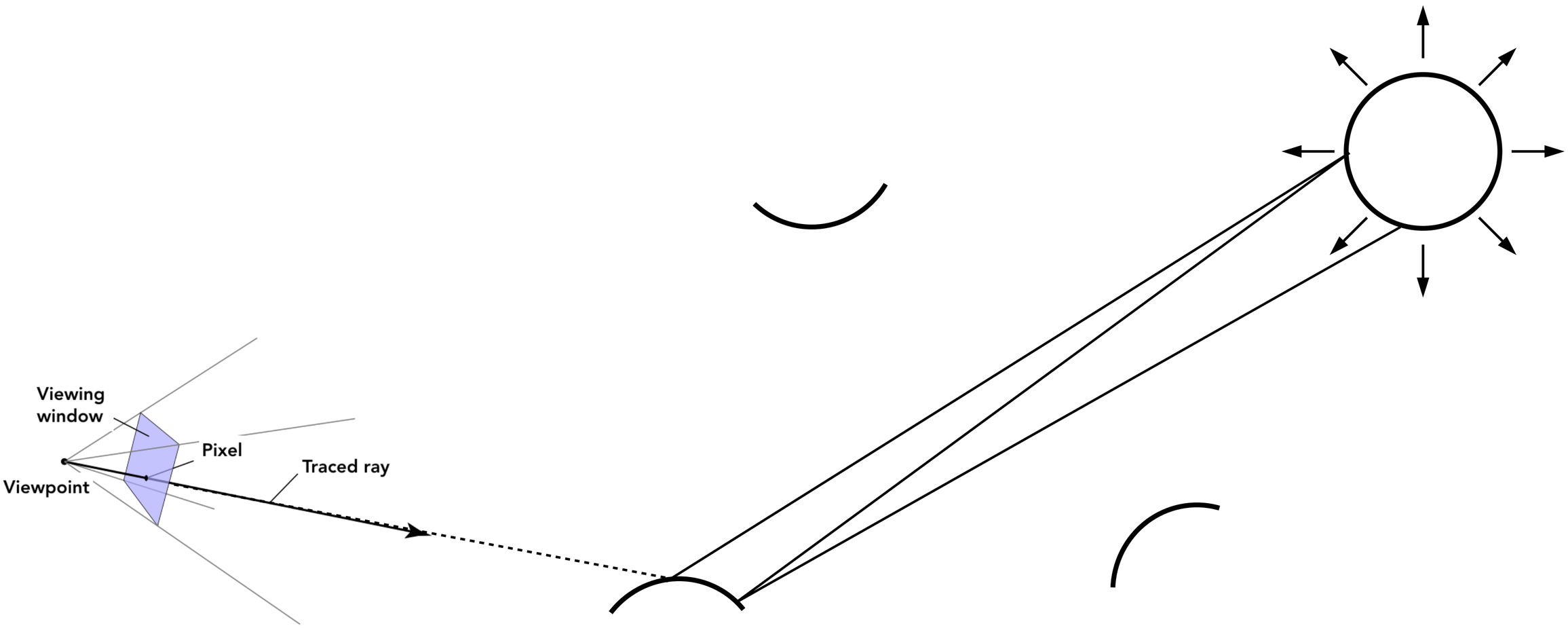

Global Illumination Rendering

Sum over all paths of all lengths

Challenges:

- How to generate all possible paths?
- How to sample space of paths efficiently?

Sum Over Paths

Try 1: Monte Carlo Sum over Paths

```
EstRadianceIn(x, ω)
    p = intersectScene(x, ω);
    L = p.emittedLight(–ω);
    ωi, pdf = p.brdf.sampleDirection(–ω);
    L += EstRadianceIn(p, ωi) * p.brdf(ωi, –ω) * costheta / pdf;
    return L;
```

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- **Note:**

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- Note:
  - Importance sampling BRDF

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- Note:
  - Importance sampling BRDF
  - Infinite recursion!

Problem: Infinite Bounces of Light

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How to integrate over infinite dimensions?

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- Note: if energy dissipates, contributions from higher bounces decrease exponentially

Idea: just use N bounces

- Problem: biased! No matter how many Monte Carlo samples, never see light taking N+1 to infinity bounces

Russian Roulette

Russian Roulette - Unbiased Random Termination

Idea: probabilistic termination of recursion

- At every recursive step (every bounce of light), probabilistically choose to stop the recursion
  - Specifically, continue with probability p_{rr}
- This goes from an infinite recursion, to a finite number of recursive function calls (how many?)
- But won't this bias our Monte Carlo integral estimate of the infinite bounces of light?
- Surprisingly, no! We can adjust the Monte Carlo estimator so that it remains unbiased (next slide)

Russian Roulette: Unbiased Random Termination

New estimator: evaluate original estimator with probability p_{rr}, reweighted. Otherwise ignore.

$$\text{Let } X_{\mathrm{rr}} = \begin{cases} \frac{X}{p_{\mathrm{rr}}}, \text{ with probability } p_{\mathrm{rr}} \\ 0, \text{ otherwise} \end{cases}$$

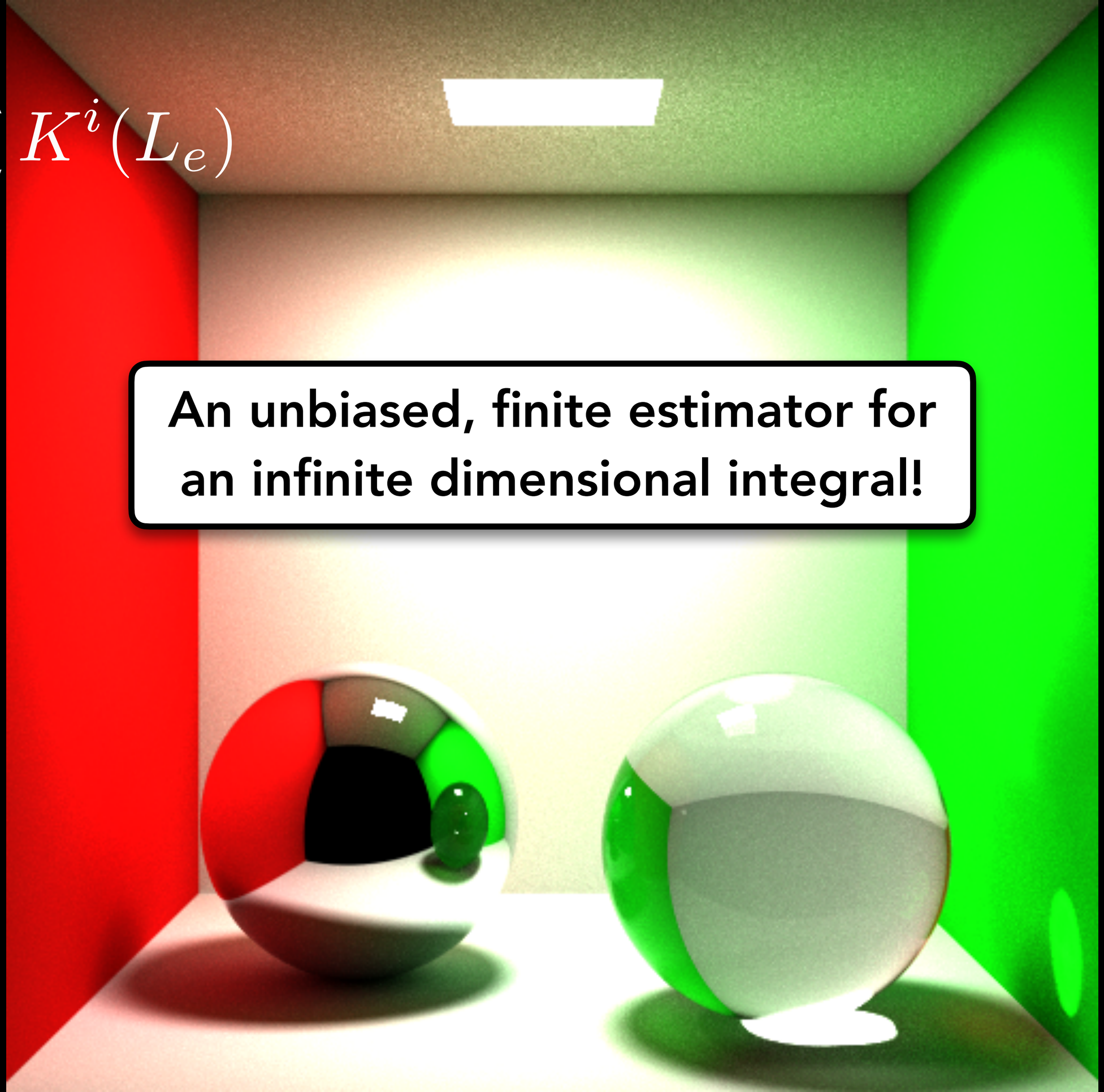
Same expected value as original estimator:

$$E[X_{\mathrm{rr}}] = p_{\mathrm{rr}}\, E\left[\frac{X}{p_{\mathrm{rr}}}\right] + (1 - p_{\mathrm{rr}})\, E[0] = E[X]$$

Want to choose p_{rr} considering Monte Carlo efficiency

- **Terminate if expensive and/or we expect low contribution**
- **In path tracing, expensive to recursively trace path. Increase termination probability if brdf is low in next bounce direction**

$$\sum_{i=0}^{\infty} K^i(L_e)$$



An unbiased, finite estimator for an infinite dimensional integral!

Try 2: Russian Roulette Monte Carlo over Paths

```
EstRadianceIn(x, ω)
    p = intersectScene(x, ω);
    L = p.emittedLight(–ω);
    ωi, pdf = p.brdf.sampleDirection(–ω);
    cpdf = continuationProbability(p.brdf, ωi);
    if (random01() < cpdf)                          // Russian Roulette
        L += EstRadianceIn(p, ωi)                   // Recursion
            * p.brdf(ωi, –ω) * costheta / pdf / cpdf;
    return L;
```

Try 2: Russian Roulette Monte Carlo over Paths

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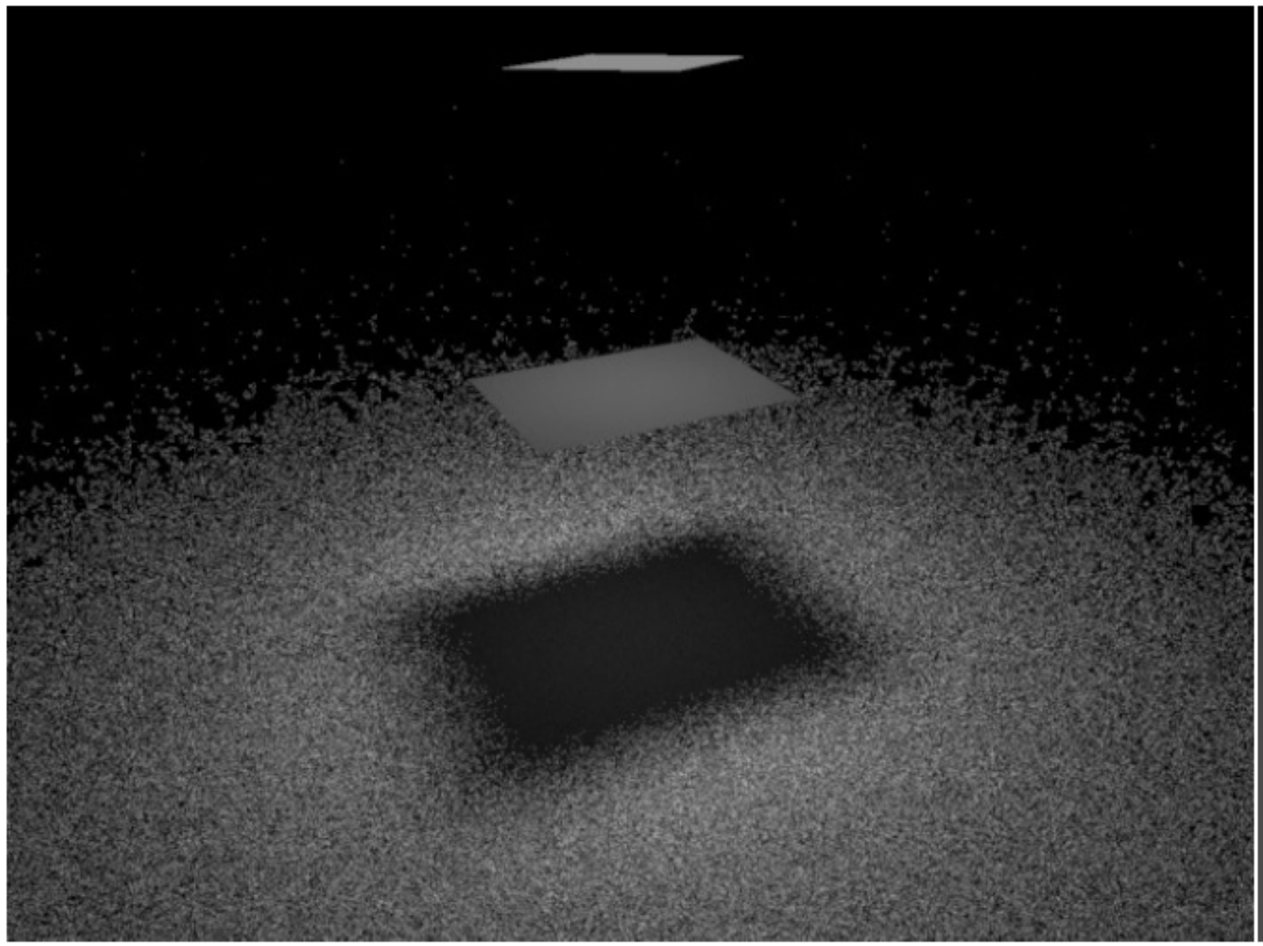
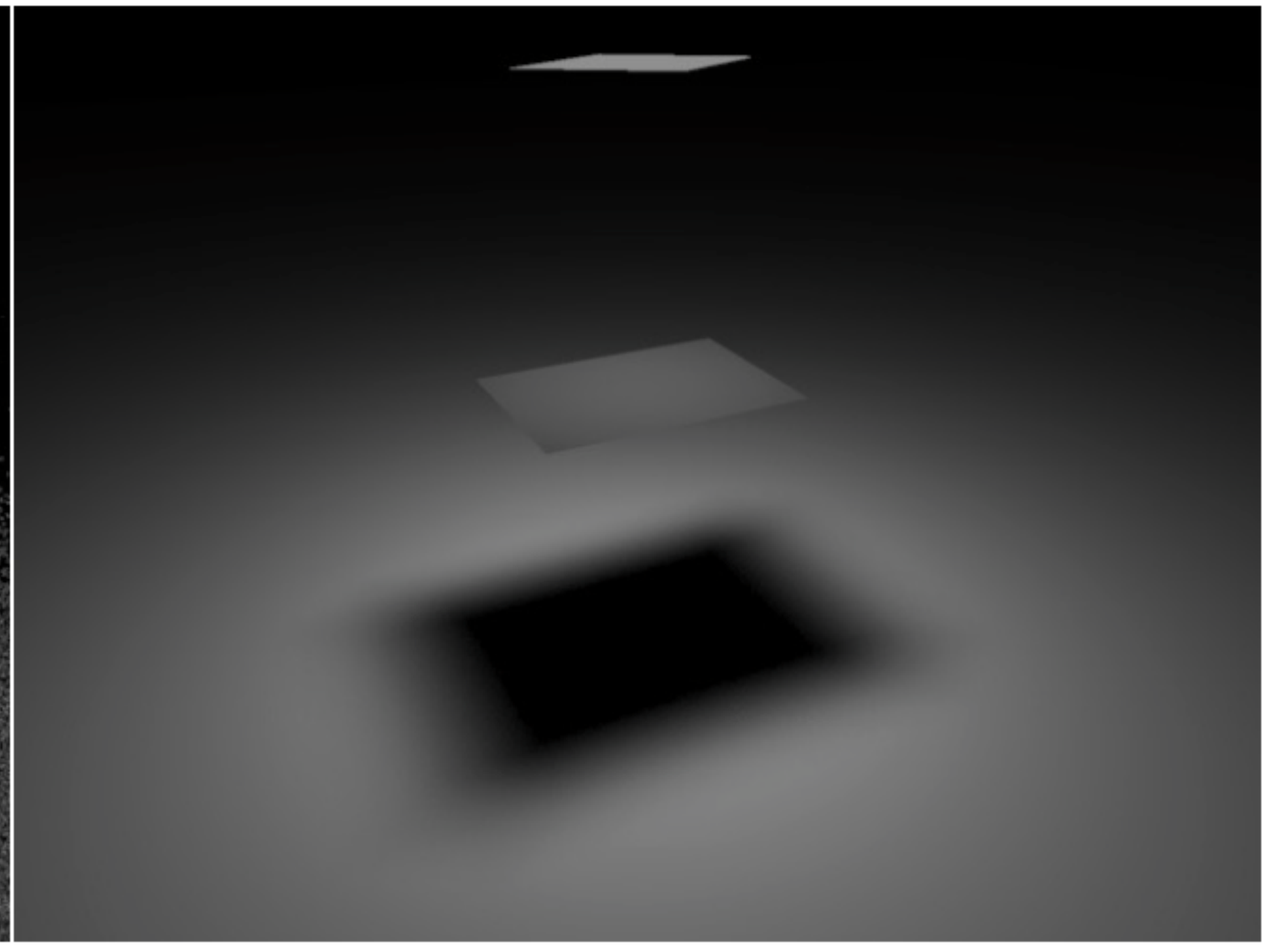
EstRadianceIn(x, ω)
    p = intersectScene(x, ω);
    L = p.emittedLight(–ω);
    ωi, pdf = p.brdf.sampleDirection(–ω);
    cpdf = continuationProbability(p.brdf, ωi);
    if (random01() < cpdf)                          // Russian Roulette
        L += EstRadianceIn(p, ωi)                   // Recursion
            * p.brdf(ωi, –ω) * costheta / pdf / cpdf;
    return L;

// Unbiased, computation terminates, but still extremely noisy!
```

Recall: Importance Sampling



Solid angle sampling



Light area sampling

Path Tracing

Path Tracing Overview

Terminate paths randomly with Russian Roulette

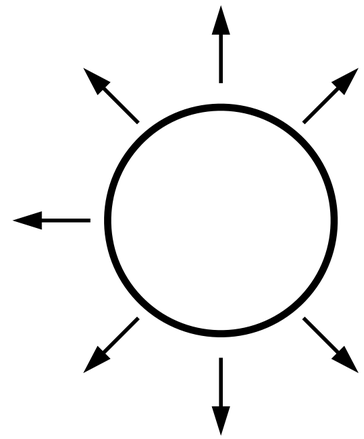
Partition the recursive radiance evaluation. At each point on light path

- Direct lighting – non-recursive, importance sample lights
- Indirect lighting – recursive, importance sample BRDF

Monte Carlo estimate for each partition separately

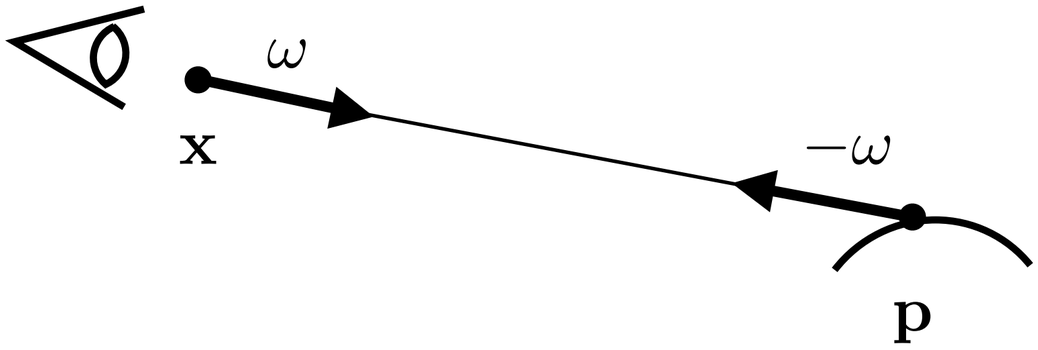
- Possible to take just one sample for each
- Assume: 100s - 1000s of paths sampled per pixel

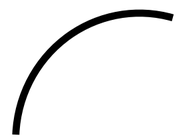
Partitioning the Rendering Equation



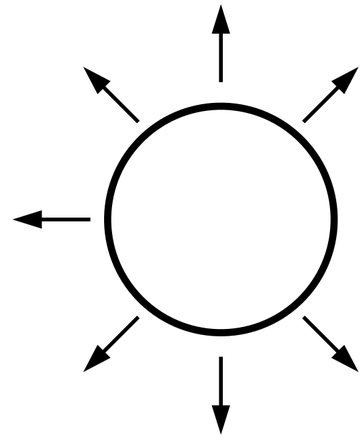
EstRadianceIn(x, ω)

= EstRadianceOut(p, –ω)

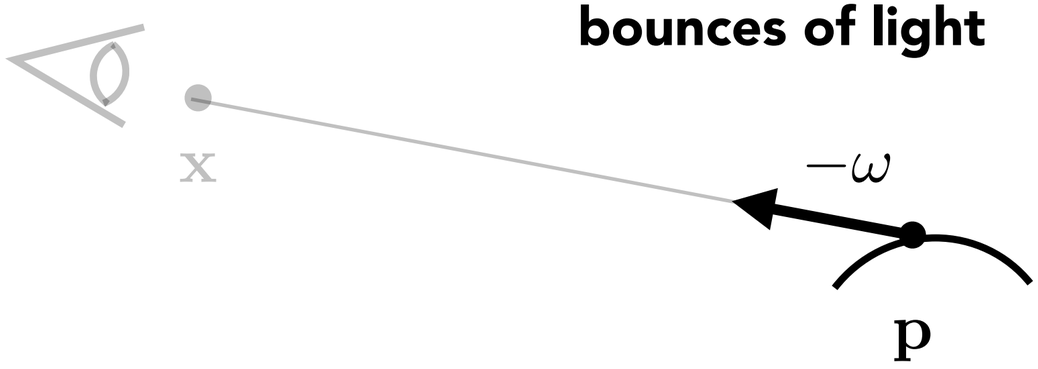


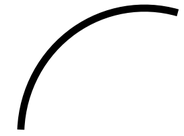


Partitioning the Rendering Equation



Need to sum paths going through p representing 0, 1, 2, 3, … bounces of light





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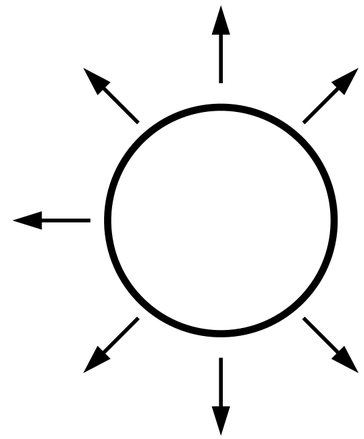
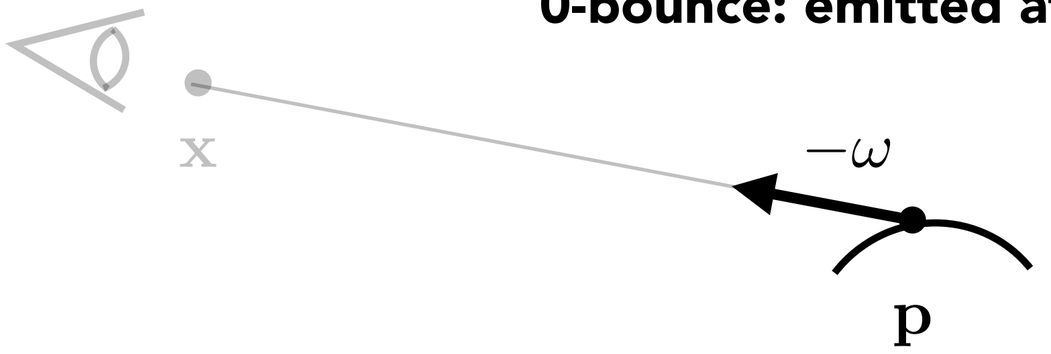
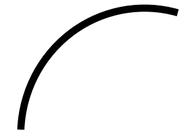


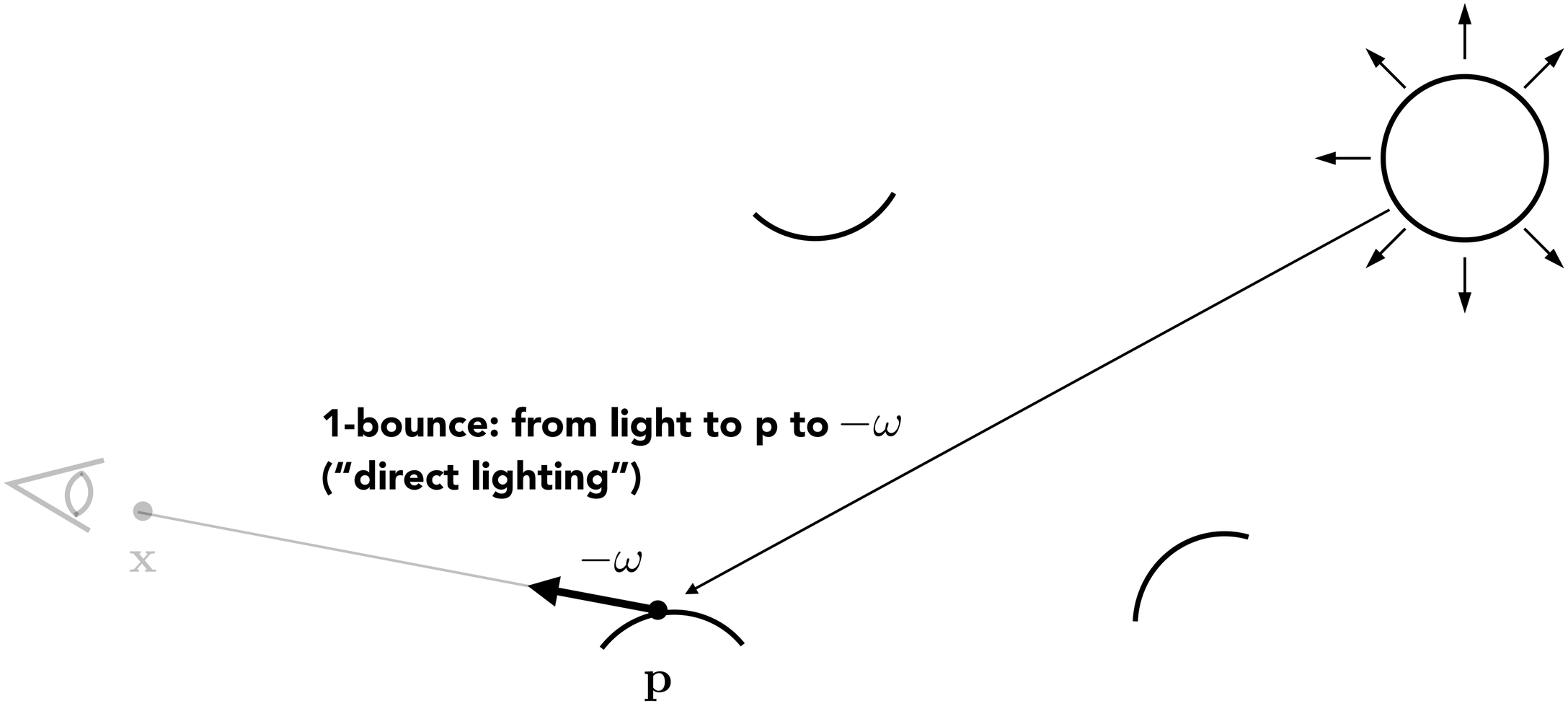




At p, consider light contributions from paths of varying bounce-length

- **0-bounce: light emitted from p (p is on a light source)**

Partitioning the Rendering Equation



At p, consider light contributions from paths of varying bounce-length

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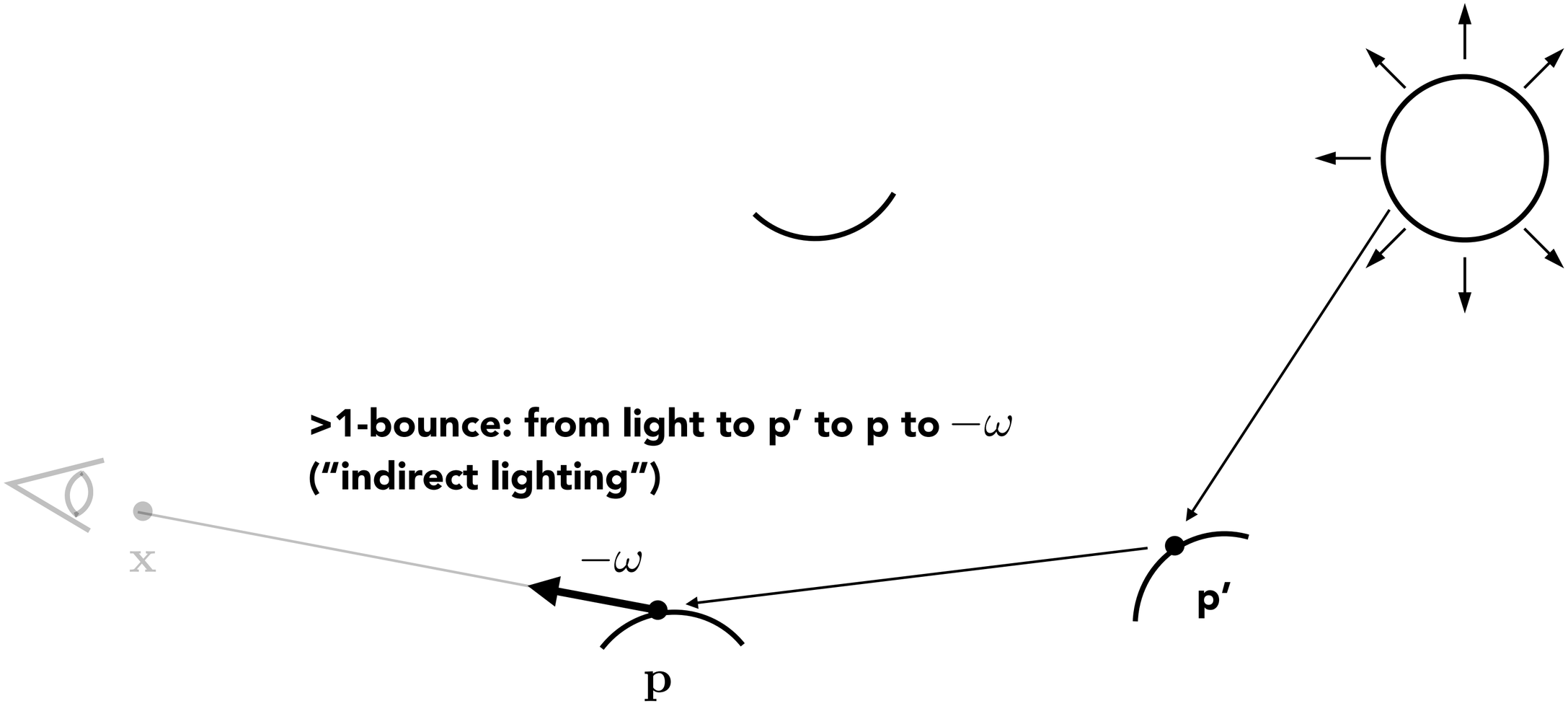
- **1-bounce: from light to p to x ("direct illumination")**

Partitioning the Rendering Equation



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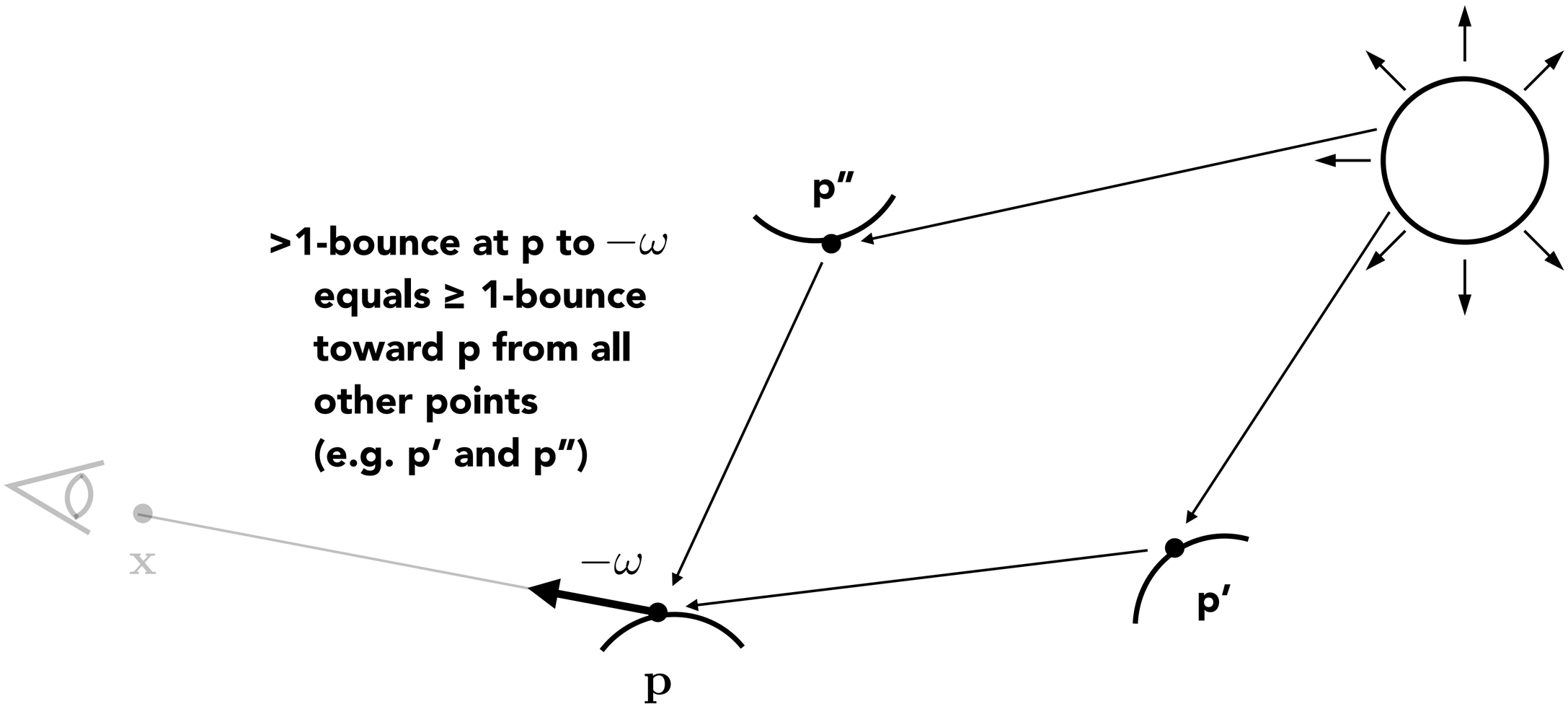
- **>1-bounce: from light to at least one other point to p to x ("indirect illumination")**

Consider Evaluation of >1 Bounce of Light



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Path Tracing Pseudocode

```
EstRadianceIn(x, ω)                          // incoming at x from dir ω
    p = intersectScene(x, ω);
    return ZeroBounceRadiance(p, –ω)
         + AtLeastOneBounceRadiance(p, –ω);

ZeroBounceRadiance(p, ωo)                    // outgoing at p in dir ωo
    return p.emittedLight(ωo);
```

Path Tracing Pseudocode

```
AtLeastOneBounceRadiance(p, ωo)              // out at p, dir ωo
    L = OneBounceRadiance(p, ωo);           // direct illum

    ωi, pdf = p.brdf.sampleDirection(ωo);     // Imp. sampling
    p' =  intersectScene(p, ωi);
    cpdf = continuationProbability(p.brdf, ωi, ωo);
    if (random01() < cpdf)                                  // Russ. Roulette
        L += AtLeastOneBounceRadiance(p', –ωi)       // Recursive est. of
        * p.brdf(ωi, ωo) * costheta / pdf / cpdf;// indirect illum
    return L;

OneBounceRadiance(p, ωo)                            // out at p, dir ωo
     return DirectLightingSampleLights(p, ωo); // direct illum
```

Direct Lighting Pseudocode (Lights)

```
DirectLightingSamplingLights(p, ωo)
    L, ωi, pdf = lights.sampleDirection(p);   // Imp. sampling

    if (scene.shadowIntersection(p, ωi))      // Shadow ray
        return 0;
    else
        return L * p.brdf(ωi, ωo) * costheta / pdf;

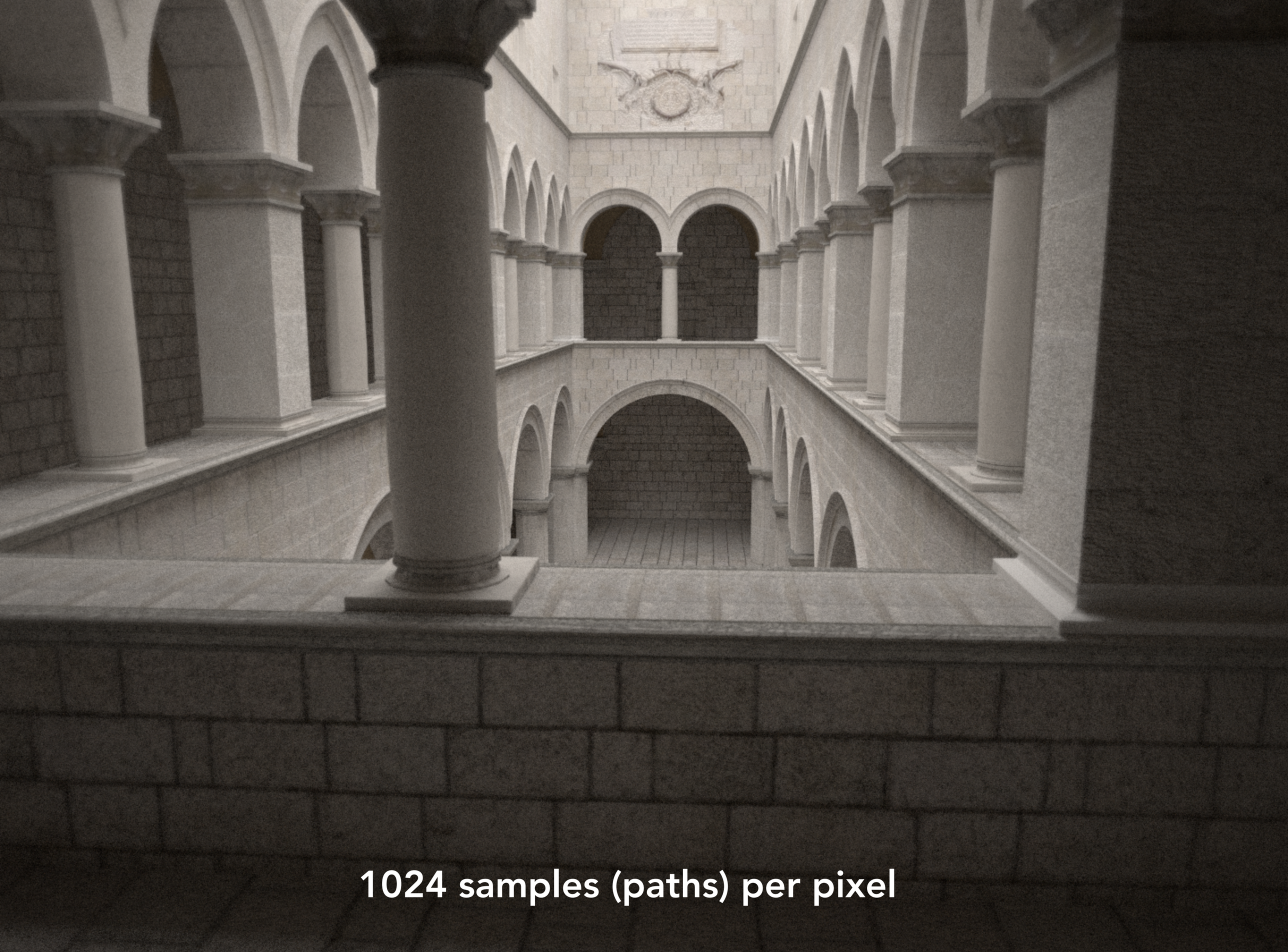
// Note: only one random sample over all lights.
// Assignment 3-1 asks you to, alternatively, loop over
// multiple lights and take multiple samples (later slide)
```

Direct Lighting (0 and 1 Bounce Only)

Path Tracing (All Bounces of Light)

One sample (path) per pixel

32 samples (paths) per pixel



1024 samples (paths) per pixel

Summary of Intuition on Global Illumination & Path Tracing

Summary of Intuition on G.I. & P.T.

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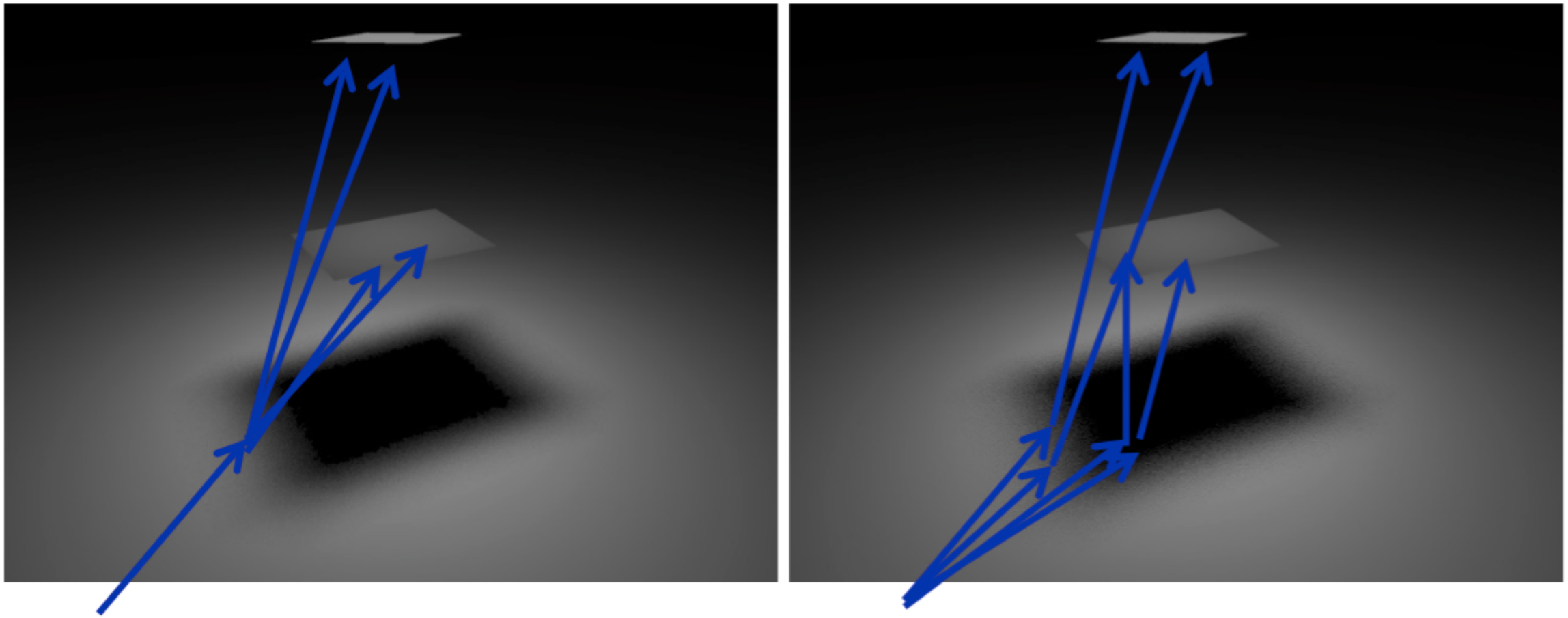
- Build paths by recursively tracing to next surface point and choosing a random reflection direction. At each surface, sum emitted light and reflected light
- How to terminate paths? We use Russian Roulette to probabilistically stop the recursion
- How to reduce noise? Use importance sampling in choosing random direction. Two ways: importance sample the lights, and importance sample the BRDF.

Implementation Notes

Paths vs Trees



4 eye rays per pixel
16 shadow rays per eye ray
(68 ray traces per pixel)

64 eye rays per pixel
1 shadow ray per eye ray
(128 ray traces per pixel)

64 image samples x 1 light sample, 13.2 seconds

1 image sample x 64 light samples, 7.0 seconds

8 image samples x 8 light samples, 7.7 seconds

Multiple Light Sources

Consider multiple lights in direct lighting estimate

One strategy:

- Loop over all N lights, sum Monte-Carlo estimates for each light
- For each light: compute Monte Carlo estimate with M samples taken with importance sampling

Needs N * M samples

This is what the assignment asks you to implement.

Multiple Light Sources (Single Sample)

Consider random sampling of multiple lights with a single sample

- Randomly choose light i, with probability pi
- Randomly sample over that light's directions, with probability pL
- Probability of choosing sample is (pi * pL)
- Weight the lighting calculation by 1/(pi * pL)
- Is this estimator unbiased? Yes!
- How would you importance sample intelligently?

Can of course average N such samples

Point Lights / Ideal Specular Materials – Issues

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Sampling problems

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- When sampling directions randomly, we have zero probability of matching exact direction of a point light or mirror reflection / specular refraction

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- In direct lighting, importance sample point lights by generating a single sample pointing directly at the light (only one sample needed)

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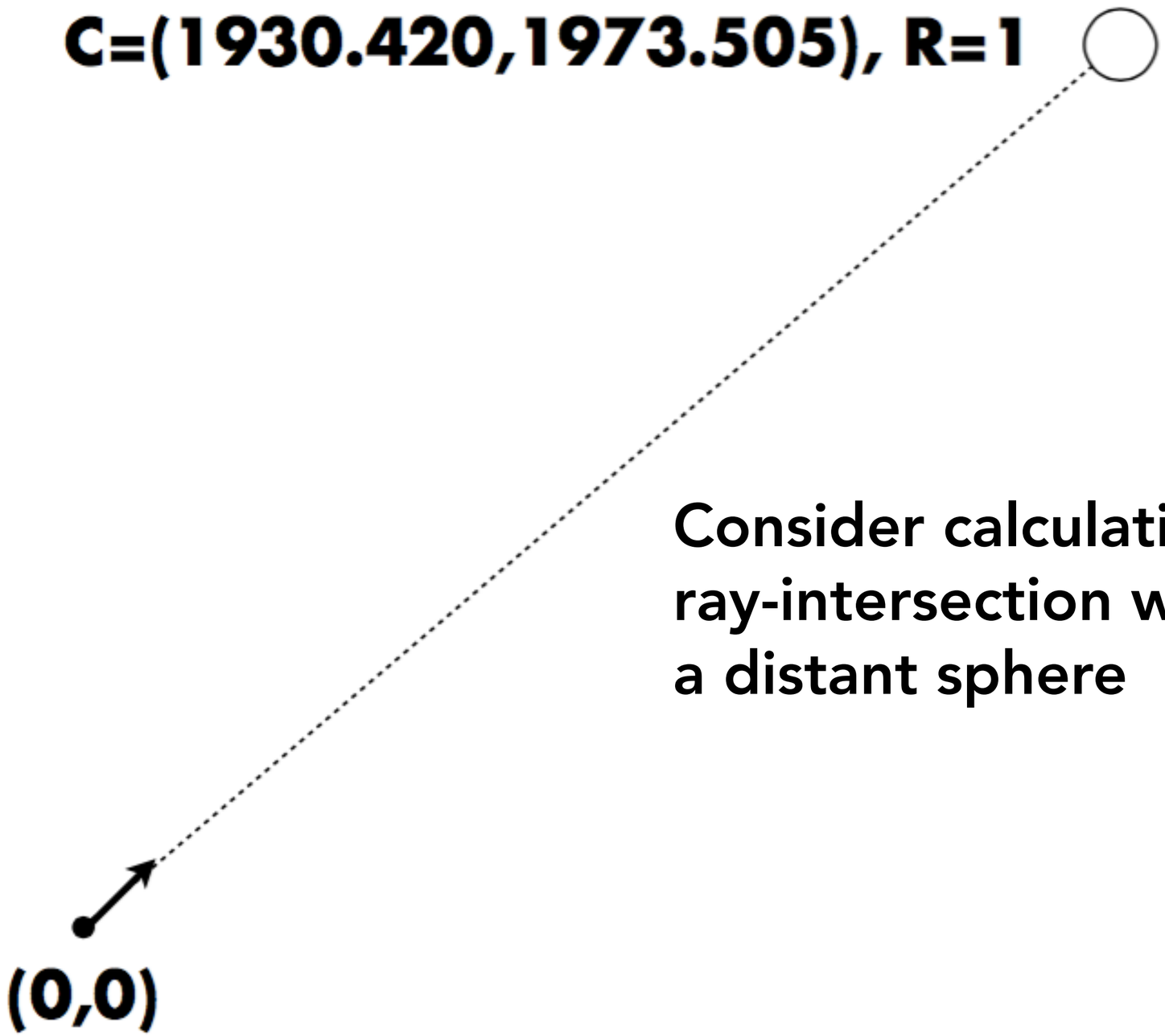
- When sampling directions randomly, we have zero probability of matching exact direction of a point light or mirror reflection / specular refraction

Remedy

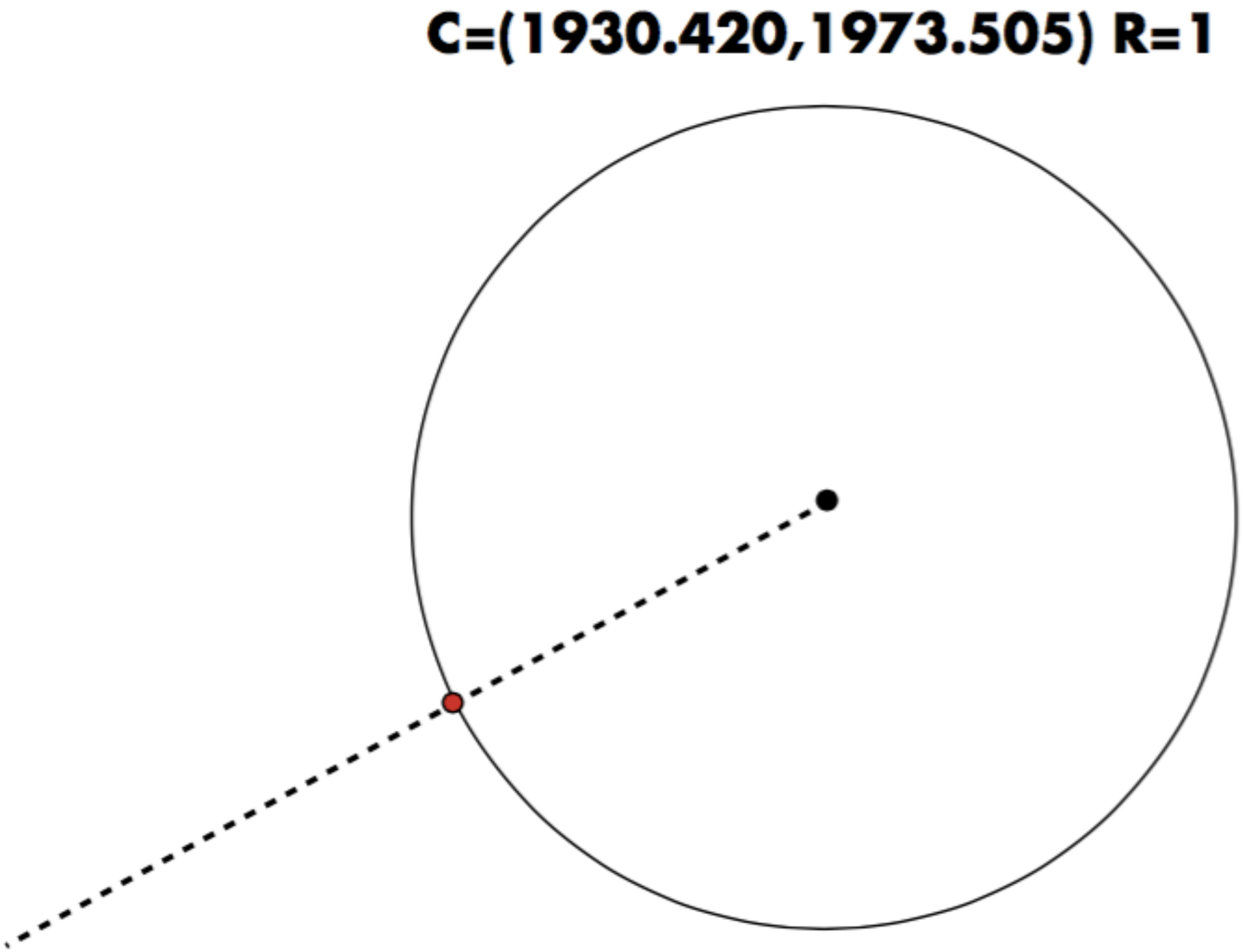
- In direct lighting, importance sample point lights by generating a single sample pointing directly at the light (only one sample needed)
- In indirect lighting, importance sample specular BRDFs by generating a single sample point directly along the specular refraction / transmission direction

Numerical Precision Issues



Consider calculating ray-intersection with a distant sphere

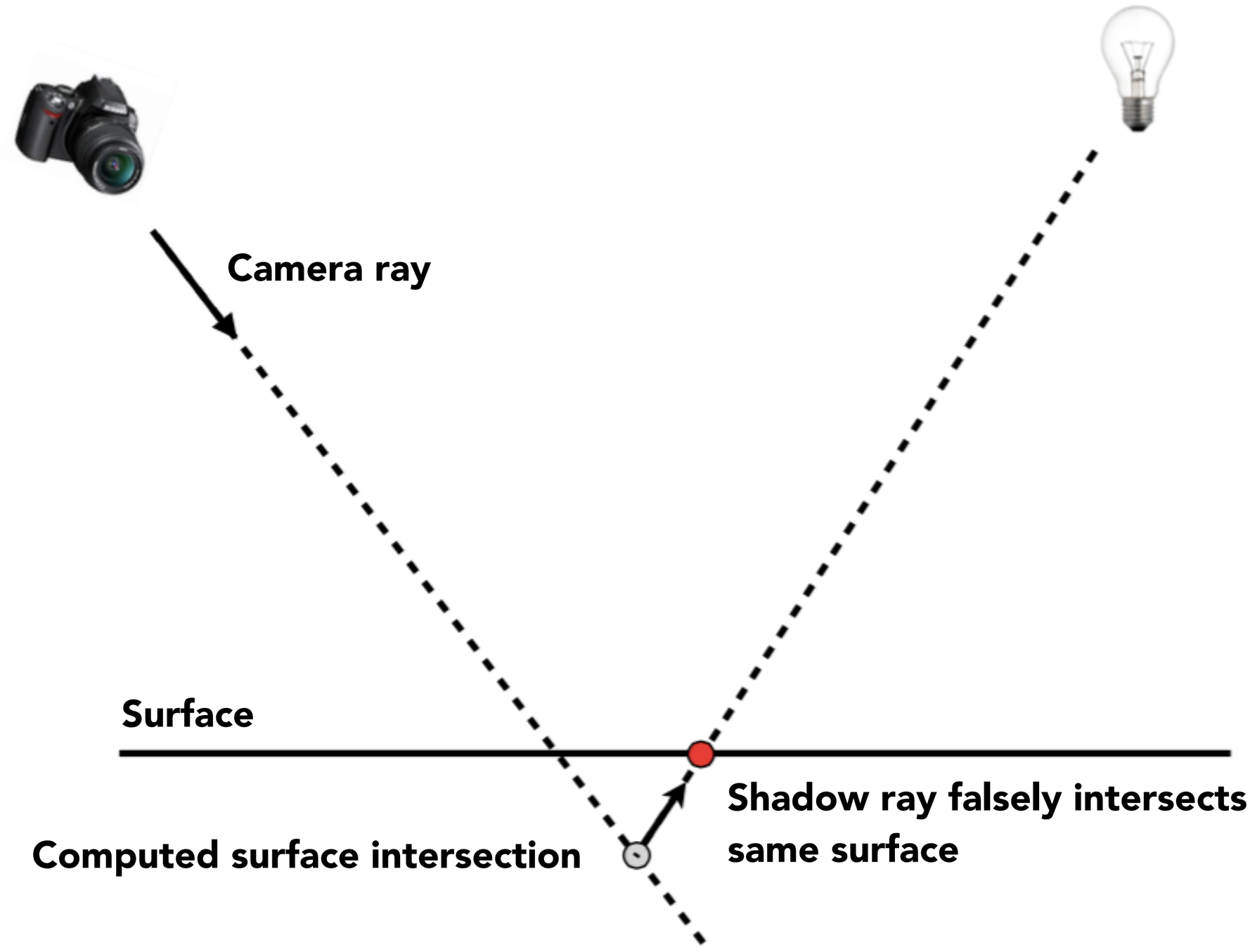
Numerical Precision Issues



True Intersection: (1929.7203..., 1972.7897...)

Computed Intersection: (1930.4196..., 1973.5054...)

Noisy Shadows



Noisy shadows because of
floating point precision problems

Floating-Point Precision Remedies

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1. double (fp64) rather than float (fp32)
   - 53-bits of precision instead of 24-bits
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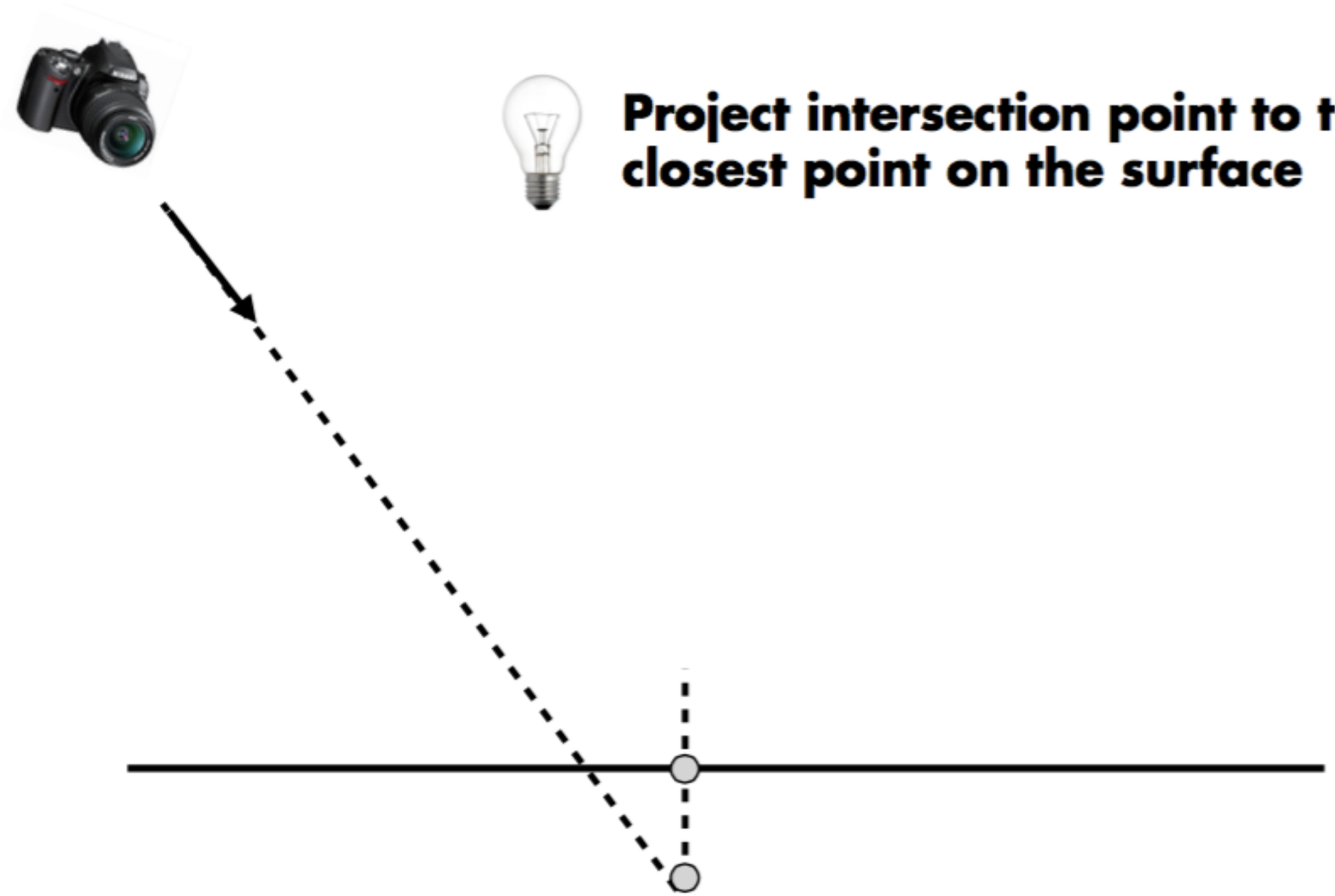
Remedy: Project Intersection Point to Surface



Project intersection point to the closest point on the surface

Good Scenes for Path Tracing (Diffuse, Sky Lighting)

M. Fajardo, Arnold Path Tracer

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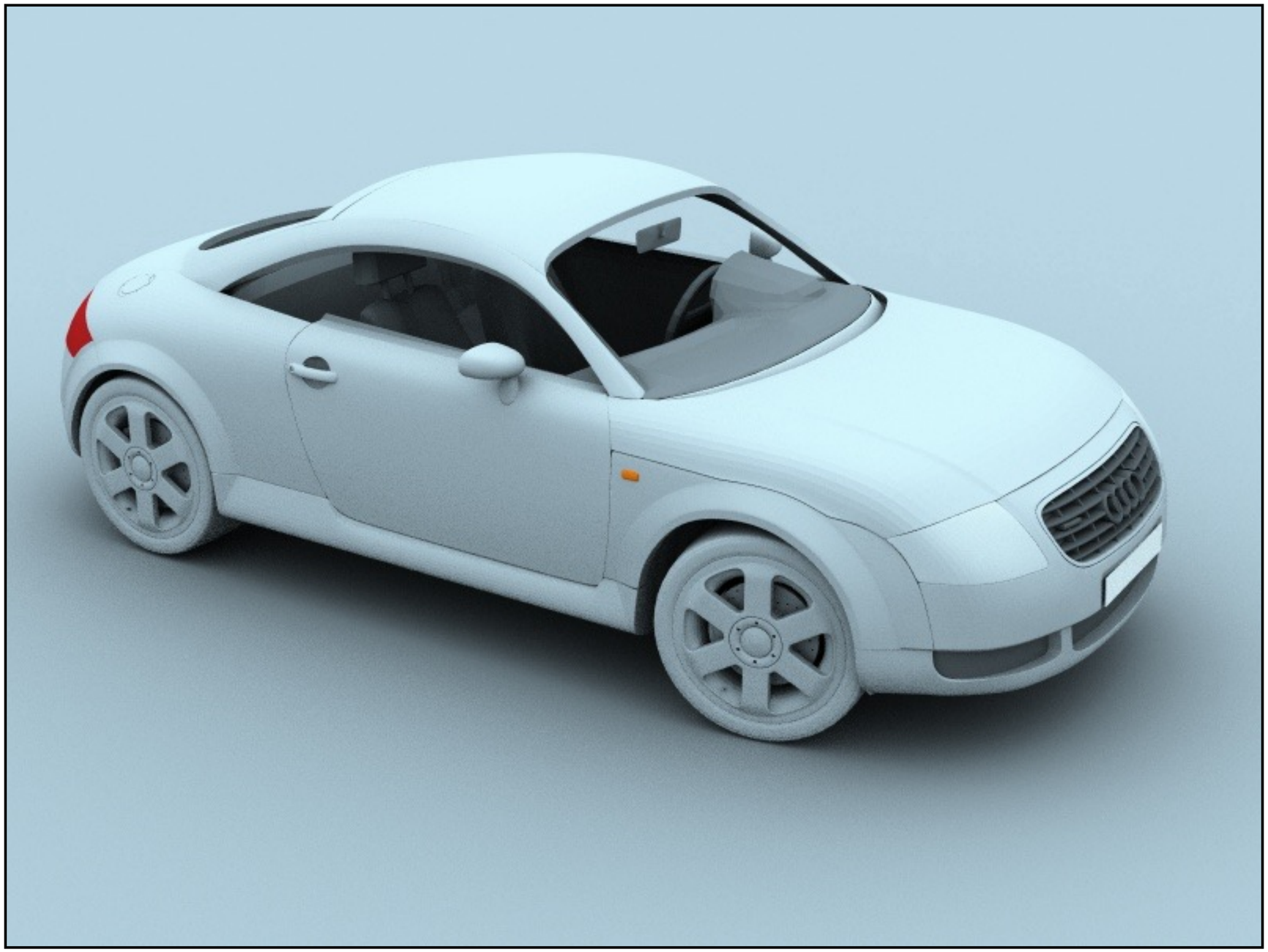

M. Fajardo, Arnold Path Tracer

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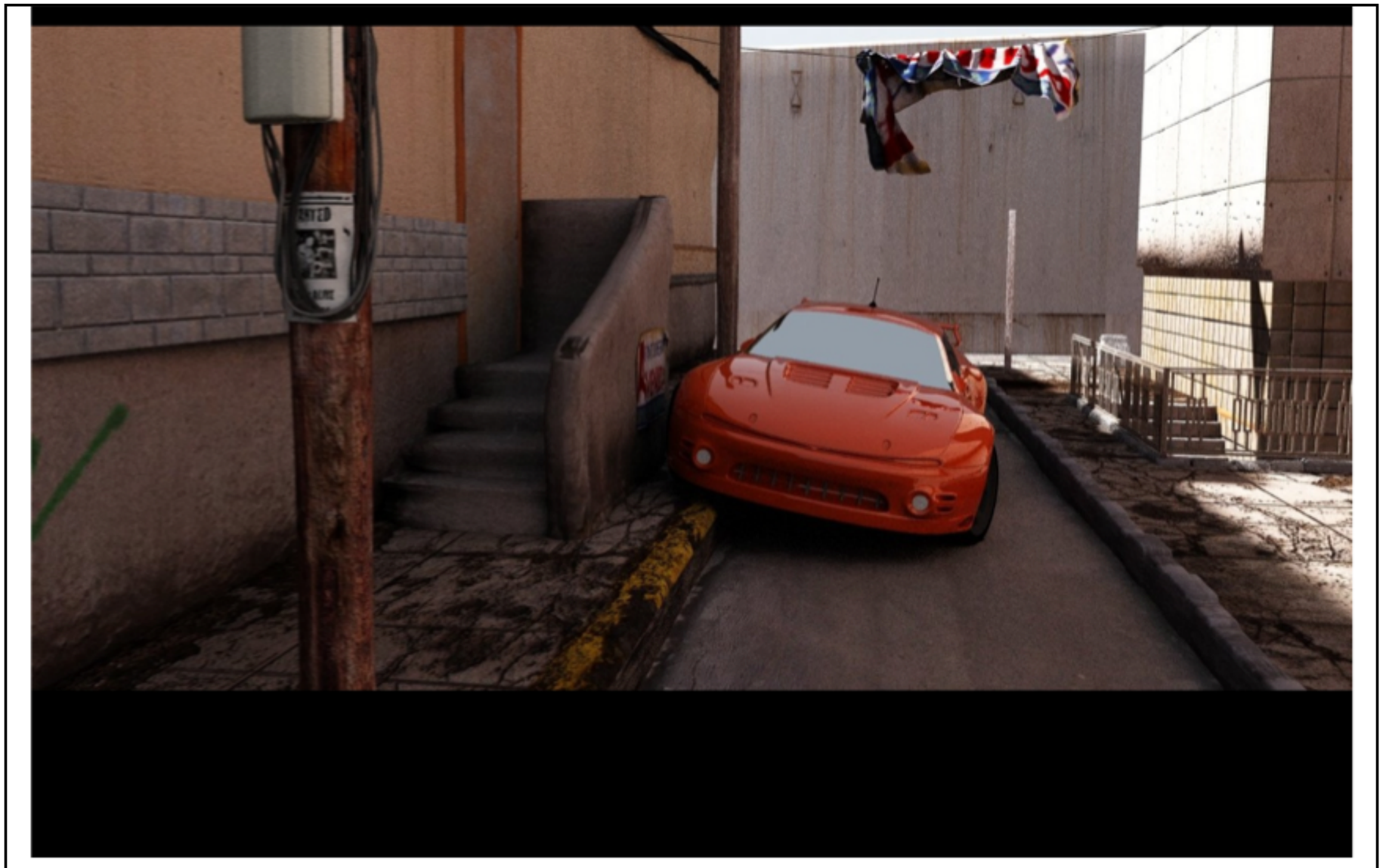
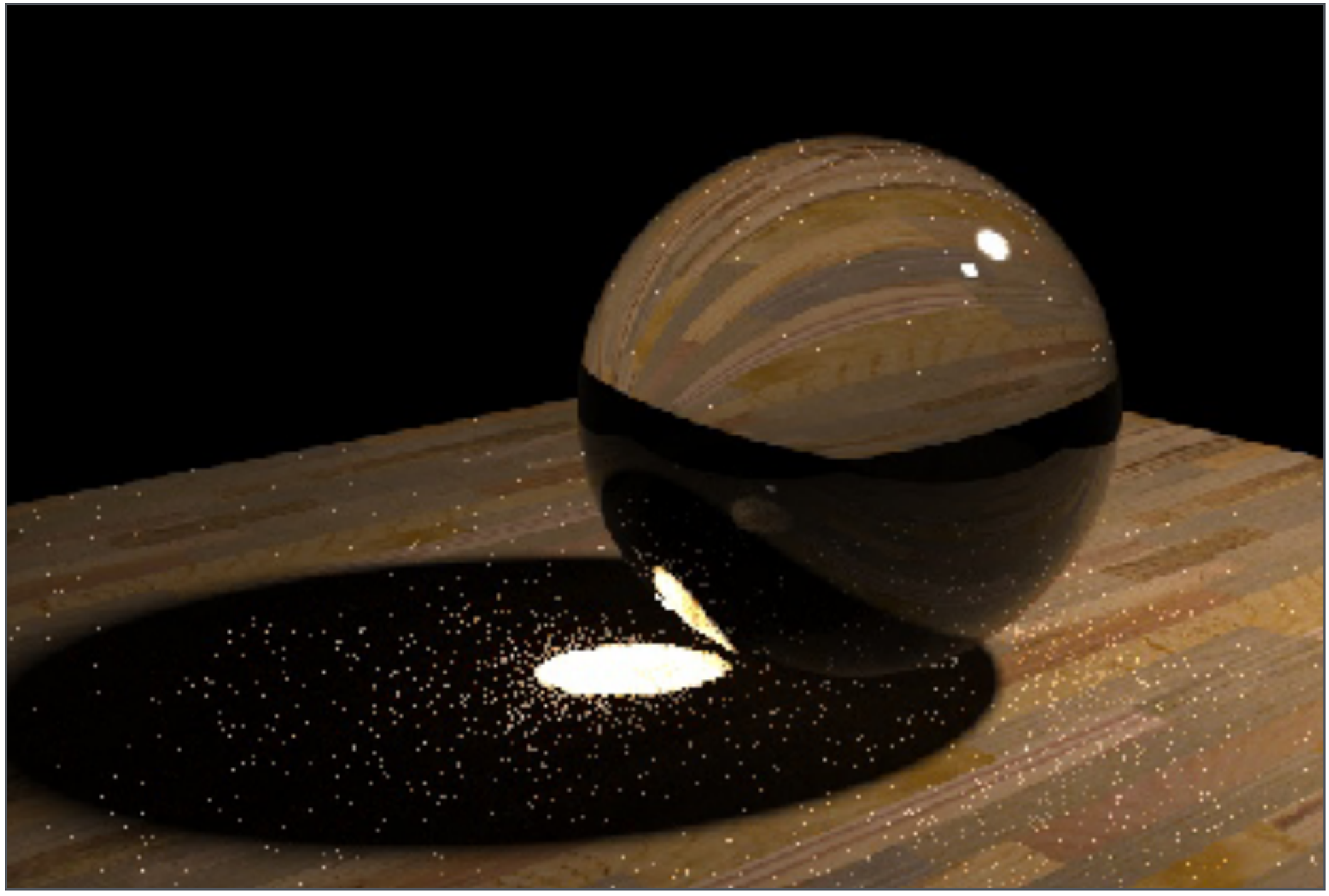


Street scene 1

1536x654, 16 paths/pixel, 2 bounces, 250,000 faces, 18 min., dual PIII-800

M. Fajardo, Arnold Path Tracer

A Challenging Scene for Path Tracing – Why?



Henrik Wann Jensen

1000 paths / pixel

Things to Remember

Global illumination challenge: recursive light transport

Reflection & rendering equations, operator notation

Neumann solution of rendering equation

- Sum successive bounces of light, infinite series

Path tracing

- Russian Roulette for unbiased finite estimate of infinite series (infinite dimensional integral)
- Partition into direct and indirect illumination
- Importance sampling of lighting and BRDF

Acknowledgments

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